DevOrSysAdminScripts

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Abstract

Some "small" useful scripts for enhancing development or system-administrators tasks

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Code lines: 10041 total lines, 9086 not empty lines, 955 empty lines.

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1 Files tree

```
.
├── build_and_checks_dependencies
│   ├── build_and_checks.exec.sh.tpl
│   ├── build_md_from_printable_md.exec.sh
│   ├── check_shell_scripts_beginnings.libr.sh
│   ├── check_shell_scripts_indentation.libr.sh
│   ├── check_URLs.libr.sh
│   ├── common_build_and_checks.exec.sh
│   ├── common_build_and_checks.exec.sh.tpl
│   ├── common_options.libr.sh
│   ├── comparisons.libr.sh
│   ├── create_PDF.exec.sh
│   ├── generate_from_template.libr.sh
│   ├── get_common_text_glob_patterns.libr.sh
│   ├── grammar_and_spell_check.libr.sh
│   ├── licenses_templates
│   │   ├── build_licenses_templates.exec.sh
│   │   ├── license_file_header_AGPLv3+.tpl
│   │   ├── license_file_header_GPLv3+.tpl
│   │   └── license_file_header_LGPLv3+.tpl
│   ├── lines_counts.libr.sh
│   ├── lines_filters.libr.sh
│   ├── lines_maps.libr.sh
│   ├── overwrite_if_not_equal.libr.sh
│   ├── python_black_complement.libr.sh
│   ├── python_isort_complement.libr.sh
│   ├── repository_data.libr.sh
│   ├── shell_checks_complement.libr.sh
│   ├── split_score.exec.php
│   ├── split_score.libr.php
│   ├── strings_functions.libr.sh
│   ├── too_long_code_lines.libr.sh
│   ├── update_common_build_and_checks.exec.sh
│   └── update_or_check_files_names_listing.exec.sh
├── build_and_checks.exec.sh
├── build_and_checks_variables
│   ├── bandit_baseline.json
│   ├── bandit.ini
│   ├── black.toml
│   ├── DevOrSysAdminScripts.html.tpl
│   ├── DevOrSysAdminScripts.tex
│   ├── DevOrSysAdminScripts.tex.tpl
│   ├── files_listing.tex.tpl
│   ├── files_names_listing.txt
```

```
│   ├── .gitignore
│   ├── licenses_notes.txt
│   ├── phpmd_baseline.xml
│   ├── pylintrc.toml
│   ├── repository_data.txt
│   ├── ruff.toml
│   ├── shellcheck.ini
│   └── shellcheck_notes.txt
├── COPYING
├── COPYING.LESSER
├── DevOrSysAdminScripts.html
├── DevOrSysAdminScripts.pdf
├── git_helpers
│   ├── git_checks.libr.sh
│   ├── post-commit
│   ├── post_commit_renaming_comment.libr.sh
│   └── pre-commit
├── IMAP_transfert.py
├── LiensUtiles.txt
├── miscellaneous
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   01234\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   012345\
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456
│   ├── 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │   0123456\
│   ├── a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │   an_see_that_it_works.txt
│   ├── funny.sh
│   ├── spampty_data.txt
│   └── yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│       that_you_can_see_that_it_works.txt
```

```
├── README.html
├── README.md
├── README.md.tpl
├── unicode.libr.php
├── upgrade_build_and_checks.exec.sh
└── wget_sha512.libr.sh

6 directories, 74 files

[ 12K Mar 10 17:49]  ./
├── [4.0K Mar 10 17:49]  build_and_checks_dependencies/
│   ├── [3.8K Mar 10 16:37]  build_and_checks.exec.sh.tpl*
│   ├── [4.1K Mar 10 16:49]  build_md_from_printable_md.exec.sh*
│   ├── [3.2K Mar 10 16:30]  check_shell_scripts_beginnings.libr.sh
│   ├── [5.2K Mar 10 17:45]  check_shell_scripts_indentation.libr.sh
│   ├── [3.4K Mar 10 16:30]  check_URLs.libr.sh
│   ├── [ 31K Mar 10 17:49]  common_build_and_checks.exec.sh*
│   ├── [ 26K Mar 10 17:42]  common_build_and_checks.exec.sh.tpl
│   ├── [9.1K Mar 10 17:08]  common_options.libr.sh
│   ├── [6.1K Mar  8 18:26]  comparisons.libr.sh
│   ├── [ 19K Mar 10 17:48]  create_PDF.exec.sh*
│   ├── [3.7K Mar 10 17:26]  generate_from_template.libr.sh
│   ├── [2.0K Mar  5 21:22]  get_common_text_glob_patterns.libr.sh
│   ├── [2.8K Mar 10 17:37]  grammar_and_spell_check.libr.sh
│   ├── [4.0K Mar 10 16:30]  licenses_templates/
│   │   ├── [ 13K Mar 10 16:30]  build_licenses_templates.exec.sh*
│   │   ├── [ 728 Feb 28 20:38]  license_file_header_AGPLv3+.tpl
│   │   ├── [ 707 Feb 28 20:38]  license_file_header_GPLv3+.tpl
│   │   └── [ 728 Feb 28 20:38]  license_file_header_LGPLv3+.tpl
│   ├── [4.8K Mar 10 16:30]  lines_counts.libr.sh
│   ├── [ 13K Mar 10 17:39]  lines_filters.libr.sh
│   ├── [2.7K Feb 19 16:16]  lines_maps.libr.sh
│   ├── [2.0K Mar  5 22:11]  overwrite_if_not_equal.libr.sh
│   ├── [2.3K Mar 10 16:30]  python_black_complement.libr.sh
│   ├── [2.9K Mar 10 17:36]  python_isort_complement.libr.sh
│   ├── [2.0K Mar  9 14:50]  repository_data.libr.sh
│   ├── [3.2K Mar 10 16:30]  shell_checks_complement.libr.sh
│   ├── [1.6K Feb  3 17:57]  split_score.exec.php
│   ├── [1.9K Feb  3 20:11]  split_score.libr.php
│   ├── [ 14K Mar  9 15:05]  strings_functions.libr.sh
│   ├── [3.8K Mar 10 16:30]  too_long_code_lines.libr.sh
│   ├── [4.2K Mar  9 16:05]  update_common_build_and_checks.exec.sh*
│   └── [5.6K Mar 10 17:44]  update_or_check_files_names_listing.exec.sh*
├── [4.0K Mar 10 17:49]  build_and_checks.exec.sh*
├── [4.0K Mar 10 17:49]  build_and_checks_variables/
│   ├── [  15 Jan  5 20:00]  bandit_baseline.json
```

```
│   ├── [  25 Jan  5 18:49]  bandit.ini
│   ├── [  30 Feb 28 20:30]  black.toml
│   ├── [ 16K Mar  8 17:00]  DevOrSysAdminScripts.html.tpl
│   ├── [ 32K Mar 10 17:26]  DevOrSysAdminScripts.tex
│   ├── [8.9K Mar  8 16:55]  DevOrSysAdminScripts.tex.tpl
│   ├── [ 14K Mar 10 16:31]  files_listing.tex.tpl
│   ├── [5.1K Mar 10 16:14]  files_names_listing.txt
│   ├── [  24 Jan 15 22:26]  .gitignore
│   ├── [5.2K Mar  8 16:52]  licenses_notes.txt
│   ├── [1.0K Mar  5 21:36]  phpmd_baseline.xml
│   ├── [ 678 Feb 24 22:38]  pylintrc.toml
│   ├── [4.4K Mar  3 20:47]  repository_data.txt
│   ├── [1.4K Feb 24 23:25]  ruff.toml
│   ├── [ 167 Feb 19 18:39]  shellcheck.ini
│   └── [3.5K Mar  9 14:46]  shellcheck_notes.txt
├── [ 34K May  6  2024]  COPYING
├── [7.5K Dec 27 21:48]  COPYING.LESSER
├── [392K Mar 10 17:26]  DevOrSysAdminScripts.html
├── [351K Mar 10 17:26]  DevOrSysAdminScripts.pdf
├── [4.0K Mar 10 17:04]  git_helpers/
│   ├── [3.8K Mar  8 21:05]  git_checks.libr.sh
│   ├── [1.8K Feb 27 20:20]  post-commit*
│   ├── [ 14K Mar 10 17:04]  post_commit_renaming_comment.libr.sh
│   └── [1.4K Feb 19 18:42]  pre-commit*
├── [8.3K Feb 28 20:31]  IMAP_transfert.py
├── [ 830 Dec 29 19:10]  LiensUtiles.txt
├── [4.0K Mar  9 15:15]  miscellaneous/
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123\
│   ├── [   0 Mar  4 10:03]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 01234\
│   ├── [   0 Mar  4 10:01]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 012345\
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456
│   ├── [   0 Mar  4 10:37]
│   │ 1234567890123456789012345678901234567890123456789012345678901234567890123456789
│   │ 0123456\
│   ├── [   0 Mar  4 09:13]
```

```
│   │ a_long_long_long_long_long_long_long_long_long_filename_so_that_you_c
│   │ an_see_that_it_works.txt
│   ├── [1.3K Mar  9 15:15]  funny.sh
│   ├── [  94 Mar  3 19:56]  spampty_data.txt
│   └── [   0 Mar  4 09:13]
│     yet_another_long_long_long_long_long_long_long_long_long_filename_so_
│     that_you_can_see_that_it_works.txt
├── [8.0K Mar 10 16:50]  README.html
├── [4.2K Mar 10 16:50]  README.md
├── [4.2K Mar  9 15:34]  README.md.tpl
├── [9.1K Feb 26 15:05]  unicode.libr.php
├── [1.4K Mar 10 16:37]  upgrade_build_and_checks.exec.sh*
└── [1.5K Mar  8 20:25]  wget_sha512.libr.sh

6 directories, 74 files
```

2 Listing of files

This repository is dual licensed. All the Bash scripts source code is covered by GPLv3+. All the PHP scripts source code is covered by LGPLv3+. All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at: `https://www.gnu.org/licenses/`. The git repository of this source code is also available at: `https://github.com/LLyaudet/DevOrSysAdminScripts/`. Clone it with this URL: `git@github.com:LLyaudet/DevOrSysAdminScripts.git`.

Here is a listing of the dependencies of the code in this repository:

- The following dependencies are GPLv3+ licensed:
  basename, bash (and its builtins), cat, chmod, cp, cut, date, diff, dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed, sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
  more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
  git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
  notify-send;
- The following dependencies are MIT licensed:
  black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
  bandit;

- The following dependencies are BSD-3-Clause licensed:
  phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD license.

Remark: Some commands listed under GPLv3+ are POSIX and can have different implementations under another license; for example, uutils coreutils is MIT licensed, whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
`https://www.gnu.org/software/bash/`,
`https://tiswww.case.edu/php/chet/bash/bashtop.html`,
`https://savannah.gnu.org/projects/bash/`.

GNU coreutils can be found at the following URLs:
`https://www.gnu.org/software/coreutils/`,
`https://savannah.gnu.org/projects/coreutils/`.

uutils coreutils can be found at the following URLs:
`https://uutils.github.io/coreutils/`,
`https://github.com/uutils/coreutils`.

diff (GNU diffutils) can be found at the following URLs:
`https://www.gnu.org/software/diffutils/`,
`https://savannah.gnu.org/projects/diffutils/`.

find, xargs (GNU findutils) can be found at the following URLs:
`https://www.gnu.org/software/findutils/`,
`https://savannah.gnu.org/projects/findutils/`.

grep (GNU grep) can be found at the following URLs:
`https://www.gnu.org/software/grep/`,
`https://savannah.gnu.org/projects/grep/`.

sed (GNU sed) can be found at the following URLs:
`https://www.gnu.org/software/sed/`,
`https://savannah.gnu.org/projects/sed/`.

ShellCheck can be found at the following URLs:
`https://www.shellcheck.net/`,
`https://github.com/koalaman/shellcheck`.

GNU Wget can be found at the following URLs:
`https://www.gnu.org/software/wget/`,
`https://savannah.gnu.org/projects/wget/`.

more of util-linux can be found at the following URLs:
`https://www.kernel.org/pub/linux/utils/util-linux/`,
`https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/`,
`https://github.com/util-linux/util-linux` official?

pandoc can be found at the following URLs:
https://pandoc.org/,
https://github.com/jgm/pandoc.

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/,
https://tug.org/texlive/svn/.

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree,
https://gitlab.com/OldManProgrammer/unix-tree,
https://github.com/Old-Man-Programmer/tree.

git can be found at the following URLs:
https://git-scm.com/about,
https://github.com/git/git.

pylint can be found at the following URLs:
https://www.pylint.org/,
https://pylint.readthedocs.io/en/stable/,
https://github.com/pylint-dev/pylint,
https://pypi.org/project/pylint/.

notify-send can be found at the following URLs:
https://gitlab.gnome.org/GNOME/libnotify/,
https://gnome.pages.gitlab.gnome.org/libnotify/.

black can be found at the following URLs:
https://github.com/psf/black,
https://black.readthedocs.io/en/stable/,
https://pypi.org/project/black/.

isort can be found at the following URLs:
https://pycqa.github.io/isort/,
https://github.com/pycqa/isort/,
https://pypi.org/project/isort/.

mypy can be found at the following URLs:
https://mypy-lang.org/,
https://mypy.readthedocs.io/en/stable/,
https://github.com/python/mypy,
https://pypi.org/project/mypy/.

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/,
https://github.com/pypa/pip,
https://pypi.org/project/pip/ (egg and chicken ;)).

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/,
https://github.com/astral-sh/ruff,
https://pypi.org/project/ruff/.

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/,
https://github.com/PyCQA/bandit,
https://pypi.org/project/bandit/.

phpmd can be found at the following URLs:
https://phpmd.org/,
https://github.com/phpmd/phpmd,
https://packagist.org/packages/phpmd/phpmd.

php can be found at the following URLs:
https://www.php.net/,
https://github.com/php/php-src.

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex.

pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/,
https://pcre2project.github.io/pcre2/,
https://github.com/PCRE2Project/pcre1,
https://github.com/PCRE2Project/pcre2,
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c.

python3 can be found at the following URLs:
https://www.python.org/,
https://docs.python.org/3/.

If you see something that I missed regarding my dependencies, please tell me/email me :).

2.1 build_and_checks_dependencies/build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
# See the GNU General Public License for more details.
#
# You should have received a copy of
# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
```

```
# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory

  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  mkdir -p "${LFBFL_subdir}/licenses_templates"
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  mkdir -p "${LFBFL_variables_directory}/temp"
  if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
    printf "temp/\n\n"\
      > "${LFBFL_variables_directory}/.gitignore"
  fi

  # LFBFL_dependencies_raw_content_download_URL
  local LFBFL_dependencies_URL
  LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
  LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
  readonly LFBFL_dependencies_URL

  declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
  # LFBFL_script_download_URL
  declare -r\
    LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
  declare -r\
    LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
  @sha512_common_build_and_checks.exec.sh@
```

```
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  "${LFBFL_file_path}" "${LFBFL_work_directory}"\
    "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
}

if [[ "$*" == *--fixed-point-build* ]]; then
  printf -- "--fixed-point-build\n"
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.2 build_and_checks_dependencies/build_md_from_printable_md.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
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#
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# This file was renamed from "build_md_from_printable_md.sh"
# to "build_md_from_printable_md.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

build_md_from_printable_md(){
  # Options:
  #   --verbose
  #   --work-directory=""
  #   --base-name="README"
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_base_name="README"
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --base-name=* ]]; then
      LFBFL_base_name=${LFBFL_arg#--base-name=}
      if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
        printf "Searching md file: %s.\n" "${LFBFL_base_name}"
      fi
      break
    fi
  done
  readonly LFBFL_base_name

  # Remove line returns here to keep lines short.
  local LFBFL_sed_expression
  LFBFL_sed_expression='s/(\n[ ]*\[[a-zA-Z0-9:-]*\]: [^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*<http[^\n\\]*)\\\n/\1/Mg'
  LFBFL_sed_expression+=';s/(\n[ ]*- <http[^\n\\]*)\\\n/\1/Mg'
  readonly LFBFL_sed_expression
  # Since Markdown can contain code, it is limited to Markdown URLs.
  # Hence it is way more complicated than 's/\\\n//Mg' expression
  # in other files.
  # Another downside is that more than one pass is needed.

  if [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
    sed -Ez -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
            -e "${LFBFL_sed_expression}"\
      "${LFBFL_base_name}.md.tpl"\
      > "${LFBFL_base_name}.md.temp"
    overwrite_if_not_equal "${LFBFL_base_name}.md"\
      "${LFBFL_base_name}.md.temp"
  else
    printf "No file %s.md.tpl\n" "${LFBFL_base_name}"
    # Unfortunately, we cannot return early, since we want to propose
    # HTML file generation with pandoc from .md file,
    # if the file .md.tpl doesn't exist.
  fi

  if [[ ! -f "${LFBFL_base_name}.md" ]]; then
    printf "No file %s.md\n" "${LFBFL_base_name}"
    return 1
  fi

  pandoc --from markdown --to html --standalone\
    --shift-heading-level-by=-1 "${LFBFL_base_name}.md"\
    -o "${LFBFL_base_name}.html.temp"
  # Correcting pandoc HTML "by hand".
  # https://github.com/jgm/pandoc/issues/10957
  # With the most open-minded answer found on Internet ;) XD.
  # This second sed expression is what I need,
```

```
  # but it may be unsuitable for you.
  # Other problems:
  # https://github.com/jgm/pandoc/issues/11484
  local LFBFL_prewrap="s~(code \{(\n|[^}])*)(    \})"
  LFBFL_prewrap+="~\1      white-space: pre-wrap;\n\3~M"
  readonly LFBFL_prewrap
  sed -i -E -e 's~(  <meta .*) />~\1>~g'\
            -e 's~<html .*>~<html lang="en">~'\
            -e '/    code\{white-space: pre-wrap;\}/d'\
    "${LFBFL_base_name}.html.temp"
  sed -i -Ez -e 's~(<img(\n|[^a-z0-9])(\n|[^>])*) />~\1>~Mg'\
             -e "${LFBFL_prewrap}"\
    "${LFBFL_base_name}.html.temp"
  overwrite_if_not_equal "${LFBFL_base_name}.html"\
    "${LFBFL_base_name}.html.temp"

  # java -Xss128M -jar "../jar_venv/vnu.jar" --exit-zero-always\
  #   --verbose ${LFBFL_base_name}.html"
}

build_md_from_printable_md "$@"
```

2.3 build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "check_shell_scripts_beginnings.sh"
# to "check_shell_scripts_beginnings.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

declare -gr LFBFL_SHELL_SCRIPT_BEGINNING="#!/usr/bin/env bash"

check_one_shell_script_beginning(){
  # declare -r LFBFL_file_path=$(basename "$1")
  # if [[ "${LFBFL_file_path}" =~ license_file_header_.*\.sh ]]; then
  #   return 0
  # fi
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  # diff <(head -n 1 "$1") <(printf '#!/usr/bin/env bash')
  # ------------------------------------------------------------------
  # is of course slower than
  # ------------------------------------------------------------------
  declare -r LFBFL_head=$(head -n 1 "$1")
  [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
    || printf "%s:File has wrong shell script beginning.\n" "$1"
}

check_shell_scripts_beginnings(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  local LFBFL_head
```

```
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_beginnings: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script beginning.\n"\
          "${LFBFL_file_path}"
    LFBFL_head=$(head -n 1 "${LFBFL_file_path}")
    [[ "${LFBFL_head}" == "${LFBFL_SHELL_SCRIPT_BEGINNING}" ]]\
      || printf "%s:File has wrong shell script beginning.\n"\
          "${LFBFL_file_path}"
  done
}
```

2.4 build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh

```
#!/usr/bin/env bash
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#
```

```
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_one_shell_script_indentation(){
  declare -i LFBFL_file_with_error=0
  local LFBFL_some_line
  local LFBFL_previous_line="workeduntilsomeasshole"
  # created a command named "workeduntilsomeasshole"
  # and called it without neither indentation, nor parameters
  # in a .sh script XD.
  # Manual justification of the lines L-4 and L-3 was a serendipity.
  # I just tried a joke and was lucky ;) :).
  local LFBFL_line_end
  local LFBFL_previous_line_end
  declare -i LFBFL_line_length
  declare -i LFBFL_line_end_length
  declare -i LFBFL_line_start_length
  local LFBFL_spacing
  declare -i LFBFL_offset
  local LFBFL_substring1
  local LFBFL_substring2
  declare -r LFBFL_pattern='^ *-e '
  # This grep would have been enough without some sed commands.
  declare -r LFBFL_s_lines=$(
    grep -EHn -B 1 '^(  )* ([^ ]|$)' "$1"
  )
  if [[ -z "${LFBFL_s_lines}" ]]; then
    return
  fi

  declare -a LFBFL_arr_lines
  mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
  readonly LFBFL_arr_lines

  for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
    if [[ "${LFBFL_some_line}" == "--" ]]; then
      # When switching between code fragments, we reinit.
      LFBFL_previous_line="workeduntilsomeasshole"
      continue
    fi
    # Note how it's funny that this loop always work because we check
```

```
    # shell scripts beginning in another script and thus we always
    # have a line before.
    if [[ "${LFBFL_previous_line}" == "workeduntilsomeasshole" ]];
    then
      LFBFL_previous_line="${LFBFL_some_line}"
      continue
    fi
    # Remove the name of the file and then the line number
    LFBFL_line_end=${LFBFL_some_line#*:}
    LFBFL_line_end=${LFBFL_line_end#*:}
    # But for the previous line '-' or ':' is used instead of ':'
    # and filenames with '-' are much more frequent than with ':'...
    LFBFL_line_length=${#LFBFL_some_line}
    LFBFL_line_end_length=${#LFBFL_line_end}
    LFBFL_line_start_length=$((
      LFBFL_line_length-LFBFL_line_end_length
    ))
    LFBFL_offset=$((LFBFL_line_start_length-2))
    # Checking if increasing number of digits in the line numbers
    # (99 -> 100).
    LFBFL_substring1=${LFBFL_previous_line:${LFBFL_offset}:1}
    # Nice puzzle below :) Look for comments above :)
    case ${LFBFL_substring1} in
      -) ((LFBFL_offset+=1));;
      :) ((LFBFL_offset+=1));;
      [0-9]) ((LFBFL_offset+=2));;
      *) printf ":-(\n";;
    esac
    LFBFL_previous_line_end=${LFBFL_previous_line:${LFBFL_offset}}
    if [[ "${LFBFL_line_end}" =~ ${LFBFL_pattern} ]]; then
      # Line starts with spacing and -e
      # Keep max spaces
      LFBFL_spacing=${LFBFL_line_end%%-*}
      LFBFL_offset=${#LFBFL_spacing}
      LFBFL_substring1=${LFBFL_line_end:${LFBFL_offset}:3}
      LFBFL_substring2=${LFBFL_previous_line_end:${LFBFL_offset}:3}
      if [[ "${LFBFL_substring1}" != "${LFBFL_substring2}" ]]; then
        printf "%s\n" "${LFBFL_some_line}"
        LFBFL_file_with_error=1
      fi
    else
      printf "%s\n" "${LFBFL_some_line}"
      LFBFL_file_with_error=1
    fi
    LFBFL_previous_line="${LFBFL_some_line}"
  done
```

```

  if [[ LFBFL_file_with_error -eq 1 ]]; then
    printf "%s:File has some lines with odd number of spaces.\n" "$1"
  fi
}

check_shell_scripts_indentation(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  # shopt -s globstar
  # for LFBFL_file_path in **/*.sh; do
  local LFBFL_file_path
  declare -r LFBFL_s_files_paths=$(
    find . -type f -name "*.sh" -printf '%P\n'\
    | relevant_find
  )
  if [[ -z "${LFBFL_s_files_paths}" ]]; then
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "check_shell_scripts_indentation: No file found.\n"
    return
  fi
  declare -a LFBFL_arr_files_paths
  mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
  readonly LFBFL_arr_files_paths
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "%s:Checking shell script indentation.\n"\
          "${LFBFL_file_path}"
    check_one_shell_script_indentation "${LFBFL_file_path}"
  done
}
```

2.5 build_and_checks_dependencies/check_URLs.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
```

```
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#
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# This file was renamed from "check_URLs.sh" to "check_URLs.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"

check_URLs(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  get_COMMON_TEXT_FILES_GLOB_PATTERNS
```

```
50
51    declare -Ar LFBFL_substitutions=(
52      ["http://www.gnu.org/licenses/"]="https://www.gnu.org/licenses/"
53    )
54
55    local LFBFL_pattern
56    local LFBFL_base_file_name
57    local LFBFL_substitution
58    local LFBFL_substitution2
59    local LFBFL_file_path
60    local LFBFL_s_files_paths
61    declare -a LFBFL_arr_files_paths
62    for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
63      [[ LFBFL_i_verbose -eq 1 ]]\
64        && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
65      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
66        | xargs grep -H 'http:'\
67        | grep -v "| xargs grep -H 'htt"\
68        | grep -vP "['\"]http(:[^'\"]*)['\"].*['\"]https\\1['\"]"
69      # Last grep just above will remove false positives from
70      # substitutions that fit on one line.
71
72      LFBFL_s_files_paths=$(
73        find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
74        | relevant_find\
75        | not_main_html_find
76      )
77      if [[ -z "${LFBFL_s_files_paths}" ]]; then
78        [[ LFBFL_i_verbose -eq 1 ]] && printf "No file found.\n"
79        continue
80      fi
81      mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
82      for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
83        [[ -f "${LFBFL_file_path}" ]] || continue
84        LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
85        [[ "${LFBFL_base_file_name}" != "check_URLs.libr.sh" ]]\
86          || continue
87        [[ LFBFL_i_verbose -eq 1 ]]\
88          && printf "Handling the file: %s.\n" "${LFBFL_file_path}"
89        for LFBFL_substitution in "${!LFBFL_substitutions[@]}"; do
90          LFBFL_substitution2=${LFBFL_substitutions[${LFBFL_substitution}]}
91          if grep -q "${LFBFL_substitution}" "${LFBFL_file_path}"; then
92            sed -i "s|${LFBFL_substitution}|${LFBFL_substitution2}|g"\
93              "${LFBFL_file_path}"
94          fi
95        done
```

```
    done
  done
}
```

2.6 build_and_checks_dependencies/common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
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# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
```

```
# declare -r LFBFL_dependencies_URL="$2" too long
declare -r LFBFL_start_URL="$2"
local LFBFL_verbose=""
declare -i LFBFL_i_verbose=0
if [[ "$*" == *--verbose* ]]; then
  printf "%s %s\n" "$0" "$*"
  LFBFL_verbose="--verbose"
  LFBFL_i_verbose=1
fi
readonly LFBFL_verbose
readonly LFBFL_i_verbose
declare -ar LFBFL_some_common_options=(
  "${LFBFL_verbose}"
  # "$--work-directory={LFBFL_work_directory}" doesn't work
  "${LFBFL_work_directory_option}"
)

source ./wget_sha512.libr.sh

LFBFL_subdir="build_and_checks_dependencies"
local LFBFL_file_name
local LFBFL_URL
local LFBFL_file_path

LFBFL_file_name="build_md_from_printable_md.exec.sh"
# LFBFL_script_download_URL
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='f94176cceb6fe386392c87e0c130611e4ac126323b6bc'
LFBFL_correct_sha512+='d5796ab82e9581f5749b7e4b0823f734085c8f6c22c7'
LFBFL_correct_sha512+='a7ed750718c040a3207a43d2321bf370ebc7287'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='883b5e6640475cda544fdb8191d444fc98a93c9c3545e'
LFBFL_correct_sha512+='5ad3769ba4187bea2addd3823a17e9d42bc1dd7056d1'
LFBFL_correct_sha512+='57517010a8094935211daad11fa7774a6651804'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

```

```
LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='ff4b303c0c7c95273c15899d699460bc0bba621c0910d'
LFBFL_correct_sha512+='b8ea5a5045331b17e203cd380410c69e9309cfa7acc3'
LFBFL_correct_sha512+='16ebb85d181c482bd54ad16170e8fb0e80846e1'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='284e3bf4f66835020141ddb7d34fe5e36131c10b548cb'
LFBFL_correct_sha512+='5d34f59ae8fccbba4c827131eb7099323bc32a71c9b5'
LFBFL_correct_sha512+='7c0b9ef80fbccb385c3ce492d7a256f4743be42'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='e2e169603341851463825fac36868e400f7a9a98addef'
LFBFL_correct_sha512+='e794cff3e4b7da9779333a95998c385844f69b1bea1b'
LFBFL_correct_sha512+='02138626d331feae5ecd0023e67c282b2a79475'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='eb4f6855e09eec27832af9f375f5ea750f08bd04f7955'
LFBFL_correct_sha512+='e4a071216ca80772469ada184cfc96821d71f80f1b42'
LFBFL_correct_sha512+='01ea9e2da4279d07aaecb98739a9c59bbbb5b0a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='97ca1d0021b7d306b2e60e36972e2fb72080fd6431808'
LFBFL_correct_sha512+='ec9c5317d01ad7b486cec3ec679df6a7f364c217bf78'
```

```
LFBFL_correct_sha512+='018bb3cd9f42549ebfeb60c477d4dfa0f715301'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='2f1187f173db3eea1c1324a57187dfd376ddd7ed67a23'
LFBFL_correct_sha512+='0a05859982bf0929a9569e17b1ca99d8f0cf2221cca5'
LFBFL_correct_sha512+='f5d017e99c9e59ef4f38002c5f6876d9553d9fd'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='fc041abf937df4c7cb1241c73e776e8bd07a5525bc9a9'
LFBFL_correct_sha512+='29c83d597ddd6aa88a6e61a99d22ca88c11561f1cdbc'
LFBFL_correct_sha512+='0fbf5df4794a2384f33e7e7dd5b65a53ab46f4a'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='a34a57c303881c42c94db9d17fde07a6578839cc9e50b'
LFBFL_correct_sha512+='07f26b1ef85f5bf19ccba74781fa83b7e69e0aab83a1'
LFBFL_correct_sha512+='5d27e3231bce411f79da77dcccf4c6e482fcfe5'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='7d5deca8c1be23f5c6cd06a2152f94c41b7e1cf8b8d0d'
LFBFL_correct_sha512+='d3b72260a51f3ae68233b5cc5f3f36cd4b82c253ae14'
LFBFL_correct_sha512+='c00f1b999e569db88deed0a1a26c2f15600d292'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="license_file_header_AGPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='9f9aa56d9b90d5003a12c5ec3370ac31aab74fc4a2524'
  LFBFL_correct_sha512+='2c5fbcb0197b0abbb8242f62dafc76eaf60faa5da173'
  LFBFL_correct_sha512+='05924b31b64b037083532fced9f98445000e2d0'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="license_file_header_GPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2bdef9a729a5a9ca5ed87ef8a518877c5ee6b31b45336'
  LFBFL_correct_sha512+='08660316a4e595fb31465a4847f4d53a2f7264118bd4'
  LFBFL_correct_sha512+='eab7c1d1892e00bc1861455ed02898f194db067'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="license_file_header_LGPLv3+.tpl"
  LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='265fd0d086b48ab798f35a072335a4976e4eda8c256fe'
  LFBFL_correct_sha512+='bba6c77c98ca86cdd844c6ac150c3445c80debbb3fc2'
  LFBFL_correct_sha512+='574b5fceeeaf79ec734a759bdef5fe798067c62'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  # ------------------------------------------------------------------

  LFBFL_file_name="lines_counts.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='ba926a14708844a2f291b348995e35f51ca2608423632'
  LFBFL_correct_sha512+='f607ca76ab5e2c84a66c09bc555362b7e3fc302a464f'
  LFBFL_correct_sha512+='76720556efa7ac83e7cf0872091b68151954339'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="lines_filters.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='7ca3e6f5f0191ed7777884c89e5a1af54aadf55d840bf'
LFBFL_correct_sha512+='d938f1b6024502c05e1ec527ef7ec3f527de10f221b5'
LFBFL_correct_sha512+='19ec6fc88a26bada4ecad363f2872b6f1014754'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='25c3e5082951260295b5f0e87ba1f2c978efdea1e8926'
LFBFL_correct_sha512+='072ce4d2062dc2a47bb92b9098f081e553eeae5d9154'
LFBFL_correct_sha512+='fdc50c9d88ae5365b776a23eb6e026b884a8374'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='1f647d6da821c5ebc6e2cc1943d85ce8067eeec571079'
LFBFL_correct_sha512+='59d37d9ee9d7d6f37be924ad278082b20284926ca5c4'
LFBFL_correct_sha512+='d31441ca910a048267968e43548c9eb2a2729c2'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='47029d60e49b85f5b96808358206176688ee0b62c6ac3'
LFBFL_correct_sha512+='5df6948ffcd53b8bdb7198170b7ce659ba44a4045d3d'
LFBFL_correct_sha512+='47dfee53cdd8149df76b12ba302fb2cfa6ec395'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='585809b2dd125cd45f4b3751fe7078f2b473fbec5af9c'
LFBFL_correct_sha512+='5d242b6dd827c680f22c913ddf9823c6d2225ffb0b29'
LFBFL_correct_sha512+='31865aa546aa5cc181f21224062bf04fc39c0f8'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
```

```
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="repository_data.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='793af27d8e658ed63148e6a3b45ba9b4eda1ddd3ede22'
  LFBFL_correct_sha512+='0b2af3c6a72e46d89b3565cef6e074cf1f9fcaccc7e4'
  LFBFL_correct_sha512+='dca98da471003bebeaabde704412ced55728094'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="shell_checks_complement.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='dbb106e7f724cf8c26f89af0aefd4394d63c3e49a5474'
  LFBFL_correct_sha512+='875508880e9c50e1ac90be08360931d106146194d32e'
  LFBFL_correct_sha512+='13752c72b878ecc18bbaf0304ffd89a94ce5f67'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.exec.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='2f7ff107a43716109b1acbd7d89870d05577af00a981a'
  LFBFL_correct_sha512+='05f162a054c96387fb99544f8225c333b1212f74326c'
  LFBFL_correct_sha512+='f9ec6196e796f6257dfbba9347ac8c57b4f09db'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="split_score.libr.php"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
  LFBFL_correct_sha512='1413898a4fb4f31ca17993fd6110f7e1f32adf31c22c4'
  LFBFL_correct_sha512+='7fa46a31d3c1511a3112a611e04fac19bb838d4476d9'
  LFBFL_correct_sha512+='ecf77a28dd48d3d2732df7392c50e38262852d7'
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

  LFBFL_file_name="strings_functions.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  local LFBFL_correct_sha512
```

```
LFBFL_correct_sha512='2bcc71452545f83cbde9f528c8c638c8026c79f0cebcb'
LFBFL_correct_sha512+='959706b17db02b04fb3101349dd15814df130c192f55'
LFBFL_correct_sha512+='d475aef749bda6905c92da97a0deb1237dc6ae8'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='aff702f2d6843cfc0fbfe2079d3760df393f548dfc711'
LFBFL_correct_sha512+='9380a2dcc83a9e216969989672c4963741ca17b09bc5'
LFBFL_correct_sha512+='51ebd44b3ce668b85daa9eac222600f2f86f561'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
local LFBFL_correct_sha512
LFBFL_correct_sha512='d39813bea25f8317280347a06afc9d49956b8584444ba'
LFBFL_correct_sha512+='a78fe15a4fd9c641776e5e97a426027634a81028942c'
LFBFL_correct_sha512+='30354906688a3631f0a62a3fce299cea65480cd'
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
```

```
  source "./${LFBFL_subdir}/lines_filters.libr.sh"
  # shellcheck source=lines_maps.libr.sh
  source "./${LFBFL_subdir}/lines_maps.libr.sh"
  # shellcheck source=overwrite_if_not_equal.libr.sh
  source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
  # shellcheck source=python_black_complement.libr.sh
  source "./${LFBFL_subdir}/python_black_complement.libr.sh"
  # shellcheck source=python_isort_complement.libr.sh
  source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
  # shellcheck source=shell_checks_complement.libr.sh
  source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
  # shellcheck source=strings_functions.libr.sh
  source "./${LFBFL_subdir}/strings_functions.libr.sh"
  # shellcheck source=too_long_code_lines.libr.sh
  source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

  enhanced_set_bash_option extglob
  # shellcheck source=repository_data.libr.sh
  source "./${LFBFL_subdir}/repository_data.libr.sh"
  # shellcheck disable=SC2154,SC2309
  [[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
    && enhanced_unset_bash_option extglob

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
```

```
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        "--work-directory=${LFBFL_some_directory}"\
        "--base-name=${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  declare -i LFBFL_will_popd
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
```

```
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options[@]}"

  printf -- "---Python---\n"
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
```

```
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
```

```
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_no_toml=0
    done
  fi
  if [[ LFBFL_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
```

```
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_pylint_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade pylint
  fi
  pylint --rcfile build_and_checks_variables/pylintrc.toml\
    --recursive=y .
  if [[ -n "${LFBFL_pylint_venv}" ]]; then
    deactivate
  fi

  printf "Running ruff\n"
  LFBFL_ruff_venv=""
  grep_variable "${LFBFL_data_file_name}" ruff_venv\
    --result-variable-prefix="LFBFL_"\
```

```
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_ruff_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade ruff
  fi
  ruff check --config build_and_checks_variables/ruff.toml
  if [[ -n "${LFBFL_ruff_venv}" ]]; then
    deactivate
  fi
  printf -- "---Python end---\n"

  printf -- "---PHP---\n"
  printf "Running PHPMD\n"
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    composer global require phpmd/phpmd
  fi

  local LFBFL_phpmd_baseline="build_and_checks_variables/"
  LFBFL_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_phpmd_baseline

  local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
  LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
  readonly LFBFL_temp_phpmd_baseline

  local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
  LFBFL_phpmd_rulesets+="design,naming,unusedcode"
  readonly LFBFL_phpmd_rulesets

  phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  # Saving new baseline in temp if necessary.
  rm "${LFBFL_temp_phpmd_baseline}"
  phpmd --color --generate-baseline\
    --baseline-file "${LFBFL_temp_phpmd_baseline}"\
    . text "${LFBFL_phpmd_rulesets}"
  sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
  diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
  printf -- "---PHP end---\n"
```

```

  printf -- "---JS---\n"
  LFBFL_npm_lint_directories=""
  grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
    --result-variable-prefix="LFBFL_"
  if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
    printf "Running ESLint\n"
    local LFBFL_JS_directory
    declare -a LFBFL_arr_paths
    mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
    for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
      (cd "${LFBFL_JS_directory}" && npm run lint)
    done
  fi
  printf -- "---JS end---\n"

  printf "Analyzing too long lines\n"
  # shellcheck disable=SC2248
  too_long_code_lines "${LFBFL_some_common_options[@]}" \
    --max-line-length=${LFBFL_max_line_length}

  printf "Analyzing shell scripts beginnings\n"
  check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©‘”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
```

```
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
```

```
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.7 build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# The file "common_build_and_checks.exec.sh.tpl" was renamed from
# "common_build_and_checks.sh.tpl" to
# "common_build_and_checks.exec.sh.tpl".
# The file "common_build_and_checks.exec.sh" generated from the file
# "common_build_and_checks.sh.tpl"
# or "common_build_and_checks.exec.sh.tpl"
# was renamed from
```

```
# "common_build_and_checks.sh" to "common_build_and_checks.exec.sh".

common_build_and_checks(){
  # $1 LFBFL_work_directory
  # $2 LFBFL_dependencies_URL
  # $3 optional --verbose
  local LFBFL_work_directory="${1:-.}"
  LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
  local LFBFL_work_directory_option="--work-directory="
  LFBFL_work_directory_option+="${LFBFL_work_directory}"
  # declare -r LFBFL_dependencies_URL="$2" too long
  declare -r LFBFL_start_URL="$2"
  local LFBFL_verbose=""
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_verbose
  readonly LFBFL_i_verbose
  declare -ar LFBFL_some_common_options=(
    "${LFBFL_verbose}"
    # "$--work-directory={LFBFL_work_directory}" doesn't work
    "${LFBFL_work_directory_option}"
  )

  source ./wget_sha512.libr.sh

  LFBFL_subdir="build_and_checks_dependencies"
  local LFBFL_file_name
  local LFBFL_URL
  local LFBFL_file_path

  LFBFL_file_name="build_md_from_printable_md.exec.sh"
  # LFBFL_script_download_URL
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
  @sha512_build_md_from_printable_md.exec.sh@
  wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
    "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
  chmod +x "./${LFBFL_file_path}"

  LFBFL_file_name="check_shell_scripts_beginnings.libr.sh"
  LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
  LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
```

```
@sha512_check_shell_scripts_beginnings.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_shell_scripts_indentation.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_shell_scripts_indentation.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="check_URLs.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_check_URLs.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="common_options.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_common_options.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="comparisons.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_comparisons.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="create_PDF.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_create_PDF.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="generate_from_template.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_generate_from_template.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```

LFBFL_file_name="get_common_text_glob_patterns.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_get_common_text_glob_patterns.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="grammar_and_spell_check.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_grammar_and_spell_check.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

# /licenses_templates/ ---------------------------------------------
declare -r LFBFL_subdir2="${LFBFL_subdir}/licenses_templates"
declare -r LFBFL_start_URL2="${LFBFL_start_URL}/licenses_templates"
LFBFL_file_name="build_licenses_templates.exec.sh"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_build_licenses_templates.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
chmod +x "./${LFBFL_file_path}"

LFBFL_file_name="license_file_header_AGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_AGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_GPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_GPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="license_file_header_LGPLv3+.tpl"
LFBFL_URL="${LFBFL_start_URL2}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir2}/${LFBFL_file_name}"
@sha512_license_file_header_LGPLv3+.tpl@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
# ------------------------------------------------------------------

LFBFL_file_name="lines_counts.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_counts.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_filters.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_filters.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="lines_maps.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_lines_maps.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="overwrite_if_not_equal.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_overwrite_if_not_equal.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_black_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_black_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="python_isort_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_python_isort_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="repository_data.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
```

```
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_repository_data.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="shell_checks_complement.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_shell_checks_complement.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.exec.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.exec.php@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="split_score.libr.php"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_split_score.libr.php@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="strings_functions.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_strings_functions.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="too_long_code_lines.libr.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_too_long_code_lines.libr.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"

LFBFL_file_name="update_or_check_files_names_listing.exec.sh"
LFBFL_URL="${LFBFL_start_URL}/${LFBFL_file_name}"
LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_file_name}"
@sha512_update_or_check_files_names_listing.exec.sh@
wget_sha512 "${LFBFL_file_path}" "${LFBFL_URL}"\
  "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
```

```
chmod +x "./${LFBFL_file_path}"

# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=check_shell_scripts_indentation.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_indentation.libr.sh"
# shellcheck source=check_URLs.libr.sh
source "./${LFBFL_subdir}/check_URLs.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=grammar_and_spell_check.libr.sh
source "./${LFBFL_subdir}/grammar_and_spell_check.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=python_black_complement.libr.sh
source "./${LFBFL_subdir}/python_black_complement.libr.sh"
# shellcheck source=python_isort_complement.libr.sh
source "./${LFBFL_subdir}/python_isort_complement.libr.sh"
# shellcheck source=shell_checks_complement.libr.sh
source "./${LFBFL_subdir}/shell_checks_complement.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"
# shellcheck source=too_long_code_lines.libr.sh
source "./${LFBFL_subdir}/too_long_code_lines.libr.sh"

enhanced_set_bash_option extglob
# shellcheck source=repository_data.libr.sh
source "./${LFBFL_subdir}/repository_data.libr.sh"
# shellcheck disable=SC2154,SC2309
[[ enhanced_set_bash_option_extglob_result -eq 0 ]]\
  && enhanced_unset_bash_option extglob

enhanced_set_shell_option pipefail\
  && trap 'enhanced_unset_shell_option pipefail' RETURN
```

```

  printf "Building license headers\n"
  "./${LFBFL_subdir2}/build_licenses_templates.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Building README.md\n"
  "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
    "--work-directory=${LFBFL_work_directory}"\
    "--base-name=README"\
    "${LFBFL_verbose}"

  printf "Building other MarkDown files\n"
  local LFBFL_some_directory
  declare -r LFBFL_readme="${LFBFL_work_directory}/README.md.tpl"
  local LFBFL_s_files_paths
  LFBFL_s_files_paths=$(
    find "${LFBFL_work_directory}" -type f -name "*.md.tpl"\
    | relevant_find
  )
  declare -a LFBFL_arr_files_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      [[ "${LFBFL_file_path}" == "${LFBFL_readme}" ]] && continue
      printf "Found template %s.\n" "${LFBFL_file_path}"
      LFBFL_some_directory=$(dirname "${LFBFL_file_path}")
      LFBFL_file_name=$(basename "${LFBFL_file_path}")
      LFBFL_file_name=${LFBFL_file_name%.md.tpl}
      "./${LFBFL_subdir}/build_md_from_printable_md.exec.sh"\
        "--work-directory=${LFBFL_some_directory}"\
        "--base-name=${LFBFL_file_name}"\
        "${LFBFL_verbose}"
    done
  fi

  declare -i LFBFL_will_popd
  pushd_to_work_directory
  LFBFL_will_popd=$?
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name=\
"build_and_checks_variables/repository_data.txt"

  declare -i LFBFL_max_line_length
  grep_variable "${LFBFL_data_file_name}" max_line_length\
    --result-variable-prefix="LFBFL_"
```

```

  declare -i LFBFL_upgrade_venvs=0
  declare -r LFBFL_upgrade_venvs_ts_file=\
"build_and_checks_variables/upgrade_venvs_ts"
  declare -i LFBFL_upgrade_venvs_ts
  declare -i LFBFL_current_ts
  local LFBFL_upgrade_venvs_answer

  local LFBFL_upgrade_venvs_time_interval_in_seconds=""
  get_upgrade_venvs_time_interval_in_seconds "${LFBFL_data_file_name}"\
    "${LFBFL_some_common_options[@]}"

  if [[ -f "${LFBFL_upgrade_venvs_ts_file}" ]]; then
    LFBFL_upgrade_venvs_ts=$(
      stat -c %Y "${LFBFL_upgrade_venvs_ts_file}"
    )
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_venvs_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      LFBFL_upgrade_venvs=1
    fi
  else
    LFBFL_upgrade_venvs=1
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    local LFBFL2_upgrade_venvs=""
    grep_variable "${LFBFL_data_file_name}" upgrade_venvs\
      --result-variable-prefix="LFBFL2_"
    if [[ "${LFBFL2_upgrade_venvs}" != "auto" ]]; then
      read -r -n 1 -t 10\
        -p "Upgrade venvs and composer global? [Y/n]"\
        LFBFL_upgrade_venvs_answer
      if [[ "${LFBFL_upgrade_venvs_answer}" == "n" ]]; then
        LFBFL_upgrade_venvs=0
      fi
    fi
  fi

  printf "Running shellcheck\n"
  LFBFL_s_files_paths=$(
    find . -type f -name "*.sh"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
```

```
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      shellcheck --rcfile=build_and_checks_variables/shellcheck.ini\
        "${LFBFL_file_path}"
    done
  fi
  printf "Running shell_checks_complement\n"
  shell_checks_complement "${LFBFL_some_common_options[@]}"

  printf -- "---Python---\n"
  printf "Running isort\n"
  local LFBFL_isort_venv=""
  grep_variable "${LFBFL_data_file_name}" isort_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_isort_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade isort
  fi
  isort .
  if [[ -n "${LFBFL_isort_venv}" ]]; then
    deactivate
  fi
  printf "Running python_isort_complement\n"
  python_isort_complement "${LFBFL_some_common_options[@]}"

  printf "Running black\n"
  # First, we update the configuration file with max_line_length.
  local LFBFL_update_max_length="s/^line-length = [0-9]*$"
  LFBFL_update_max_length+="/line-length = ${LFBFL_max_line_length}/"
  sed -E -e "${LFBFL_update_max_length}"\
    build_and_checks_variables/black.toml\
    > build_and_checks_variables/black.toml.temp
  overwrite_if_not_equal build_and_checks_variables/black.toml\
    build_and_checks_variables/black.toml.temp

  local LFBFL_black_venv=""
  grep_variable "${LFBFL_data_file_name}" black_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
  if [[ -n "${LFBFL_black_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_black_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade black
  fi
  black --config build_and_checks_variables/black.toml .
  if [[ -n "${LFBFL_black_venv}" ]]; then
    deactivate
  fi
  printf "Running python_black_complement\n"
  python_black_complement "${LFBFL_some_common_options[@]}"

  printf "Probing if mypy should be runned\n"
  local LFBFL_mypy_venv=""
  grep_variable "${LFBFL_data_file_name}" mypy_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_mypy_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade mypy
  fi
  shopt -s lastpipe
  local LFBFL_directory_path
  declare -i LFBFL_no_toml=1
  LFBFL_s_files_paths=$(
    find . -type f -name "pyproject.toml"\
    | relevant_find
  )
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      if grep -q "Typing :: Typed" "${LFBFL_file_path}"; then
        printf "Running mypy\n"
```

```
        LFBFL_directory_path=$(dirname "${LFBFL_file_path}")
        mypy "${LFBFL_directory_path}"
      fi
      LFBFL_no_toml=0
    done
  fi
  if [[ LFBFL_no_toml -eq 1 ]]; then
    printf "Running mypy\n"
    mypy .
  fi
  if [[ -n "${LFBFL_mypy_venv}" ]]; then
    deactivate
  fi

  printf "Running bandit\n"
  LFBFL_bandit_venv=""
  grep_variable "${LFBFL_data_file_name}" bandit_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    if command -v deactivate
    then
      deactivate
    fi
    # shellcheck disable=SC1090,SC1091
    source "${LFBFL_bandit_venv}/bin/activate"
  fi
  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    pip install --upgrade bandit
  fi
  bandit --ini build_and_checks_variables/bandit.ini\
    -b build_and_checks_variables/bandit_baseline.json\
    -r .
  # Saving new baseline in temp if necessary.
  bandit --ini build_and_checks_variables/bandit.ini\
    -f json -o build_and_checks_variables/temp/bandit_baseline.json\
    -r .
  if [[ -n "${LFBFL_bandit_venv}" ]]; then
    deactivate
  fi

  printf "Running pylint\n"
  LFBFL_pylint_venv=""
  grep_variable "${LFBFL_data_file_name}" pylint_venv\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
```

```
537   if [[ -n "${LFBFL_pylint_venv}" ]]; then
538     if command -v deactivate
539     then
540       deactivate
541     fi
542     # shellcheck disable=SC1090,SC1091
543     source "${LFBFL_pylint_venv}/bin/activate"
544   fi
545   if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
546     pip install --upgrade pylint
547   fi
548   pylint --rcfile build_and_checks_variables/pylintrc.toml\
549     --recursive=y .
550   if [[ -n "${LFBFL_pylint_venv}" ]]; then
551     deactivate
552   fi
553
554   printf "Running ruff\n"
555   LFBFL_ruff_venv=""
556   grep_variable "${LFBFL_data_file_name}" ruff_venv\
557     --result-variable-prefix="LFBFL_"\
558     --replace-line-returns-by=""
559   if [[ -n "${LFBFL_ruff_venv}" ]]; then
560     if command -v deactivate
561     then
562       deactivate
563     fi
564     # shellcheck disable=SC1090,SC1091
565     source "${LFBFL_ruff_venv}/bin/activate"
566   fi
567   if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
568     pip install --upgrade ruff
569   fi
570   ruff check --config build_and_checks_variables/ruff.toml
571   if [[ -n "${LFBFL_ruff_venv}" ]]; then
572     deactivate
573   fi
574   printf -- "---Python end---\n"
575
576   printf -- "---PHP---\n"
577   printf "Running PHPMD\n"
578   if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
579     composer global require phpmd/phpmd
580   fi
581
582   local LFBFL_phpmd_baseline="build_and_checks_variables/"
```

```
LFBFL_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_phpmd_baseline

local LFBFL_temp_phpmd_baseline="build_and_checks_variables/temp/"
LFBFL_temp_phpmd_baseline+="phpmd_baseline.xml"
readonly LFBFL_temp_phpmd_baseline

local LFBFL_phpmd_rulesets="cleancode,codesize,controversial,"
LFBFL_phpmd_rulesets+="design,naming,unusedcode"
readonly LFBFL_phpmd_rulesets

phpmd --color --baseline-file "${LFBFL_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
# Saving new baseline in temp if necessary.
rm "${LFBFL_temp_phpmd_baseline}"
phpmd --color --generate-baseline\
  --baseline-file "${LFBFL_temp_phpmd_baseline}"\
  . text "${LFBFL_phpmd_rulesets}"
sed -i -e 's/" file="/"\n    file="/g' "${LFBFL_temp_phpmd_baseline}"
diff "${LFBFL_phpmd_baseline}" "${LFBFL_temp_phpmd_baseline}"
printf -- "---PHP end---\n"

printf -- "---JS---\n"
LFBFL_npm_lint_directories=""
grep_variable "${LFBFL_data_file_name}" npm_lint_directories\
  --result-variable-prefix="LFBFL_"
if [[ -n "${LFBFL_npm_lint_directories}" ]]; then
  printf "Running ESLint\n"
  local LFBFL_JS_directory
  declare -a LFBFL_arr_paths
  mapfile -t LFBFL_arr_paths <<< "${LFBFL_npm_lint_directories}"
  for LFBFL_JS_directory in "${LFBFL_arr_paths[@]}"; do
    (cd "${LFBFL_JS_directory}" && npm run lint)
  done
fi
printf -- "---JS end---\n"

printf "Analyzing too long lines\n"
# shellcheck disable=SC2248
too_long_code_lines "${LFBFL_some_common_options[@]}" \
  --max-line-length=${LFBFL_max_line_length}

printf "Analyzing shell scripts beginnings\n"
check_shell_scripts_beginnings "${LFBFL_some_common_options[@]}" \
  | relevant_grep

```

```
  printf "Analyzing shell scripts indentation\n"
  check_shell_scripts_indentation "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing URLs\n"
  check_URLs "${LFBFL_some_common_options[@]}" \
    | relevant_grep

  printf "Analyzing strange characters: hover over in doubt\n"
  # shellcheck disable=SC1111
  LFBFL_usual_characters="\x00-\x7Fàâéèêëîïôûç©“”└├│«»"
  grep --exclude-dir .git --binary-files=without-match --color=always\
    -nPr "[^${LFBFL_usual_characters}]" .

  [[ LFBFL_will_popd -eq 0 ]] && popd_from_work_directory

  printf "Checking listed files\n"
  "./${LFBFL_subdir}/update_or_check_files_names_listing.exec.sh"\
    "${LFBFL_some_common_options[@]}"

  printf "Creating the PDF file of the listing of the source code\n"
  "./${LFBFL_subdir}/create_PDF.exec.sh" "${LFBFL_some_common_options[@]}"

  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  if [[ -f "build_and_checks_variables/post_build.sh" ]]; then
    ./build_and_checks_variables/post_build.sh "${LFBFL_verbose}"
  fi

  # ------------------------------------------------------------------
  # It is at the end that all the HTML files are generated and that
  # we can check there are no mistakes.
  printf "Running Nu W3C HTML CSS and SVG validator\n"
  local LFBFL_upgrade_vnu_jar_time_interval_in_seconds=""
  grep_variable "${LFBFL_data_file_name}"\
    upgrade_vnu_jar_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  if [[ "${LFBFL_upgrade_vnu_jar_time_interval_in_seconds}" == "wat" ]];
  then
    LFBFL_upgrade_vnu_jar_time_interval_in_seconds=${RANDOM}
  fi

  declare -i LFBFL_vnu_jar_ts
```

```
  local LFBFL_vnu_jar_path=""
  grep_variable "${LFBFL_data_file_name}" vnu_jar_path\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  if [[ -f "${LFBFL_vnu_jar_path}" ]]; then
    LFBFL_vnu_jar_ts=$(stat -c %Y "${LFBFL_vnu_jar_path}")
    LFBFL_current_ts=$(date +%s)
    ((LFBFL_current_ts-=LFBFL_vnu_jar_ts))
    ((LFBFL_current_ts-=LFBFL_upgrade_vnu_jar_time_interval_in_seconds))
    if [[ LFBFL_current_ts -gt 0 ]]; then
      printf $'/!\\ATTENTION: Nu W3C validator is too old./!\\\n'
      printf "Go get the latest version here:\n"
      printf "https://github.com/validator/validator/releases\n"
    fi
    declare -r LFBFL_files_for_Nu=$(
      find . -iregex ".*\.\(css\|html\|svg\|xhtml\)"\
      | relevant_find
    )
    if [[ -n "${LFBFL_files_for_Nu}" ]]; then
      declare -a LFBFL_files_for_Nu2
      mapfile -t LFBFL_files_for_Nu2 <<< "${LFBFL_files_for_Nu}"
      readonly LFBFL_files_for_Nu2
      java -Xss128M -jar "${LFBFL_vnu_jar_path}" --exit-zero-always\
        --verbose "${LFBFL_files_for_Nu2[@]}"
    fi
  fi
  # ------------------------------------------------------------------

  if [[ LFBFL_upgrade_venvs -eq 1 ]]; then
    touch "${LFBFL_upgrade_venvs_ts_file}"
  fi
}

common_build_and_checks "$@"
```

2.8 build_and_checks_dependencies/common_options.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
# DevOrSysAdminScripts is free software:
# you can redistribute it and/or modify it under the terms
# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
```

```
#
# DevOrSysAdminScripts is distributed in the hope
# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
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get_where_was_i(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=offset
  LFBFL_where_was_i="$0"
  if [[ "${BASH_SOURCE[$1]}" != "" && "${BASH_SOURCE[$1]}" != "$0" ]]; then
    LFBFL_where_was_i+=" ${BASH_SOURCE[$1]}"
  fi
  if [[ "${FUNCNAME[$1]}" != "" ]]; then
    LFBFL_where_was_i+=" ${FUNCNAME[$1]}()"
  fi
  readonly LFBFL_where_was_i
  # Comment out line above in case someone would use distinct
  # offsets in the same function.
}

is_top_dirstack_directory(){
  # This function should not be needed since when testing "${DIRSTACK[0]}"
  # is already a realpath. But I keep it, in case it is needed.
  # https://lists.gnu.org/archive/html/help-bash/2026-03/msg00005.html
  # $1=to_directory needs to be "realpathed" already
  if [[ -z "$1" ]]; then
    return 0
  fi
  local LFBFL_real_path
  LFBFL_real_path=$(realpath "${DIRSTACK[0]}")
  readonly LFBFL_real_path
  [[ "${LFBFL_real_path}" == "$1" ]]
}

enhanced_pushd(){
```

```
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory must be sanitized with realpath before call
  # $2=offset_for_where_was_i
  # $3=error_message_intermediate_complement
  # returns an error code whenever no pushd happened
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_pushd requested: %s\n" "$1"
  declare -ig enhanced_pushd_result=0
  if [[ -z "$1" ]]; then
    enhanced_pushd_result=110
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  fi
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated1\n"
  is_top_dirstack_directory "$1" && {
    enhanced_pushd_result=111
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd validated2\n"

  pushd "$1" || {
    enhanced_pushd_result=$?
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s %s%s no such directory.\n" "${LFBFL_where_was_i}" "$3" "$1"
    # shellcheck disable=SC2248
    return ${enhanced_pushd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_pushd executed\n"
  # shellcheck disable=SC2248
  return ${enhanced_pushd_result}
}

enhanced_popd(){
  # Auxiliary function needed below.
  # See usages below if you want to reuse it.
  # $1=to_directory
  # $2=offset_for_where_was_i
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "enhanced_popd requested: %s\n" "$1"
  if [[ -z "$1" ]]; then
    return 0
  fi

```

```
  declare -i LFBFL_i_popd_result
  popd || {
    LFBFL_i_popd_result=$?
    readonly LFBFL_i_popd_result
    local LFBFL_where_was_i
    get_where_was_i "$2"
    printf "%s popd failed.\n" "${LFBFL_where_was_i}"
    # shellcheck disable=SC2248
    return ${LFBFL_i_popd_result}
  }
  [[ LFBFL_i_verbose -eq 1 ]] && printf "enhanced_popd executed\n"
  return 0
}

can_continue_after_enhanced_pushd(){
  [[ enhanced_pushd_result -eq 0
  || enhanced_pushd_result -eq 110
  || enhanced_pushd_result -eq 111
  ]]
}

# Main usage for the next functions relative to verbose mode:
# Add the two following lines at the start of a function.
#   declare -i LFBFL_i_verbose=0
#   get_verbose_option "$@"
# /!\ You really need all of the two lines. /!\
# Otherwise, the call to get_verbose_option may impact a function above;
# in case you didn't copy all of "$@" in nested function calls...
# /!\ The same warning applies for other functions below. /!\
# /!\ I will not repeat it below. /!\
# Use get_verbose_options_array, if needed.

get_verbose_options_array(){
  # Use this constant with code like:
  # "${LFBFL_VERBOSE_OPTIONS[LFBFL_i_verbose]}"
  # It could be extended if there is more than one level of verboseness.
  if [[ -n "${LFBFL_VERBOSE_OPTIONS}" ]]; then
    return
  fi

  declare -agr LFBFL_VERBOSE_OPTIONS=(
    ""
    "--verbose"
  )
}

```

```
get_verbose_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --verbose
  # This test is faster than:
  #   for LFBFL_arg in "$@"; do
  #     if [[ "${LFBFL_arg}" == "--verbose" ]]; then
  # But it is not bug free.
  if [[ "$*" == *--verbose* ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    readonly LFBFL_where_was_i
    printf "%s %s\n" "${LFBFL_where_was_i}" "$*"
    LFBFL_i_verbose=1
  fi
  # shellcheck disable=SC2034
  readonly LFBFL_i_verbose
}


# Main usage for the next functions relative to work_directory:
# Add the five following lines at the start of a function.
#   local LFBFL_work_directory=""
#   get_work_directory_option "$@"
#   pushd_to_work_directory\
#     && trap 'popd_from_work_directory' RETURN
#   can_continue_after_enhanced_pushd || return 1

get_work_directory_option(){
  # This command is to be called in another one with same arguments.
  # Options:
  #   --work-directory=""
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --work-directory=* ]]; then
      LFBFL_work_directory=${LFBFL_arg#--work-directory=}
      LFBFL_work_directory=${LFBFL_work_directory:-.}
      # LFBFL_work_directory=${LFBFL_work_directory/#~/${HOME}} line below
      LFBFL_work_directory=$(realpath "${LFBFL_work_directory}")
      break
    fi
  done
  readonly LFBFL_work_directory
}

pushd_to_work_directory(){
```

```
  enhanced_pushd "${LFBFL_work_directory}" 3 " --work-directory="
}

popd_from_work_directory(){
  enhanced_popd "${LFBFL_work_directory}" 3
}

work_directory_is_top_dirstack_directory(){
  is_top_dirstack_directory "${LFBFL_work_directory}"
}


# Functions relative to shell and bash options
# (See SHELLOPTS and BASHOPTS, set and shopt):
# - shell options usage, example with pipefail:
#     enhanced_set_shell_option pipefail\
#       && trap 'enhanced_unset_shell_option pipefail' RETURN
#   or maybe:
#     enhanced_unset_shell_option pipefail\
#       && trap 'enhanced_set_shell_option pipefail' RETURN
#   if you need to unset it in some function.
# - bash options usage, example with globstar:
#     enhanced_set_bash_option globstar\
#       && trap 'enhanced_unset_bash_option globstar' RETURN
#   or maybe:
#     enhanced_unset_bash_option globstar\
#       && trap 'enhanced_set_bash_option globstar' RETURN

enhanced_set_shell_option(){
  # $1=optname some shell option for set
  declare -r LFBFL_result_name="enhanced_set_shell_option_$1_result"
  declare -gi "${LFBFL_result_name}"=0
  if [[ -o "$1" ]]; then
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  set -o "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s shell option activated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

```
240
241 enhanced_unset_shell_option(){
242   # $1=optname some shell option for set
243   declare -r LFBFL_result_name="enhanced_unset_shell_option_$1_result"
244   declare -gi "${LFBFL_result_name}"=0
245   if [[ ! -o "$1" ]]; then
246     printf -v "${LFBFL_result_name}" "%d" "1"
247     # shellcheck disable=SC2086
248     return ${!LFBFL_result_name}
249   fi
250   set +o "$1"
251   if [[ LFBFL_i_verbose -eq 1 ]]; then
252     local LFBFL_where_was_i
253     get_where_was_i 2
254     printf "%s %s shell option unactivated.\n" "${LFBFL_where_was_i}" "$1"
255   fi
256   # shellcheck disable=SC2086
257   return ${!LFBFL_result_name}
258 }
259
260 enhanced_set_bash_option(){
261   # $1=optname some bash option for shopt
262   declare -r LFBFL_result_name="enhanced_set_bash_option_$1_result"
263   declare -gi "${LFBFL_result_name}"=0
264   # declare -r LFBFL_regexp="^(.*:)?$1(:.*)?$" funny but brainfucked
265   # if [[ "${BASHOPTS}" =~ ${LFBFL_regexp} ]]; then
266   if shopt -q "$1"; then
267     printf -v "${LFBFL_result_name}" "%d" "1"
268     # shellcheck disable=SC2086
269     return ${!LFBFL_result_name}
270   fi
271   shopt -s "$1"
272   if [[ LFBFL_i_verbose -eq 1 ]]; then
273     local LFBFL_where_was_i
274     get_where_was_i 2
275     printf "%s %s bash option activated.\n" "${LFBFL_where_was_i}" "$1"
276   fi
277   # shellcheck disable=SC2086
278   return ${!LFBFL_result_name}
279 }
280
281 enhanced_unset_bash_option(){
282   # $1=optname some bash option for shopt
283   declare -r LFBFL_result_name="enhanced_unset_bash_option_$1_result"
284   declare -gi "${LFBFL_result_name}"=0
285   if ! shopt -q "$1"; then
```

```
    printf -v "${LFBFL_result_name}" "%d" "1"
    # shellcheck disable=SC2086
    return ${!LFBFL_result_name}
  fi
  shopt -u "$1"
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    local LFBFL_where_was_i
    get_where_was_i 2
    printf "%s %s bash option unactivated.\n" "${LFBFL_where_was_i}" "$1"
  fi
  # shellcheck disable=SC2086
  return ${!LFBFL_result_name}
}
```

2.9 build_and_checks_dependencies/comparisons.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
# as published by the Free Software Foundation,
# either version 3 of the License,
# or (at your option) any later version.
#
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# that it will be useful,
# but WITHOUT ANY WARRANTY;
# without even the implied warranty of
# MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
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# the GNU General Public License
# along with DevOrSysAdminScripts.
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# The file "comparisons.sh" was renamed to "comparisons.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

equal(){
  # Returns 1 if all the arguments strings are equal.
```

```
31    # Renvoie 1 dans $? si toutes les chaînes en arguments sont égales.
32    # J'ai rerepensé au 3-way-merge
33    # et à mon article sur le principe de première différence
34    # appliqué à la largeur modulaire, de clique ou de rang avec
35    # l'idée d'un principe de première différence ternaire.
36    declare -i LFBFL_first=1
37    local LFBFL_current
38    local LFBFL_arg
39    for LFBFL_arg in "$@"; do
40      if [[ LFBFL_first -eq 1 ]]; then
41        LFBFL_current="${LFBFL_arg}"
42        LFBFL_first=0
43        continue
44      fi
45      if [[ "${LFBFL_current}" != "${LFBFL_arg}" ]]; then
46        return 0
47      fi
48    done
49    return 1
50  }
51
52  some_distinct(){
53    ! equal "$@"
54  }
55
56  all_distinct(){
57    # Returns 1 if all the arguments strings are distinct.
58    declare -ir LFBFL_ij_max=$#
59    declare -i LFBFL_i
60    declare -i LFBFL_j
61    for ((LFBFL_i=1; LFBFL_i<=LFBFL_ij_max; ++LFBFL_i)); do
62      for ((LFBFL_j=1; LFBFL_j<=LFBFL_ij_max; ++LFBFL_j)); do
63        [[ LFBFL_i -eq LFBFL_j ]] && continue
64        [[ "${!LFBFL_i}" == "${!LFBFL_j}" ]] && return 0
65      done
66    done
67    return 1
68  }
69
70  all_distinct2(){
71    # Returns 1 if all the arguments strings are distinct.
72    local LFBFL_element
73    local LFBFL_previous_element
74    if [[ $# -le 1 ]]; then
75      return 1
76    fi
```

```
  declare -r LFBFL_s_sorted_arguments=$(
    printf "%s\n" "$@" \
    | sort
  )
  declare -a LFBFL_arr_sorted_arguments
  mapfile -t LFBFL_arr_sorted_arguments <<< "${LFBFL_s_sorted_arguments}"
  declare -i LFBFL_i
  declare -ir LFBFL_i_max=${#LFBFL_arr_sorted_arguments[@]}
  LFBFL_previous_element=${LFBFL_arr_sorted_arguments[0]}
  for ((LFBFL_i=1; LFBFL_i<LFBFL_i_max; ++LFBFL_i)); do
    LFBFL_element="${LFBFL_arr_sorted_arguments[${LFBFL_i}]}"
    if [[ "${LFBFL_previous_element}" == "${LFBFL_element}" ]]; then
      return 0
    fi
    LFBFL_previous_element="${LFBFL_element}"
  done
  return 1
}

all_distinct3(){
  # Returns 1 if all the arguments strings are distinct.
  # all_distinct3 is slower than old all_distinct2 since a call to sort is
  # needed:
  # the manual of uniq says it doesn't detect non-adjacent repeated lines.
  # A few weeks ago, when testing without sort, I had no bug...
  # I may have tested all_distinct3 a a b and not all_distinct3 a b a
  # There is a sort also in all_distinct2, but uniq could have an option to
  # sort before "applying -d", or sort could have an option for printing
  # duplicates only. It would complement -u option of sort.
  # Both solutions would avoid one pipe and one more shell command call.
  # Note that old all_distinct2 was faster with pipes and read -r commands,
  # but read -r has small drawbacks. Using mapfile is a cleaner way to do
  # it, since it handles arguments with spaces from scratch.
  # Arguments with line-returns are still not handled.
  # Use C, PHP or Python for example for that.
  # Somehow Bash was made to push people to stop being lazy and do actual C
  # code, even if it needs compiling something, adding a makefile, etc. XD
  # Note that C code would be faster with all_distinct2 blueprint,
  # that's the reason I kept all 3 versions, see the (asymptotic)
  # complexity and simple algorithms ideas on this toy problem.
  if [[ $# -le 1 ]]; then
    return 1
  fi
  declare -r LFBFL_s_duplicated_arguments=$(
    printf "%s\n" "$@" \
    | sort\
```

```
    | uniq -d
  )
  [[ -n "${LFBFL_s_duplicated_arguments}" ]]
}

some_equal(){
  ! all_distinct "$@"
}

some_equal2(){
  ! all_distinct2 "$@"
}

some_equal3(){
  ! all_distinct3 "$@"
}

max(){
  # Returns the maximum string among the arguments.
  # Renvoie la chaîne la plus grande parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ max sort a b c
  # c
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1 -r"\
    | head -1
}

min(){
  # Returns the minimum string among the arguments.
  # Renvoie la chaîne la plus petite parmi celles proposées.
  # $1=$sort command with the flags you want to be applied.
  # $ min sort a b c
  # a
  if [[ $# -eq 0 ]]; then
    return 1
  fi
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  printf "%s\n" "${@:2}" \
    | eval "$1"\
```

```
    | head -1
}

is_substring(){
  # $1=$string
  # $2=$substring
  declare -r LFBFL_var_1=$(
    printf "%s" "$1"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
  )
  declare -r LFBFL_var_2=$(
    printf "%s" "$2"\
    | sed -e 's/\\/\\&/g' -e "s/'/\\\\&/g"
  )
  local LFBFL_var_3="\$a = '${LFBFL_var_1}';"
  LFBFL_var_3+=" \$b = preg_quote('${LFBFL_var_2}');"
  LFBFL_var_3+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_3+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_3
  php -r "${LFBFL_var_3}"
  return $?
}

is_subfile(){
  # $1=$file_name
  # $2=$sub_file_name
  local LFBFL_var_1="\$a = file_get_contents('$1');"
  LFBFL_var_1+=" \$b = preg_quote(file_get_contents('$2'));"
  LFBFL_var_1+=" \$c = '/'.addcslashes(\$b, '/').'/';"
  LFBFL_var_1+=" die(1 - preg_match(\$c, \$a));"
  readonly LFBFL_var_1
  php -r "${LFBFL_var_1}"
  return $?
}

alias min_from_int_sort="min 'sort --numeric-sort'"
alias max_from_int_sort="max 'sort --numeric-sort'"
```

2.10 build_and_checks_dependencies/create_PDF.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# of the GNU General Public License
```

```
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#
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#
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# This file was renamed from "create_PDF.sh" to "create_PDF.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=generate_from_template.libr.sh
source "./${LFBFL_subdir}/generate_from_template.libr.sh"
# shellcheck source=lines_counts.libr.sh
source "./${LFBFL_subdir}/lines_counts.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=lines_maps.libr.sh
source "./${LFBFL_subdir}/lines_maps.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

create_PDF(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1
```

```
53
54    enhanced_set_shell_option pipefail\
55      && trap 'enhanced_unset_shell_option pipefail' RETURN
56
57    local LFBFL_subdir2="build_and_checks_variables"
58    local LFBFL_data_file_name="${LFBFL_subdir2}/repository_data.txt"
59    local LFBFL_repository_name=""
60    grep_variable "${LFBFL_data_file_name}" repository_name\
61      --result-variable-prefix="LFBFL_"\
62      --replace-line-returns-by=""
63
64    local LFBFL_repository_web_url=""
65    grep_variable "${LFBFL_data_file_name}" repository_web_url\
66      --result-variable-prefix="LFBFL_"\
67      --replace-line-returns-by=""
68
69    local LFBFL_repository_git_url=""
70    grep_variable "${LFBFL_data_file_name}" repository_git_url\
71      --result-variable-prefix="LFBFL_"\
72      --replace-line-returns-by=""
73
74    local LFBFL_abstract=""
75    grep_variable "${LFBFL_data_file_name}" abstract\
76      --result-variable-prefix="LFBFL_"
77
78    local LFBFL_acknowledgments=""
79    grep_variable "${LFBFL_data_file_name}" acknowledgments\
80      --result-variable-prefix="LFBFL_"
81
82    local LFBFL_author_email=""
83    grep_variable "${LFBFL_data_file_name}" author_email\
84      --result-variable-prefix="LFBFL_"\
85      --replace-line-returns-by=""
86
87    local LFBFL_author_full_name=""
88    grep_variable "${LFBFL_data_file_name}" author_full_name\
89      --result-variable-prefix="LFBFL_"\
90      --replace-line-returns-by=" "
91
92    local LFBFL_author_website=""
93    grep_variable "${LFBFL_data_file_name}" author_website\
94      --result-variable-prefix="LFBFL_"\
95      --replace-line-returns-by=""
96
97    declare -i LFBFL_max_line_length
98    grep_variable "${LFBFL_data_file_name}" max_line_length\
```

```
    --result-variable-prefix="LFBFL_"

  declare -r LFBFL_current_date=$(date -I"date")

  declare -r LFBFL_current_git_SHA1=$(git rev-parse HEAD)

  declare -ir LFBFL_number_of_commits=$(
    git shortlog\
    | space_starting_lines\
    | wc -l
  )

  local LFBFL_redirect="/dev/stdout"
  if [[ LFBFL_i_verbose -eq 0 ]]; then
    LFBFL_redirect="/dev/null"
  fi
  {
    code_lines_count_all "$@"
    code_lines_count_not_empty "$@"
    code_lines_count_empty "$@"
  } &> "${LFBFL_redirect}"

  local LFBFL_number_of_lines="${code_lines_count_all_result}"
  LFBFL_number_of_lines+=" total lines,"
  LFBFL_number_of_lines+=" ${code_lines_count_not_empty_result}"
  LFBFL_number_of_lines+=" not empty lines,"
  LFBFL_number_of_lines+=" ${code_lines_count_empty_result}"
  LFBFL_number_of_lines+=" empty lines."
  readonly LFBFL_number_of_lines

  local LFBFL_temp_path
  # LFBFL_temp_path=$(realpath "./${LFBFL_subdir2}/temp") for pushd
  LFBFL_temp_path="./${LFBFL_subdir2}/temp"
  readonly LFBFL_temp_path

  # see https://gitlab.com/OldManProgrammer/unix-tree/-/issues
  # ?show=eyJpaWQiOiI0MyIsImZ1bGxfcGF0aCI6Ik9sZE1hblByb2dyYW1t
  # ZXIvdW5peC10cmVlIiwiaWQiOjE4NTE0NTgzOH0%3D
  tree -a --gitignore\
    -I "node_modules/"\
    -I "__pycache__/"\
    -I ".mypy_cache/"\
    -I ".ruff_cache/"\
    -I ".git/"\
    | replace_non_ascii_spaces\
    > "${LFBFL_temp_path}/current_tree_light.txt"
```

```
145
146   tree -a -DFh --gitignore\
147     -I "node_modules/"\
148     -I "__pycache__/"\
149     -I ".mypy_cache/"\
150     -I ".ruff_cache/"\
151     -I ".git/"\
152     | replace_non_ascii_spaces\
153     > "${LFBFL_temp_path}/current_tree.txt.temp"
154
155   local LFBFL_s_files_paths
156   LFBFL_s_files_paths=$(
157     sed -Ez 's/(.)\\\n/\1/Mg' "./${LFBFL_subdir2}/files_names_listing.txt"\
158     | grep -v '^//'
159   )
160   readonly LFBFL_s_files_paths
161   declare -a LFBFL_arr_files_paths
162   if [[ -n "${LFBFL_s_files_paths}" ]]; then
163     mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
164   fi
165   readonly LFBFL_arr_files_paths
166
167   local LFBFL_temp_files_listing="${LFBFL_temp_path}/"
168   LFBFL_temp_files_listing+="files_listing.tex.tpl.temp"
169   readonly LFBFL_temp_files_listing
170   : > "${LFBFL_temp_files_listing}"
171   local LFBFL_temp_files_listing2="${LFBFL_temp_path}/"
172   LFBFL_temp_files_listing2+="files_listing.html.tpl.temp"
173   readonly LFBFL_temp_files_listing2
174   : > "${LFBFL_temp_files_listing2}"
175   # HTML <li> elements, hence "lis".
176   local LFBFL_temp_files_lis
177   LFBFL_temp_files_lis="${LFBFL_temp_path}/files_lis.html.tpl"
178   readonly LFBFL_temp_files_lis
179   : > "${LFBFL_temp_files_lis}"
180   # shellcheck disable=SC2248
181   get_split_score_simple 1 ${LFBFL_max_line_length} /
182   declare -r LFBFL_score_command="${get_split_score_result}"
183   local LFBFL_score_command_properties="${get_split_score_result2}"
184   readonly LFBFL_score_command_properties
185   # shellcheck disable=SC2248
186   get_split_score_simple 1 ${LFBFL_max_line_length} ':'
187   declare -r LFBFL_score_command2="${get_split_score_result}"
188   local LFBFL_score_command_properties2="${get_split_score_result2}"
189   readonly LFBFL_score_command_properties2
190   declare -r LFBFL_suffix='%'
```

```
  declare -i LFBFL_i=0
  local LFBFL_cleaned_path1
  local LFBFL_cleaned_path2
  local LFBFL_new_lines
  local LFBFL_new_lines2
  local LFBFL_new_lines3
  local LFBFL_file_path
  local LFBFL_s_format
  # Remove line returns here to keep lines short.
  for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
    printf "Listing file for tex/HTML : %s\n" "${LFBFL_file_path}"
    # printf "Listing file for tex : %s\n" "${LFBFL_file_path}"
    # LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    LFBFL_cleaned_path1="${LFBFL_file_path//_/\\_}"
    LFBFL_cleaned_path2=$(
      printf "%s" "${LFBFL_file_path}"\
      | sed -e 's/\//:/g' -e 's/\.//g'
    )
    LFBFL_new_lines="  ${LFBFL_cleaned_path1}"
    if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"\
        "\\"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines2="  ${LFBFL_cleaned_path2}"
    if [[ ${#LFBFL_new_lines2} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines2}"\
        ""\
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command2}"\
        "${LFBFL_score_command_properties2}"
      LFBFL_new_lines2=${repeated_split_last_line_result}
    fi
    LFBFL_new_lines3="${LFBFL_file_path}"
    if [[ ${#LFBFL_new_lines3} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines3}"\
        ""\
```

```
        ${LFBFL_max_line_length}\
        "${LFBFL_suffix}"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines3=${repeated_split_last_line_result}
    fi
    LFBFL_s_format="\\subsection{\n%s\n}\n\\label{\n%s\n}\n\n"
    LFBFL_s_format+="\\VerbatimInput[numbers=left,xleftmargin=-5mm]"
    LFBFL_s_format+="{\n%s\n}\n\n\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_new_lines}"\
      "${LFBFL_new_lines2}"\
      "${LFBFL_new_lines3}"\
      >> "${LFBFL_temp_files_listing}"

    # printf "Listing file for HTML : %s\n" "${LFBFL_file_path}"
    ((++LFBFL_i))
    LFBFL_new_lines="${LFBFL_file_path}"
    if [[ ${#LFBFL_file_path} -gt LFBFL_max_line_length ]]; then
      # shellcheck disable=SC2248
      repeated_split_last_line "${LFBFL_new_lines}"\
        "-->"\
        ${LFBFL_max_line_length}\
        "<!--"\
        "${LFBFL_score_command}"\
        "${LFBFL_score_command_properties}"
      LFBFL_new_lines=${repeated_split_last_line_result}
    fi
    {
      LFBFL_s_format="<h3 id=\"subsection2.%s\">2.%s\n%s\n</h3>\n"
      LFBFL_s_format+="<pre class=\"numbered_lines\">\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_i}"\
        "${LFBFL_i}"\
        "${LFBFL_new_lines}"
      sed -e 's/&/\&amp;/g' -e 's/</\&lt;/g'\
        < "${LFBFL_file_path}"
      printf "</pre>\n\n\n"
    } >> "${LFBFL_temp_files_listing2}"
    LFBFL_s_format="      <li><a href=\"#subsection2.%s\">\n%s\n"
    LFBFL_s_format+="      </a></li>\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}" "${LFBFL_i}" "${LFBFL_new_lines}"\
      >> "${LFBFL_temp_files_lis}"
```

```
  done
  overwrite_if_not_equal "./${LFBFL_subdir2}/files_listing.tex.tpl"\
    "${LFBFL_temp_files_listing}"
  sed -i -e "s|\\n</pre>|</pre>|g" "${LFBFL_temp_files_listing2}"
  overwrite_if_not_equal\
    "${LFBFL_temp_path}/files_listing.html.tpl"\
    "${LFBFL_temp_files_listing2}"

  # We verify if some lines are beyond max_line_length characters
  # in current_tree_light.txt and current_tree.txt.
  declare -ar LFBFL_trees=(
    "current_tree_light.txt"
    "current_tree.txt.temp"
  )
  local LFBFL_tree_path
  local LFBFL_line
  local LFBFL_prefix
  declare -i LFBFL_prefix_last_position
  local LFBFL_char
  local LFBFL_line_start
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  local LFBFL_file_name
  local LFBFL_s_lines
  declare -a LFBFL_arr_lines
  declare -i LFBFL_keep_length
  for LFBFL_tree in "${LFBFL_trees[@]}"; do
    LFBFL_tree_path="${LFBFL_temp_path}/${LFBFL_tree}"
    # shellcheck disable=SC2248
    LFBFL_s_lines=$(grep '.\{'${LFBFL_overlength}'\}' "${LFBFL_tree_path}")
    if [[ -z "${LFBFL_s_lines}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
    for LFBFL_line in "${LFBFL_arr_lines[@]}"; do
      # printf "LFBFL_line: %s\n" "${LFBFL_line}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E -e 's/(.*)——[^—]+$/\1/g'
      )
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_prefix_last_position=$((${#LFBFL_prefix}-1))
      LFBFL_char="${LFBFL_prefix:${LFBFL_prefix_last_position}:1}"
      # printf "LFBFL_char: %s\n" "${LFBFL_char}"
      LFBFL_prefix=$(
        printf "%s" "${LFBFL_prefix}"\
        | sed -E -e 's/[^ ]+$//g'
```

```
      )
      if [[ "${LFBFL_char}" == "└" ]]; then
        LFBFL_prefix+="  "
      else
        LFBFL_prefix+="│ "
      fi
      # printf "LFBFL_prefix: %s\n" "${LFBFL_prefix}"
      LFBFL_file_name=$(
        printf "%s" "${LFBFL_line}"\
        | sed -E 's|.* (([a-zA-Z0-9\._/-]+).)$|\1|g'
      )
      # printf "LFBFL_file_name: %s\n" "${LFBFL_file_name}"
      LFBFL_keep_length=$((${#LFBFL_line} - ${#LFBFL_file_name}))
      LFBFL_line_start=${LFBFL_line:0:${LFBFL_keep_length}}
      LFBFL_line_start=$(
        printf "%s" "${LFBFL_line_start}"\
        | sed -e 's/ *$//g'
      )
      # printf "LFBFL_line_start: %s\n" "${LFBFL_line_start}"
      LFBFL_new_lines="${LFBFL_prefix}${LFBFL_file_name}"
      if [[ ${#LFBFL_new_lines} -gt LFBFL_max_line_length ]]; then
        # shellcheck disable=SC2248
        repeated_split_last_line "${LFBFL_new_lines}"\
          "${LFBFL_prefix}"\
          ${LFBFL_max_line_length}\
          ""\
          ""\
          ""
        LFBFL_new_lines=${repeated_split_last_line_result}
      fi
      printf "%s\n%s\n" "${LFBFL_line_start}" "${LFBFL_new_lines}"\
        > "${LFBFL_tree_path}.some_line.temp"
      insert_file_at_token "${LFBFL_tree_path}"\
        "${LFBFL_line}"\
        "${LFBFL_tree_path}.some_line.temp"\
        ""\
        --quiet
    done
    rm -f "${LFBFL_tree_path}.some_line.temp"
  done

  overwrite_if_not_equal "${LFBFL_temp_path}/current_tree.txt"\
    "${LFBFL_temp_path}/current_tree.txt.temp" 1 1

  declare -r LFBFL_tex_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.tex"
```

```
  declare -r LFBFL_html_path_start=\
"${LFBFL_temp_path}/${LFBFL_repository_name}.html"

  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl" ]];
  then
    # If there is a template, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex.tpl"\
      "${LFBFL_tex_path_start}.1"
  elif [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex" ]];
  then
    # Otherwise if there is a tex file, we init the process from it.
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.1"
  else
    printf "Neither .tex.tpl, nor .tex in ./%s/\n" "${LFBFL_subdir2}"
  fi

  # Same logic with repository HTML file.
  if [[ -f "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl" ]];
  then
    cp "./${LFBFL_subdir2}/${LFBFL_repository_name}.html.tpl"\
      "${LFBFL_html_path_start}.1"
  elif [[ -f "./${LFBFL_repository_name}.html" ]]; then
    cp "./${LFBFL_repository_name}.html" "${LFBFL_html_path_start}.1"
  else
    printf "Neither .html.tpl in ./%s/, nor .html in ./\n"\
      "${LFBFL_subdir2}"
  fi

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_abstract}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/abstract_temp"

  # shellcheck disable=SC2001
  printf "%s\n" "${LFBFL_acknowledgments}"\
    | sed -e 's/\\n/\n/g'\
    > "${LFBFL_temp_path}/acknowledgments_temp"

  # But the filling still occurs, in case the dev want to refill
  # part of the tex/html file that he modified with @token@
  # using some computed results.
  # Tex filling:
  if [[ -f "${LFBFL_tex_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
           -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
```

```
          -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
          -e "s|@author_email@|${LFBFL_author_email}|g"\
          -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
          -e "s|@author_website@|${LFBFL_author_website}|g"\
          -e "s|@current_date@|${LFBFL_current_date}|g"\
          -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
          -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
          -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_tex_path_start}.1"

    insert_file_at_token "${LFBFL_tex_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_tex_path_start}.2"

    insert_file_at_token "${LFBFL_tex_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_tex_path_start}.3"

    insert_file_at_token "${LFBFL_tex_path_start}.3"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_tex_path_start}.4"

    insert_file_at_token "${LFBFL_tex_path_start}.4"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_tex_path_start}.5"

    insert_file_at_token "${LFBFL_tex_path_start}.5"\
      @files_listing_VerbatimInput@\
      "./${LFBFL_subdir2}/files_listing.tex.tpl"\
      "${LFBFL_tex_path_start}.6"

    overwrite_if_not_equal\
      "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
      "${LFBFL_tex_path_start}.6" 1
  fi

  # HTML filling:
  declare -i LFBFL_HTML_updated=0
  if [[ -f "${LFBFL_html_path_start}.1" ]]; then
    sed -i -e "s|@repository_name@|${LFBFL_repository_name}|g"\
          -e "s|@repository_web_url@|${LFBFL_repository_web_url}|g"\
          -e "s|@repository_git_url@|${LFBFL_repository_git_url}|g"\
          -e "s|@author_email@|${LFBFL_author_email}|g"\
```

```
            -e "s|@author_full_name@|${LFBFL_author_full_name}|g"\
            -e "s|@author_website@|${LFBFL_author_website}|g"\
            -e "s|@current_date@|${LFBFL_current_date}|g"\
            -e "s|@current_git_SHA1@|${LFBFL_current_git_SHA1}|g"\
            -e "s|@number_of_commits@|${LFBFL_number_of_commits}|g"\
            -e "s|@number_of_lines@|${LFBFL_number_of_lines}|g"\
      "${LFBFL_html_path_start}.1"

    insert_file_at_token "${LFBFL_html_path_start}.1" @abstract@\
      "${LFBFL_temp_path}/abstract_temp"\
      "${LFBFL_html_path_start}.2"

    insert_file_at_token "${LFBFL_html_path_start}.2"\
      @acknowledgments@\
      "${LFBFL_temp_path}/acknowledgments_temp"\
      "${LFBFL_html_path_start}.3"

    insert_file_at_token "${LFBFL_html_path_start}.3"\
      @files_lis@ "${LFBFL_temp_files_lis}"\
      "${LFBFL_html_path_start}.4"

    insert_file_at_token "${LFBFL_html_path_start}.4"\
      @current_tree_light@\
      "${LFBFL_temp_path}/current_tree_light.txt"\
      "${LFBFL_html_path_start}.5"

    insert_file_at_token "${LFBFL_html_path_start}.5"\
      @current_tree@\
      "${LFBFL_temp_path}/current_tree.txt"\
      "${LFBFL_html_path_start}.6"

    insert_file_at_token "${LFBFL_html_path_start}.6"\
      @files_listing_HTMLPreInput@\
      "${LFBFL_temp_path}/files_listing.html.tpl"\
      "${LFBFL_html_path_start}.7"

    overwrite_if_not_equal "./${LFBFL_repository_name}.html"\
      "${LFBFL_html_path_start}.7" 1
    LFBFL_HTML_updated=$?
  fi

  # The HTML contains everything including listings with source code.
  # If it doesn't get updated,
  # then it is useless to recompute the PDF.
  if [[ LFBFL_HTML_updated -eq 0 ]]; then
    return
```

```
  fi

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"
    done
  else
    for ((i=0; i<3; i++)); do
      pdflatex "./${LFBFL_subdir2}/${LFBFL_repository_name}.tex"\
        > /dev/null
    done
  fi

  LFBFL_files_to_temp=(\
    "${LFBFL_repository_name}.aux"\
    "${LFBFL_repository_name}.log"\
    "${LFBFL_repository_name}.out"\
    "${LFBFL_repository_name}.toc"
  )
  for LFBFL_file_name in "${LFBFL_files_to_temp[@]}"; do
    mv "${LFBFL_file_name}" "${LFBFL_temp_path}/"
  done
}

create_PDF "$@"
```

2.11 build_and_checks_dependencies/generate_from_template.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```
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# This file was renamed from "generate_from_template.sh" to
# "generate_from_template.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=overwrite_if_not_equal.libr.sh
source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"

generate_from_template_with_block_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=enter_block_comment
  # $4=exit_block_comment
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_block_comments.temp"
  readonly LFBFL_temp
  {
    printf "%s\n" "$3"
    cat "$1"
    printf "%s\n" "$4"
  } > "${LFBFL_temp}"
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

generate_from_template_with_line_comments(){
  # $1=base_file_name
  # $2=target_file_name
  # $3=line_comment_prefix
  # $4=optional_post_processing
  # $5=file_prefix
  local LFBFL_temp
  LFBFL_temp="$2.generate_from_template_with_line_comments.temp"
  readonly LFBFL_temp
  if [[ -n "$5" ]]; then
    printf "%s\n" "$5"\
      > "${LFBFL_temp}"
  fi
  sed -e "s/^/$3/g" "$1"\
    >> "${LFBFL_temp}"
  if [[ -n "$4" ]]; then
```

```
    eval "$4"
  fi
  overwrite_if_not_equal "$2" "${LFBFL_temp}"
}

split_file_in_two(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_part1
  # $4=$file_name_part2
  # Other arguments are options for grep.
  declare -a LFBFL_grep_options=()
  declare -i LFBFL_i=0
  local LFBFL_arg
  for ((LFBFL_i=5; LFBFL_i<=$#; ++LFBFL_i)); do
    LFBFL_arg="${!LFBFL_i}"
    LFBFL_grep_options+=("${LFBFL_arg}")
  done
  declare -ir LFBFL_line_number=$(
    grep -n "${LFBFL_grep_options[@]}" -- "$2" "$1"\
    | head --lines=1\
    | cut -f 1 -d ':'
  )
  declare -ir LFBFL_line_count=$(ll_wc -l -n "$1")
  declare -ir LFBFL_lines_after=$((
    LFBFL_line_count - LFBFL_line_number
  ))
  head --lines="$((LFBFL_line_number - 1))" "$1"\
    > "$3"
  tail --lines="${LFBFL_lines_after}" "$1"\
    > "$4"
}

insert_file_at_token(){
  # $1=$file_name
  # $2=$token assume that token is the only thing on his line.
  # $3=$file_name_to_insert
  # $4=$file_name_result
  # Options:
  #   $5=--quiet to suppress diff output in overwrite_if_not_equal
  declare -r LFBFL_start_file_name="$1.insert_file_at_token1.temp"
  declare -r LFBFL_end_file_name="$1.insert_file_at_token2.temp"
  local LFBFL_result_file_name
  if [[ -n "$4" ]]; then
    LFBFL_result_file_name="$4"
  else
```

```
    LFBFL_result_file_name="$1.insert_file_at_token3.temp"
  fi
  readonly LFBFL_result_file_name
  split_file_in_two "$1" "$2" "${LFBFL_start_file_name}"\
    "${LFBFL_end_file_name}" --fixed-strings
  cat "${LFBFL_start_file_name}" "$3" "${LFBFL_end_file_name}"\
    > "${LFBFL_result_file_name}"
  rm "${LFBFL_start_file_name}" "${LFBFL_end_file_name}"
  if [[ -z "$4" ]]; then
    overwrite_if_not_equal "$1" "${LFBFL_result_file_name}" "" "" "$5"
  fi
}
```

2.12 build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "get_common_text_glob_patterns.sh"
# to "get_common_text_glob_patterns.libr.sh".

get_COMMON_TEXT_PATHS_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_PATHS_GLOB_PATTERNS}" ]]; then
    return
  fi

```

```
  declare -agr COMMON_TEXT_PATHS_GLOB_PATTERNS=(
    '**/*.c'
    '**/*.css'
    '**/*.h'
    '**/*.htm'
    '**/*.html'
    '**/*.ini'
    '**/*.js'
    '**/*.json'
    '**/*.md'
    '**/*.php'
    '**/*.py'
    '**/*.sh'
    '**/*.sql'
    '**/*.tex'
    '**/*.toml'
    '**/*.tpl'
    '**/*.ts'
    '**/*.txt'
    '**/*.xml'
    '**/*.yml'
    '**/COPYING'
    '**/COPYING.LESSER'
    '**/pre-commit'
    '**/post-commit'
    '**/.gitignore'
    '**/py.typed'
  )
}


get_COMMON_TEXT_FILES_GLOB_PATTERNS(){
  if [[ -n "${COMMON_TEXT_FILES_GLOB_PATTERNS}" ]]; then
    return
  fi

  declare -agr COMMON_TEXT_FILES_GLOB_PATTERNS=(
    '*.c'
    '*.css'
    '*.h'
    '*.htm'
    '*.html'
    '*.ini'
    '*.js'
    '*.json'
    '*.md'
```

```
    '*.php'
    '*.py'
    '*.sh'
    '*.sql'
    '*.tex'
    '*.toml'
    '*.tpl'
    '*.ts'
    '*.txt'
    '*.xml'
    '*.yml'
    'COPYING'
    'COPYING.LESSER'
    'pre-commit'
    'post-commit'
    '.gitignore'
    'py.typed'
  )
}
```

2.13 build_and_checks_dependencies/grammar_and_spell_check.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "grammar_and_spell_check.sh"
```

```
25 # to "grammar_and_spell_check.libr.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.libr.sh
29 source "./${LFBFL_subdir}/common_options.libr.sh"
30 # shellcheck source=get_common_text_glob_patterns.libr.sh
31 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
32 # shellcheck source=lines_filters.libr.sh
33 source "./${LFBFL_subdir}/lines_filters.libr.sh"
34
35 grammar_and_spell_check(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   declare -i LFBFL_i_verbose=0
40   get_verbose_option "$@"
41   local LFBFL_work_directory=""
42   get_work_directory_option "$@"
43   pushd_to_work_directory\
44     && trap 'popd_from_work_directory' RETURN
45   can_continue_after_enhanced_pushd || return 1
46
47   enhanced_set_shell_option pipefail\
48     && trap 'enhanced_unset_shell_option pipefail' RETURN
49
50   get_COMMON_TEXT_FILES_GLOB_PATTERNS
51   local LFBFL_grammar_or_spell_checker_command=""
52   grep_variable "$1" grammar_or_spell_checker_command\
53     --result-variable-prefix="LFBFL_"\
54     --replace-line-returns-by=""
55   declare -r LFBFL_command="${LFBFL_grammar_or_spell_checker_command}"
56   if [[ -z "${LFBFL_command}" ]]; then
57     return
58   fi
59   local LFBFL_file_path
60   local LFBFL_s_files_paths
61   declare -a LFBFL_arr_files_paths
62   local LFBFL_pattern
63   local LFBFL_eval_string
64   for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
65     [[ LFBFL_i_verbose -eq 1 ]]\
66       && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
67     LFBFL_s_files_paths=$(
68       find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
69       | relevant_find
70     )
```

```
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "grammar_and_spell_check: No file found.\n"
      return
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      LFBFL_eval_string="${LFBFL_command} ${LFBFL_file_path}"
      eval "${LFBFL_eval_string}"
    done
  done
}
```

2.14 build_and_checks_dependencies/licenses_templates/build_licenses_templates.exec.sh

```
#!/usr/bin/env bash
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# This file was renamed from "build_licenses_templates.sh"
# to "build_licenses_templates.exec.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=check_shell_scripts_beginnings.libr.sh
source "./${LFBFL_subdir}/check_shell_scripts_beginnings.libr.sh"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
```

```
32 # shellcheck source=comparisons.libr.sh
33 source "./${LFBFL_subdir}/comparisons.libr.sh"
34 # shellcheck source=generate_from_template.libr.sh
35 source "./${LFBFL_subdir}/generate_from_template.libr.sh"
36 # shellcheck source=lines_filters.libr.sh
37 source "./${LFBFL_subdir}/lines_filters.libr.sh"
38 # shellcheck source=overwrite_if_not_equal.libr.sh
39 source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
40
41 build_licenses_templates(){
42   # Options:
43   #   --verbose
44   #   --work-directory=""
45   declare -i LFBFL_i_verbose=0
46   get_verbose_option "$@"
47   local LFBFL_work_directory=""
48   get_work_directory_option "$@"
49
50   enhanced_set_shell_option pipefail\
51     && trap 'enhanced_unset_shell_option pipefail' RETURN
52
53   # Source
54   local LFBFL_license_source_subdir
55   LFBFL_license_source_subdir="./${LFBFL_subdir}/licenses_templates/"
56   readonly LFBFL_license_source_subdir
57   local LFBFL_source_license_prefix="${LFBFL_license_source_subdir}"
58   LFBFL_source_license_prefix+="license_file_header_"
59   readonly LFBFL_source_license_prefix
60   # Target
61   local LFBFL_license_target_subdir
62   LFBFL_license_target_subdir="${LFBFL_work_directory}/"
63   LFBFL_license_target_subdir="build_and_checks_variables/temp/"
64   readonly LFBFL_license_target_subdir
65   local LFBFL_target_license_prefix="${LFBFL_license_target_subdir}"
66   LFBFL_target_license_prefix+="license_file_header_"
67   readonly LFBFL_target_license_prefix
68
69   declare -ar LFBFL_licenses=(
70     "AGPLv3+"
71     "GPLv3+"
72     "LGPLv3+"
73   )
74
75   # ------------------------------------------------------------------
76   # Various language definitions for text file formats
77   declare -ar LFBFL_block_comment_languages=(
```

```
    "c"
    "py"
  )
  declare -ar LFBFL_block_comment_enters=(
    '/*'
    '"""'
  )
  declare -ar LFBFL_block_comment_exits=(
    '*/'
    '"""'
  )
  equal "${#LFBFL_block_comment_languages[@]}"\
        "${#LFBFL_block_comment_enters[@]}"\
        "${#LFBFL_block_comment_exits[@]}"
  declare -i LFBFL_equal_result=$?
  if [[ LFBFL_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 1/!\\\n'
    return 1
  fi

  declare -ar LFBFL_line_comment_languages=(
    "sh"
    "tex"
  )
  declare -ar LFBFL_line_comment_prefixes=(
    '# '
    '% '
  )
  equal "${#LFBFL_line_comment_languages[@]}"\
        "${#LFBFL_line_comment_prefixes[@]}"
  LFBFL_equal_result=$?
  if [[ LFBFL_equal_result -eq 0 ]]; then
    printf $'/!\\Problem with the definitions of languages arrays 2/!\\\n'
    return 1
  fi
  # ------------------------------------------------------------------

  # ------------------------------------------------------------------
  # We generate the license header templates adapted to various text
  # files formats.
  # Kamoulox do endfor, Jacques Beauregard XD
  # (C'est toujours mieux que l'inverse :) !)
  # for do endfor -> for do done (Fort Doux Donne).
  local LFBFL_some_license
  local LFBFL_license_prefix
  local LFBFL_license_prefix2
```

```
124     local LFBFL_extension
125     local LFBFL_license_file_name
126     local LFBFL_enter_string
127     local LFBFL_exit_string
128     local LFBFL_temp2
129     local LFBFL_prefix_string
130     local LFBFL_intermediate_file_name
131     local LFBFL_file_prefix
132     declare -i LFBFL_generate_from_template_result
133     for LFBFL_some_license in "${LFBFL_licenses[@]}"; do
134       LFBFL_license_prefix="${LFBFL_source_license_prefix}"
135       LFBFL_license_prefix+="${LFBFL_some_license}"
136       LFBFL_license_prefix2="${LFBFL_target_license_prefix}"
137       LFBFL_license_prefix2+="${LFBFL_some_license}"
138       for ((i=0; i<${#LFBFL_block_comment_languages[@]}; i++)); do
139         LFBFL_extension=${LFBFL_block_comment_languages[i]}
140         LFBFL_license_file_name="${LFBFL_license_prefix2}"
141         LFBFL_license_file_name+=".${LFBFL_extension}"
142         LFBFL_enter_string=${LFBFL_block_comment_enters[i]}
143         LFBFL_exit_string=${LFBFL_block_comment_exits[i]}
144         generate_from_template_with_block_comments\
145           "${LFBFL_license_prefix}.tpl"\
146           "${LFBFL_license_file_name}.tpl"\
147           "${LFBFL_enter_string}" "${LFBFL_exit_string}"
148         LFBFL_generate_from_template_result=$?
149         if [[ LFBFL_i_verbose -eq 1 ]]; then
150           if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
151             printf "License template %s.tpl updated.\n"\
152               "${LFBFL_license_file_name}"
153           elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
154             printf "License template %s.tpl created.\n"\
155               "${LFBFL_license_file_name}"
156           fi
157         fi
158       done
159       LFBFL_temp2=".generate_from_template_with_line_comments.temp"
160       for ((i=0; i<${#LFBFL_line_comment_languages[@]}; i++)); do
161         LFBFL_extension=${LFBFL_line_comment_languages[i]}
162         LFBFL_license_file_name="${LFBFL_license_prefix2}"
163         LFBFL_license_file_name+=".${LFBFL_extension}"
164         LFBFL_prefix_string=${LFBFL_line_comment_prefixes[i]}
165         LFBFL_intermediate_file_name="${LFBFL_license_file_name}.tpl"
166         LFBFL_intermediate_file_name+="${LFBFL_temp2}"
167         LFBFL_file_prefix=""
168         if [[ "${LFBFL_extension}" == "sh" ]]; then
169           LFBFL_file_prefix="${LFBFL_SHELL_SCRIPT_BEGINNING}"
```

```
      fi
      # The sed command is here to remove trailing spaces of previously
      # empty lines after comment prefix is added to the line.
      generate_from_template_with_line_comments\
        "${LFBFL_license_prefix}.tpl"\
        "${LFBFL_license_file_name}.tpl"\
        "${LFBFL_prefix_string}"\
        "sed -Ei -e 's/\s*$//g' '${LFBFL_intermediate_file_name}'"\
        "${LFBFL_file_prefix}"
      LFBFL_generate_from_template_result=$?
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        if [[ LFBFL_generate_from_template_result -eq 1 ]]; then
          printf "License template %s.tpl updated.\n"\
            "${LFBFL_license_file_name}"
        elif [[ LFBFL_generate_from_template_result -eq 2 ]]; then
          printf "License template %s.tpl created.\n"\
            "${LFBFL_license_file_name}"
        fi
      fi
    done
  done
  # ------------------------------------------------------------------

  # Now that base license templates generated languages license
  # templates in working directory, we can go there.
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_data_file_name="build_and_checks_variables/"
  LFBFL_data_file_name+="repository_data.txt"
  readonly LFBFL_data_file_name
  local LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  local LFBFL_license=""
  grep_variable "${LFBFL_data_file_name}" license\
    --result-variable-prefix="LFBFL_"
  local LFBFL_license2=""
  grep_variable "${LFBFL_data_file_name}" license2\
    --result-variable-prefix="LFBFL_"
  local LFBFL_author_full_name=""
  grep_variable "${LFBFL_data_file_name}" author_full_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=" "
```

```
LFBFL_license_prefix="build_and_checks_variables/temp/"
LFBFL_license_prefix+="license_file_header_"
readonly LFBFL_license_prefix
LFBFL_license_prefix2="${LFBFL_license_prefix}${LFBFL_license}"
readonly LFBFL_license_prefix2
declare -r\
  LFBFL_license_prefix3="${LFBFL_license_prefix}${LFBFL_license2}"

# First year according to current state of git repository.
declare -ir LFBFL_first_year=$(
  git log\
  | grep 'Date:'\
  | cut -f 8 -d ' '\
  | tail -1
)
# Last year according to current state of git repository.
declare -ir LFBFL_last_year=$(
  git log\
  | grep 'Date:'\
  | cut -f 8 -d ' '\
  | head -1
)
local LFBFL_copyright_string
LFBFL_copyright_string="${LFBFL_first_year}-${LFBFL_last_year}"
LFBFL_copyright_string+=" ${LFBFL_author_full_name}"
readonly LFBFL_copyright_string

prepare_filled_license_file_block(){
  # $1=license_file_name
  sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
      -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
      "$1.tpl"\
    > "$1.temp"
  overwrite_if_not_equal "$1" "$1.temp"
  declare -r LFBFL_result=$?
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    if [[ LFBFL_result -eq 1 ]]; then
      printf "License template %s updated.\n" "$1"
    elif [[ LFBFL_result -eq 2 ]]; then
      printf "License template %s created.\n" "$1"
    fi
  fi
  head --lines=-1 "$1"\
    | tail --lines=+2\
    > "$1.temp"
```

```
    # Strange world: tail --lines=-1 was working
    # to exclude first line.
    # I tested it months ago, and saw with my eyes the start and end
    # block comments removed with:
    # head --lines=-1 "$1" | tail --lines=-1.
    # Now, man page says to use --lines=+2 instead.
    # Welcome in the world where AI change your environment and your
    # documentation on the fly...
  }

  declare -Ar LFBFL_all_block_comment_languages=(
    ["c"]="c"
    ["php"]="c"
    ["py"]="py"
  )

  local LFBFL_key
  local LFBFL_dest
  local LFBFL_license_file_name2
  declare -i LFBFL_not_subfile
  declare -i LFBFL_not_subfile2
  local LFBFL_file_path
  local LFBFL_s_files_paths
  declare -a LFBFL_arr_files_paths
  for LFBFL_key in "${!LFBFL_all_block_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_block_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_block "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_block "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}"\
        "${LFBFL_license_file_name}.temp"
      LFBFL_not_subfile=$?
```

```
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}"\
          "${LFBFL_license_file_name2}.temp"
        LFBFL_not_subfile2=$?
      fi
      if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
    rm "${LFBFL_license_file_name}.temp"
    if [[ -n "${LFBFL_license2}" ]]; then
      rm "${LFBFL_license_file_name2}.temp"
    fi
  done

  prepare_filled_license_file_line(){
    # $1=LFBFL_license_file_name
    sed -e "s/@repository_name@/${LFBFL_repository_name}/g"\
        -e "s/@copyright_string@/${LFBFL_copyright_string}/g"\
        "$1.tpl"\
      > "$1.temp"
    overwrite_if_not_equal "$1" "$1.temp"
    declare -r LFBFL_result=$?
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      if [[ LFBFL_result -eq 1 ]]; then
        printf "License template %s updated." "$1"
      elif [[ LFBFL_result -eq 2 ]]; then
        printf "License template %s created." "$1"
      fi
    fi
  }

  declare -Ar LFBFL_all_line_comment_languages=(
    ["sh"]="sh"
    ["tex"]="tex"
  )

  for LFBFL_key in "${!LFBFL_all_line_comment_languages[@]}"; do
    LFBFL_dest=${LFBFL_all_line_comment_languages[${LFBFL_key}]}
    LFBFL_license_file_name="${LFBFL_license_prefix2}.${LFBFL_dest}"
    prepare_filled_license_file_line "${LFBFL_license_file_name}"
    if [[ -n "${LFBFL_license2}" ]]; then
      LFBFL_license_file_name2="${LFBFL_license_prefix3}"
```

```
      LFBFL_license_file_name2+=".${LFBFL_dest}"
      prepare_filled_license_file_line "${LFBFL_license_file_name2}"
    fi

    LFBFL_s_files_paths=$(
      find . -type f -name "*.${LFBFL_key}" -printf '%P\n'\
      | relevant_find
    )
    if [[ -z "${LFBFL_s_files_paths}" ]]; then
      continue
    fi
    mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
    for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
      is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name}"
      LFBFL_not_subfile=$?
      LFBFL_not_subfile2=1
      if [[ -n "${LFBFL_license2}" ]]; then
        is_subfile "${LFBFL_file_path}" "${LFBFL_license_file_name2}"
        LFBFL_not_subfile2=$?
      fi
      if [[ LFBFL_not_subfile -ge 1 && LFBFL_not_subfile2 -ge 1 ]];
      then
        printf "File %s has no/wrong license header?\n"\
          "${LFBFL_file_path}"
      fi
    done
  done
}

build_licenses_templates "$@"
```

2.15 build_and_checks_dependencies/licenses_templates/license_file_header_AGPLv3+.tpl

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```

2.16 build_and_checks_dependencies/licenses_templates/license_file_header_GPLv3+.tpl

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```

2.17 build_and_checks_dependencies/licenses_templates/license_file_header_LGPLv3+.tpl

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```

2.18 build_and_checks_dependencies/lines_counts.libr.sh

```
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# This file was renamed from "lines_counts.sh" to
# "lines_counts.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=get_common_text_glob_patterns.libr.sh
```

```
31 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
32 # shellcheck source=lines_filters.libr.sh
33 source "./${LFBFL_subdir}/lines_filters.libr.sh"
34
35 all_code_lines(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   declare -i LFBFL_i_verbose=0
40   get_verbose_option "$@"
41   local LFBFL_work_directory=""
42   get_work_directory_option "$@"
43   pushd_to_work_directory\
44     && trap 'popd_from_work_directory' RETURN
45   can_continue_after_enhanced_pushd || return 1
46
47   enhanced_set_shell_option pipefail\
48     && trap 'enhanced_unset_shell_option pipefail' RETURN
49
50   get_COMMON_TEXT_FILES_GLOB_PATTERNS
51   local LFBFL_pattern
52   for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
53     [[ LFBFL_i_verbose -eq 1 ]]\
54       && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
55     find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
56       | xargs grep -H -v 'a(?!a)a'
57   done
58 }
59
60 all_self_code_lines(){
61   # Options:
62   #   --verbose
63   #   --work-directory=""
64   declare -i LFBFL_i_verbose=0
65   get_verbose_option "$@"
66   local LFBFL_work_directory=""
67   get_work_directory_option "$@"
68   pushd_to_work_directory\
69     && trap 'popd_from_work_directory' RETURN
70   can_continue_after_enhanced_pushd || return 1
71
72   enhanced_set_shell_option pipefail\
73     && trap 'enhanced_unset_shell_option pipefail' RETURN
74
75   # below and another lines, see SC2145 false positive
76   all_code_lines "$@" \
```

```
    | relevant_grep\
    | not_license_grep\
    | not_main_tex_grep\
    | not_main_html_grep
}

# shellcheck disable=SC2120
all_self_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | empty_lines_after_file_name
}

# shellcheck disable=SC2120
all_self_not_empty_code_lines(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  all_self_code_lines "$@" \
    | not_empty_lines_after_file_name
}

code_lines_count_all(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

```

```
  declare -gi code_lines_count_all_result
  code_lines_count_all_result=$(
    all_self_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${code_lines_count_all_result}
}

code_lines_count_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_empty_result
  # shellcheck disable=SC2119
  code_lines_count_empty_result=$(
    all_self_empty_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${code_lines_count_empty_result}
}

code_lines_count_not_empty(){
  # Options:
  #   --verbose
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -gi code_lines_count_not_empty_result
  # shellcheck disable=SC2119
  code_lines_count_not_empty_result=$(
```

```
    all_self_not_empty_code_lines\
    | wc -l
  )
  # The printf is for command line use.
  # A script should use the result value instead.
  # shellcheck disable=SC2248
  printf "%s\n" ${code_lines_count_not_empty_result}
}
```

2.19 build_and_checks_dependencies/lines_filters.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
# This file was renamed from "lines_filters.sh" to
# "lines_filters.libr.sh".

# The goal of this file is to have some common line filters
# for bash processing and pipes.
# As is the case for other .sh files in this repository,
# it can be copied somewhere in your home directory
# and sourced in your .bashrc.

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

```

```
ll_wc(){
  # current options to wc:
  # -c --bytes
  # -m --chars
  # -l --lines
  # --files0-from # Incompatible with cat,
  #               # another thing intentional by ?
  #               # I don't see the meaning of "files0".
  # -L --max-line-length
  # -w --words
  # --help
  # --version
  # New option by Laurent Lyaudet (and probably other before):
  # -n --no-filenames display only number(s)
  # I think I did that the first time many years ago (Teliae ?).
  # Since then, they added it as ignored... (-n)
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  declare -a ll_wc_var_args=()
  declare -i ll_wc_var_number_only=0
  declare -i ll_wc_var_i=0
  for ll_wc_var_arg in "$@"; do
    # printf "%s\n" "${ll_wc_var_arg}"
    if [[ "${ll_wc_var_arg}" == "-n" ]]; then
      ll_wc_var_number_only=1
    elif [[ "${ll_wc_var_arg}" == "--no-filenames" ]]; then
      ll_wc_var_number_only=1
    elif [[ ll_wc_var_number_only -eq 0 ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    elif [[ "${ll_wc_var_arg}" =~ -.* ]]; then
      ll_wc_var_args[ll_wc_var_i]="${ll_wc_var_arg}"
      ll_wc_var_i=$((ll_wc_var_i + 1))
    fi
  done
  # typeset -p ll_wc_var_args
  # printf "%s\n" "${ll_wc_var_number_only}"
  if [[ ll_wc_var_number_only -gt 0 ]]; then
    # Too simple code, only the base use case is handled now.
    # Convenient but incomplete.
    # With more time,
    # the simpler would be to submit a patch to GNU coreutils,
    # unless blockade from assholes.
    # Otherwise, it may be more simply handled in PHP or Python
    # (argparse, etc.).
    # You'll need to start by improving cat first,
```

```
    # with --files0-from handling.
    # wc does not display file_name when stream...
    # hence cat of last arg "${!#}"
    cat "${!#}"\
      | wc "${ll_wc_var_args[@]}"
  else
    # printf "normal\n"
    wc "${ll_wc_var_args[@]}"
  fi
}

# I submitted a patch for wc :).
# printf "https://domain/list-path/2024-05/msg00013.html" | \
#   sed -e 's|domain|lists.gnu.org|' \
#       -e 's|list-path|archive/html/bug-coreutils|'
# I found another way to have a long URL fit on 70 characters.
# If it is merged, the following would be better:
# If --no-filenames option available,
# we redefine ll_wc command as an alias.
wc_help_text=$(wc --help)
if [[ "${wc_help_text}" == *" --no-filenames "* ]]; then
  alias ll_wc='wc'
fi

# Trace avant le (2e ?) renommage
# de wget_sha512.sh à wget_sha512.libr.sh.
# $ ll_wc wget_sha512.sh
#  41  189 1224 wget_sha512.sh
# $ ll_wc -n wget_sha512.sh
#     41     189    1224
# $ ll_wc -l wget_sha512.sh
# 41 wget_sha512.sh
# $ ll_wc -l -n wget_sha512.sh
# 41

in_place_grep(){
  declare -r LFBFL_temp="${!#}.in_place_grep.temp"
  grep "$@" \
    > "${LFBFL_temp}"
  # declare -ir LFBFL_lines_before=$(ll_wc -l -n "${!#}")
  # declare -ir LFBFL_lines_after=$(ll_wc -l -n "${LFBFL_temp}")
  # printf "%s\n" "${LFBFL_lines_before}"
  # printf "%s\n" "${LFBFL_lines_after}"
  # if [[ LFBFL_lines_before -eq LFBFL_lines_after ]]; then
  # Almost as fast (on large files) and correct in all cases :).
  # Thanks Paul Eggert for the suggestion :).
```

```
  # https://debbugs.gnu.org/cgi/bugreport.cgi?bug=80547
  if cmp --silent "${LFBFL_temp}" "${!#}"; then
    rm "${LFBFL_temp}"
    return
  fi
  mv "${LFBFL_temp}" "${!#}"
}

grep_fixed_string_with_anchor(){
  # $1=filepath
  # $2=some_fixed_string
  # Options:
  #   --enforce-line-starts-with-fixed-string
  #   --enforce-line-ends-with-fixed-string
  #   Apart from $1 and $2 and the options above,
  #   all other arguments are considered grep_options like --quiet
  #   for example.
  #   But note that --fixed-strings will be added below, you don't need to
  #   supply it.
  declare -a LFBFL_grep_options=()
  declare -a LFBFL_grep_options2=()
  declare -i LFBFL_i=0
  declare -i LFBFL_enforce_line_starts=0
  declare -i LFBFL_enforce_line_ends=0
  declare -i LFBFL_quiet=0
  local LFBFL_arg
  for LFBFL_arg in "$@"; do
    ((++LFBFL_i))
    if [[ LFBFL_i -lt 3 ]]; then
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-starts-with-fixed-string" ]];
    then
      LFBFL_enforce_line_starts=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "--enforce-line-ends-with-fixed-string" ]];
    then
      LFBFL_enforce_line_ends=1
      continue
    fi
    if [[ "${LFBFL_arg}" == "-q" || "${LFBFL_arg}" == "--quiet" ]];
    then
      LFBFL_quiet=1
      LFBFL_grep_options+=("${LFBFL_arg}")
      continue
```

```
174       fi
175       LFBFL_grep_options2+=("${LFBFL_arg}")
176     done
177
178     if [[
179       LFBFL_enforce_line_starts -eq 0
180       && LFBFL_enforce_line_ends -eq 0
181       ]];
182     then
183       grep "${LFBFL_grep_options[@]}" --fixed-strings -- "$2" "$1"
184       return
185     fi
186     declare -r LFBFL_s_lines=$(
187       grep "${LFBFL_grep_options2[@]}" --color=never --fixed-strings --\
188         "$2" "$1"
189     )
190     declare -i LFBFL_i_result=1
191     if [[ -z "${LFBFL_s_lines}" ]]; then
192       # shellcheck disable=SC2248
193       return ${LFBFL_i_result}
194     fi
195     declare -a LFBFL_arr_lines
196     mapfile -t LFBFL_arr_lines <<< "${LFBFL_s_lines}"
197     readonly LFBFL_arr_lines
198
199     declare -i LFBFL_fixed_string_length=${#2}
200     local LFBFL_some_line
201     local LFBFL_line_part
202     for LFBFL_some_line in "${LFBFL_arr_lines[@]}"; do
203       if [[ "${LFBFL_some_line}" == "$2" ]]; then
204         [[ LFBFL_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
205         LFBFL_i_result=0
206         continue
207       fi
208       if [[
209         LFBFL_enforce_line_starts -eq 1
210         && LFBFL_enforce_line_ends -eq 1
211         ]];
212       then
213         continue
214       fi
215       if [[ LFBFL_enforce_line_starts -eq 1 ]]; then
216         LFBFL_line_part=${LFBFL_some_line:0:${LFBFL_fixed_string_length}}
217       fi
218       if [[ LFBFL_enforce_line_ends -eq 1 ]]; then
219         LFBFL_line_part=${LFBFL_some_line: -${LFBFL_fixed_string_length}}
```

```
    fi
    if [[ "${LFBFL_line_part}" == "$2" ]]; then
      [[ LFBFL_quiet -eq 1 ]] || printf "%s\n" "${LFBFL_some_line}"
      LFBFL_i_result=0
    fi
  done
  LFBFL_i_result=$((LFBFL_i_result | $?))
  # shellcheck disable=SC2248
  return ${LFBFL_i_result}
}

grep_variable(){
  # $1=$file
  # $2=$variable_name
  # Options :
  #   --replace-line-returns-by=""
  #   --result-variable-prefix="LFBFL_" for example
  declare -r LFBFL_regexp="(?<=^$2=).*$"
  local LFBFL_arg
  # printf "%s\n" "${LFBFL_regexp}"
  local LFBFL_variable_value
  if [[ "$*" == *--replace-line-returns-by=* ]]; then
    local LFBFL_s_replace=""
    for LFBFL_arg in "$@"; do
      if [[ "${LFBFL_arg}" == --replace-line-returns-by=* ]]; then
        LFBFL_s_replace=${LFBFL_arg#--replace-line-returns-by=}
        break
      fi
    done
    LFBFL_variable_value=$(
      grep -oP "${LFBFL_regexp}" "$1"\
      | sed -Ez -e 's/\n/'"${LFBFL_s_replace}"'/Mg'
    )
    # We remove the trailing replace string,
    # since the intent is only to replace the intermediate \n.
    if [[ -n "${LFBFL_s_replace}" ]]; then
    LFBFL_variable_value=${LFBFL_variable_value%"${LFBFL_s_replace}"}
    fi
  else
    LFBFL_variable_value=$(grep -oP "${LFBFL_regexp}" "$1")
  fi
  local LFBFL_prefix=""
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --result-variable-prefix=* ]]; then
      LFBFL_prefix=${LFBFL_arg#--result-variable-prefix=}
      break
```

```
    fi
  done
  declare -r LFBFL_variable_name="${LFBFL_prefix}$2"
  # printf "%s : %s\n " "${LFBFL_variable_name}" "${LFBFL_variable_value}"
  # The declare line may be dropped by a local or declare
  # in outer scope.
  # declare -g "${LFBFL_variable_name}"
  # So the assignment must be done on another line.
  printf -v "${LFBFL_variable_name}" "%s" "${LFBFL_variable_value}"
  # printf "%s\n" "${!LFBFL_variable_name}"
}

empty_lines(){
  grep '^$'
}

not_empty_lines(){
  grep -v '^$'
}

space_starting_lines(){
  grep '^ '
}

not_space_starting_lines(){
  grep -v '^ '
}

# Space starting or empty lines = spampty
spampty_lines(){
  grep -E '^( |$)'
}

not_spampty_lines(){
  grep '^[^ ]'
  # grep -v -E '^( |$)'
}

all_space_lines(){
  grep '^[ ]\+$'
}

not_all_space_lines(){
  grep -v '^[ ]\+$'
}

```

```
# All space or empty lines = aspampty
aspampty_lines(){
  grep '^[ ]*$'
}

not_aspampty_lines(){
  grep '[^ ]'
  # grep -v '^[ ]*$'
}

empty_lines_after_file_name(){
  grep '^[^:]\+:$'
}

not_empty_lines_after_file_name(){
  grep -v '^[^:]\+:$'
}

space_starting_lines_after_file_name(){
  grep '^[^:]\+: '
}

not_space_starting_lines_after_file_name(){
  grep -v '^[^:]\+: '
}

spampty_lines_after_file_name(){
  grep -E '^[^:]+:( |$)'
}

not_spampty_lines_after_file_name(){
  # The two variants results differ on lines without filename before.
  # Since the intent of the function is only on lines "blabla.sh:whatever",
  # use the fastest one. I didn't do extensive testing and just assumed
  # that the "positive" one is faster as usual when querying database.
  # But there is no index here ;) XD.
  grep '^[^:]\+:[^ ]'
  # grep -v -E '^[^:]+:( |$)'
}

all_space_lines_after_file_name(){
  grep '^[^:]\+:[ ]\+$'
}

not_all_space_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]\+$'
```

```
}

aspampty_lines_after_file_name(){
  grep '^[^:]\+:[ ]*$'
}

not_aspampty_lines_after_file_name(){
  grep -v '^[^:]\+:[ ]*$'
}

# If you need to test the previous 20 functions,
# look at file spampty_data.txt in miscellaneous.
# And give a try at something like:
# cat -E <(cat miscellaneous/spampty_data.txt | empty_lines)
# :). Nice command :).

not_JS_dependencies_find(){
  grep -vE -e "(^|/)node_modules/"\
           -e "(^|/)package-lock\.json$"
}

not_JS_dependencies_grep(){
  grep -vE -e "^([^:]+/)?node_modules/"\
           -e "^([^:]+/)?package-lock\.json:"
}

not_dependencies_find(){
  not_JS_dependencies_find
}

not_dependencies_grep(){
  not_JS_dependencies_grep
}

not_python_cache_find(){
  grep -vE -e "(^|/)__pycache__/"\
           -e "(^|/)\.mypy_cache/"\
           -e "(^|/)\.ruff_cache/"
}

not_python_cache_grep(){
  grep -vE -e "^([^:]+/)?__pycache__/"\
           -e "^([^:]+/)?\.mypy_cache/"\
           -e "^([^:]+/)?\.ruff_cache/"
}
```

```
not_cache_find(){
  not_python_cache_find
}

not_cache_grep(){
  not_python_cache_grep
}

not_git_find(){
  grep -vE "(^|/)\.git/"
}

not_git_grep(){
  grep -vE "^([^:]+/)?\.git/"
}

not_archive_find(){
  grep -vE "(\.gz|\.rar|\.tar|\.tgz|\.whl)$"
}

not_archive_grep(){
  grep -vE "^[^:]*(\.gz|\.rar|\.tar|\.tgz|\.whl):"
}

not_license_find(){
  grep -vE -e "(^|/)COPYING$"\
           -e "(^|/)COPYING\.LESSER$"
}

not_license_grep(){
  grep -vE -e "^([^:]+/)?COPYING:"\
           -e "^([^:]+/)?COPYING\.LESSER:"
}

not_main_tex_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.tex$"
}

not_main_tex_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
```

```
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.tex:"
}

not_main_html_find(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "(^|/)${LFBFL_repository_name}\.html$"
}

not_main_html_grep(){
  local LFBFL_repository_name=""
  grep_variable build_and_checks_variables/repository_data.txt\
    repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  grep -vE "^([^:]+/)?${LFBFL_repository_name}\.html:"
}

not_temp_file_find(){
  grep -vE "(^|/)build_and_checks_variables/temp/.*$"
}

not_temp_file_grep(){
  grep -vE "^([^:]+/)?build_and_checks_variables/temp/[^:]*:"
}

relevant_find(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_find\
    | not_cache_find\
    | not_git_find\
    | not_archive_find\
    | not_temp_file_find
}

relevant_grep(){
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN
  not_dependencies_grep\
```

```
    | not_cache_grep\
    | not_git_grep\
    | not_archive_grep\
    | not_temp_file_grep
}
```

2.20 build_and_checks_dependencies/lines_maps.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "lines_maps.sh" to "lines_maps.libr.sh".

# https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
# Some invisible unicode characters that are a problem:
# 2000-200F 2028-202F 205F 2060-206F
# [
#   '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
#   '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
#   '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
#   '205F',
#   '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
#   '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
# ]
replace_non_ascii_spaces(){
  sed -e 's/\xc2\xa0/ /g'\
```

```
39          -e 's/\xe2\x80\x80/ /g'\
40          -e 's/\xe2\x80\x81/ /g'\
41          -e 's/\xe2\x80\x82/ /g'\
42          -e 's/\xe2\x80\x83/ /g'\
43          -e 's/\xe2\x80\x84/ /g'\
44          -e 's/\xe2\x80\x85/ /g'\
45          -e 's/\xe2\x80\x86/ /g'\
46          -e 's/\xe2\x80\x87/ /g'\
47          -e 's/\xe2\x80\x88/ /g'\
48          -e 's/\xe2\x80\x89/ /g'\
49          -e 's/\xe2\x80\x8a/ /g'\
50          -e 's/\xe2\x80\x8b/ /g'\
51          -e 's/\xe2\x80\x8c/ /g'\
52          -e 's/\xe2\x80\x8d/ /g'\
53          -e 's/\xe2\x80\x8e/ /g'\
54          -e 's/\xe2\x80\x8f/ /g'\
55          -e 's/\xe2\x80\xa8/ /g'\
56          -e 's/\xe2\x80\xa9/ /g'\
57          -e 's/\xe2\x80\xaa/ /g'\
58          -e 's/\xe2\x80\xab/ /g'\
59          -e 's/\xe2\x80\xac/ /g'\
60          -e 's/\xe2\x80\xad/ /g'\
61          -e 's/\xe2\x80\xae/ /g'\
62          -e 's/\xe2\x80\xaf/ /g'\
63          -e 's/\xe2\x81\x9f/ /g'\
64          -e 's/\xe2\x81\xa0/ /g'\
65          -e 's/\xe2\x81\xa1/ /g'\
66          -e 's/\xe2\x81\xa2/ /g'\
67          -e 's/\xe2\x81\xa3/ /g'\
68          -e 's/\xe2\x81\xa4/ /g'\
69          -e 's/\xe2\x81\xa5/ /g'\
70          -e 's/\xe2\x81\xa6/ /g'\
71          -e 's/\xe2\x81\xa7/ /g'\
72          -e 's/\xe2\x81\xa8/ /g'\
73          -e 's/\xe2\x81\xa9/ /g'\
74          -e 's/\xe2\x81\xaa/ /g'\
75          -e 's/\xe2\x81\xab/ /g'\
76          -e 's/\xe2\x81\xac/ /g'\
77          -e 's/\xe2\x81\xad/ /g'\
78          -e 's/\xe2\x81\xae/ /g'\
79          -e 's/\xe2\x81\xaf/ /g'
80  }
```

2.21 build_and_checks_dependencies/overwrite_if_not_equal.libr.sh

```
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# This file was renamed from "overwrite_if_not_equal.sh"
# to "overwrite_if_not_equal.libr.sh".

overwrite_if_not_equal(){
  # $1=$target_file_name
  # $2=$temp_file_name
  # $3=$keep_temp
  # $4=$tree_mode : when the two files are the output of tree command,
  # we ignore if all the differences are only on directories.
  # $5=--quiet : we even don't want the message of diff saying that
  #              files are distinct
  # Return 0 target was the same
  #        1 target was different
  #        2 target did not already exist
  if [[ ! -f "$1" ]]; then
    if [[ -n "$3" ]]; then
      cp -p "$2" "$1"
    else
      mv "$2" "$1"
    fi
    return 2
```

```
  fi
  declare -i LFBFL_is_equal
  if [[ -n "$4" ]]; then
    # Not setting pipefail since the result would still be incorrect.
    # shellcheck disable=SC2312
    diff "$1" "$2"\
      | grep -E "^(>|<)"\
      | grep -v "/$"
    LFBFL_is_equal=${PIPESTATUS[2]}
  else
    if [[ -n "$5" ]]; then
      diff -q "$1" "$2" > /dev/null
    else
      diff -q "$1" "$2"
    fi
    LFBFL_is_equal=1-$?
  fi
  if [[ LFBFL_is_equal -eq 1 ]]; then
    if [[ -z "$3" ]]; then
      rm "$2"
    fi
    return 0
  fi
  if [[ -n "$3" ]]; then
    cp -p "$2" "$1"
  else
    mv "$2" "$1"
  fi
  return 1
}
```

2.22 build_and_checks_dependencies/python_black_complement.libr.sh

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

# It was rejected to enhance black with this currently.
# https://github.com/psf/black/issues/2370
# And moreover, at some point, I saw the reverse rule
# in black or pylint to add an empty line...
# This regexp may give false positives,
# but that's not the end of the world.
check_no_empty_line_after_python_function_docstring(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "Checking empty lines after Python function docstrings\n"
  pcre2grep -M --\
    $'def ([^"]|"(?!"))*"""([^"]|"(?!"))*"""\n\n(?!\s*def)'\
    **/*.py
}

python_black_complement(){
  # Options:
```

```
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  check_no_empty_line_after_python_function_docstring "$@"
}
```

2.23 build_and_checks_dependencies/python_isort_complement.libr.sh

```
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# This file was renamed from "python_isort_complement.sh"
# to "python_isort_complement.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
```

```
31 source "./${LFBFL_subdir}/lines_filters.libr.sh"
32 # shellcheck source=overwrite_if_not_equal.libr.sh
33 source "./${LFBFL_subdir}/overwrite_if_not_equal.libr.sh"
34
35 check_collections_abc_place(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   declare -i LFBFL_i_verbose=0
40   get_verbose_option "$@"
41   local LFBFL_work_directory=""
42   get_work_directory_option "$@"
43   pushd_to_work_directory\
44     && trap 'popd_from_work_directory' RETURN
45   can_continue_after_enhanced_pushd || return 1
46
47   [[ LFBFL_i_verbose -eq 1 ]]\
48     && printf "Checking import of _collections_abc is at the right place\n"
49
50   enhanced_set_shell_option pipefail\
51     && trap 'enhanced_unset_shell_option pipefail' RETURN
52
53   declare -r LFBFL_temp=".check_collections_abc_place.temp"
54   local LFBFL_file_path
55   declare -r LFBFL_s_files_paths=$(
56     find . -type f -name "*.py" -printf '%P\n'\
57     | relevant_find
58   )
59   if [[ -z "${LFBFL_s_files_paths}" ]]; then
60     [[ LFBFL_i_verbose -eq 1 ]]\
61       && printf "check_collections_abc_place: No file found.\n"
62     return
63   fi
64   declare -a LFBFL_arr_files_paths
65   mapfile -t LFBFL_arr_files_paths <<< "${LFBFL_s_files_paths}"
66   readonly LFBFL_arr_files_paths
67   for LFBFL_file_path in "${LFBFL_arr_files_paths[@]}"; do
68     [[ -f "${LFBFL_file_path}" ]] || continue
69     sed -Ez 's/\n(\nfrom _collections_abc[^\n]*)/\1\n/Mg'\
70       "${LFBFL_file_path}"\
71       > "${LFBFL_file_path}${LFBFL_temp}"
72     overwrite_if_not_equal "${LFBFL_file_path}"\
73       "${LFBFL_file_path}${LFBFL_temp}"
74   done
75 }
76
```

```
python_isort_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  check_collections_abc_place "$@"
}
```

2.24 build_and_checks_dependencies/repository_data.libr.sh

```
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LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
```

```
source "./${LFBFL_subdir}/lines_filters.libr.sh"

# /!\ You need to activate extglob before parsing this file. /!\
# And I'm kind enough to activate it inside functions that need it below.

get_upgrade_venvs_time_interval_in_seconds(){
  # $1=data_file_name
  # Options:
  #   --verbose
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"

  enhanced_set_bash_option extglob\
    && trap 'enhanced_unset_bash_option extglob' RETURN

  grep_variable "$1"\
    upgrade_venvs_time_interval_in_seconds\
    --result-variable-prefix="LFBFL_"

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "You say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"

  case ${LFBFL_upgrade_venvs_time_interval_in_seconds} in
    +([0-9])) : ;;
    wat) LFBFL_upgrade_venvs_time_interval_in_seconds=${RANDOM};;
    watyouwant\?) LFBFL_upgrade_venvs_time_interval_in_seconds=${SRANDOM};;
    *) printf "No wat for you?\n";; #TempsDeCerveauDisponible XD SC2249 ;)
  esac

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "I say \"%s\".\n"\
        "${LFBFL_upgrade_venvs_time_interval_in_seconds}"
}
```

2.25 build_and_checks_dependencies/shell_checks_complement.libr.sh

```
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```

```
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# This file was renamed from "python_black_complement.sh"
# to "python_black_complement.libr.sh".

# shellcheck source=common_options.libr.sh
source "./build_and_checks_dependencies/common_options.libr.sh"

check_no_size_of_array_first_element(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  local LFBFL_s="Checking that shell scripts do not"
  LFBFL_s+=" contain wrong array size expansion."
  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf "%s\n" "${LFBFL_s}"

  pcre2grep -M -- $'\$\{#.*(\b|_)arr_(?!.*\[@\])' **/*.sh
}

check_no_misplaced_then(){
  # Options:
  #   --verbose
```

```
  #   --work-directory=""
  #   --max-line-length
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  local LFBFL_arg
  declare -i LFBFL_max_line_length=70
  for LFBFL_arg in "$@"; do
    if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
      LFBFL_arg=${LFBFL_arg#--max-line-length=}
      LFBFL_max_line_length=$((LFBFL_arg))
      break
    fi
  done
  declare -i LFBFL_max_previous_line_length=$((LFBFL_max_line_length - 5))

  enhanced_set_bash_option globstar\
    && trap 'enhanced_unset_bash_option globstar' RETURN

  [[ LFBFL_i_verbose -eq 1 ]]\
    && printf\
        "Checking that shell scripts do not contain misplaced 'then'.\n"

  local LFBFL_regexp
  LFBFL_regexp="^[^\n]{1,${LFBFL_max_previous_line_length}}"
  LFBFL_regexp+="(?<=\]\];)"
  LFBFL_regexp+="(?<!^\s{0,${LFBFL_max_previous_line_length}}\]\];)"
  LFBFL_regexp+="\n\s*then"

  pcre2grep -M -- "${LFBFL_regexp}" **/*.sh
}

shell_checks_complement(){
  # Options:
  #   --verbose
  #   --work-directory=""
  # shellcheck disable=SC2034
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
```

```
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

  check_no_size_of_array_first_element "$@"
  check_no_misplaced_then "$@"
}
```

2.26 build_and_checks_dependencies/split_score.exec.php

```
#!/usr/bin/php
<?php
/*
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2026-01-04T13:38+01:00: This file was renamed from
"split_line_at_most.exec.php"
to
"split_score.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.exec.php" to
"split_line_at_most.exec.php".
This file was renamed from
"call_split_score_after_before_or_before_after.php" to
"call_split_score_after_before_or_before_after.exec.php".
*/
```

```

require_once('./build_and_checks_dependencies/split_score.libr.php');

/*
php -r\
 $'require_once(\'./build_and_checks_dependencies/'\
$'split_score.libr.php\');'\
$' echo(generate_split_score'\
$'(true, 70, [\'/\'])(\'/\', 60, true));'
*/

// The first cast "(int) generate_split_score" is to avoid false
// positive errors on possible XSS if output is unescaped.
echo(
  (int) generate_split_score(
    (bool) $argv[1], (int) $argv[2], explode(',', $argv[3]),
  )($argv[4], (int) $argv[5], (bool) $argv[6])
);
```

2.27 build_and_checks_dependencies/split_score.libr.php

```
<?php
/*
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```

```
"split_line_at_most.libr.php"
to
"split_score.libr.php".
This file was renamed from "split_line_at_most.php" to
"split_line_at_most.libr.php".
*/

function generate_split_score(
  $b_after_before,
  $i_max_length,
  $a_delimiter_strings_domain,
){
  return function(
    $s_delimiter_string,
    $i_cut_position,
    $b_is_cut_after,
  ) use ($b_after_before, $i_max_length, $a_delimiter_strings_domain){
    if(
      !in_array(
        $s_delimiter_string, $a_delimiter_strings_domain, true
      )
    ){
      // I do not handle regexps for the moment.
      // I'm too brainfucked. I just want to finish this feature fast
      // for HTML and Tex/PDF listings generation with HTML and Tex on
      // at most 70 characters.
      return 0;
    }
    if($b_is_cut_after === $b_after_before){
      // When $b_after_before,
      // always way better to cut after '/' than before it.
      // When !$b_after_before (aka before_after),
      // always way better to cut before '/' than after it.
      return 1 + $i_max_length + $i_cut_position;
    }
    return 1 + $i_cut_position;
  };
}
```

2.28 build_and_checks_dependencies/strings_functions.libr.sh

```
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```

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# I think I created a file strings_functions.sh or renamed to
# strings_functions.sh at some point, before committing.
# Hence, someone must have changed or reversed the name of this file
# to string_functions.sh at some point.
# This file was renamed from "string_functions.sh" to
# "strings_functions.libr.sh".

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=get_common_text_glob_patterns.libr.sh
source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
# shellcheck source=comparisons.libr.sh
source "./${LFBFL_subdir}/comparisons.libr.sh"

split_line_at(){
  # $1=$line
  # $2=$position : number of characters in split_line_at_result_beginning
  declare -g split_line_at_result_beginning="${1:0:$2}"
  declare -g split_line_at_result_end="${1:$2}"
  # printf "%s\n" "$split_line_at_result_beginning"
  # printf "%s\n" "$split_line_at_result_end"
}

declare -gir SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM=1
declare -gir SPLIT_SCORE_PROPERTY_POSITION_UNIFORM=2
declare -gir SPLIT_SCORE_PROPERTY_LARGER_AFTER=4
declare -gir SPLIT_SCORE_PROPERTY_LARGER_BEFORE=8

split_score_simple(){
```

```
52    # $1=$after_before
53    # $2=$delimiters_strings_domain concatenated characters/delimiters
54    # $3=$delimiter_string a single character
55    # $4=$is_cut_after
56    declare -gi split_score_result
57    if [[ "$2" != *$3* ]]; then
58      split_score_result=0
59      return
60    fi
61    if [[ "$4" == "$1" ]]; then
62      split_score_result=2
63      return
64    fi
65    split_score_result=1
66  }
67
68  get_split_score_simple(){
69    # $1=$after_before
70    # $2=$max_length
71    # $3=$delimiters_strings_domain
72    declare -g get_split_score_result="split_score_simple $1 '$3'"
73    declare -gi get_split_score_result2
74    # get_split_score_result2=$((11-$1*4))
75    # if [[ $1 -eq 1 ]]; then
76    #   get_split_score_result2=7
77    # else
78    #   get_split_score_result2=11
79    # fi
80    get_split_score_result2=$((
81      SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
82      + SPLIT_SCORE_PROPERTY_POSITION_UNIFORM
83      + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
84      + (
85        $1 * (
86          SPLIT_SCORE_PROPERTY_LARGER_AFTER
87          - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
88        )
89      )
90    ))
91  }
92
93  get_split_score_exec(){
94    declare -g SPLIT_SCORE_EXEC="./build_and_checks_dependencies"
95    SPLIT_SCORE_EXEC+="/split_score.exec.php"
96  }
97
```

```
98  split_score(){
99    # $1=$after_before
100   # $2=$max_length
101   # $3=$delimiters_strings_domain concatenated characters/delimiters
102   # $4=$delimiter_string a single character
103   # $5=$cut_position
104   # $6=$is_cut_after
105   get_split_score_exec
106   declare -r LFBFL_command="${SPLIT_SCORE_EXEC} $1 $2 '$3' '$4' $5 $6"
107   declare -gi split_score_result
108   split_score_result=$(eval "${LFBFL_command}")
109 }
110
111 # This function is currently not called,
112 # hence we need that shellcheck disable.
113 # shellcheck disable=SC2034
114 get_split_score(){
115   # $1=$after_before
116   # $1=$max_length
117   # $2=$delimiters_strings_domain
118   declare -g get_split_score_result="split_score $1 $2 '$3'"
119   declare -gi get_split_score_result2
120   # get_split_score_result2=$((9-$1*4))
121   get_split_score_result2=$((
122     SPLIT_SCORE_PROPERTY_DELIMITER_UNIFORM
123     + SPLIT_SCORE_PROPERTY_LARGER_BEFORE
124     + (
125       $1 * (
126         SPLIT_SCORE_PROPERTY_LARGER_AFTER
127         - SPLIT_SCORE_PROPERTY_LARGER_BEFORE
128       )
129     )
130   ))
131 }
132
133 split_line_at_most(){
134   # $1=$line
135   # $2=$max_length of beginning result string
136   # $3=$split_score_command
137   #   negative score means do not consider as a possible split
138   # $4=$split_score_command_properties :
139   #   These properties are flags.
140   #   If someday some property is not a flag,
141   #   then create split_score_command_properties2.
142   #   - null or 0 no property;
143   #   - 1 split score doesn't depend on which delimiter matches;
```

```
144     #   - 2 split score doesn't depend on the current position;
145     #   - 4 any split score after is always larger
146     #     than any split score before;
147     #   - 8 any split score before is always larger
148     #     than any split score after;
149     # $5=$forbidden_previous_character
150     # $6=$split_score_command_delimiter_strings_domain
151     #   optimisation if small domain
152     #   - this argument can be "null", then we will use $3 only instead,
153     #     in that case, only mono-character strings are considered,
154     #   - this argument can be an array of characters,
155     #     optimize accordingly,
156     #   - this argument can be an array of strings,
157     #     optimize accordingly,
158     #   - this argument can be an array of regexps,
159     #     optimize accordingly.
160     #     Not sure an array of regexps makes more sense than a single
161     #     one. Maybe they could be both cases with corresponding
162     #     optimizations. Same for a single character or substring.
163     #
164     # Note that we're always looking for the last position to split
165     # among maximum score;
166     # if some position gets as score from 2 delimiters,
167     # its score is always the maximum of the two corresponding scores.
168     declare -g split_line_at_most_result_start
169     declare -g split_line_at_most_result_end
170     declare -A LFBFL_positions
171     # printf "%s %s %s %s\n" "$1" "$2" "$3" "$4"
172     # For my use case in bash scripts, I will need only an array of
173     # characters. See get_split_score().
174     if [[ -n "$6" ]]; then
175       printf $'split_line_at_most() $6 NOT IMPLEMENTED YET\n'
176     fi
177     if [[ $2 -ge ${#1} ]]; then
178       split_line_at_most_result_start=$1
179       split_line_at_most_result_end=""
180       return
181     fi
182     declare -r LFBFL_sort_command="sort --numeric-sort"
183     declare -ir LFBFL_length_minus_1=$(($(#1} - 1))
184     declare -ir LFBFL_i_max=$(
185       min "${LFBFL_sort_command}" "${LFBFL_length_minus_1}" "$2"
186     )
187     declare -i LFBFL_i
188     declare -i LFBFL_j
189     local LFBFL_current_char
```

```
190
191   # At the beginning, each position has score 0.
192   for ((LFBFL_i=0; LFBFL_i<=LFBFL_i_max; ++LFBFL_i)) do
193     LFBFL_positions["${LFBFL_i}"]="0"
194   done
195   LFBFL_j=$((LFBFL_i+1))
196   LFBFL_positions["${LFBFL_j}"]="0"
197
198   # If we have a forbidden previous character,
199   # we mark the forbidden positions with -1.
200   if [[ -n "$5" ]]; then
201     for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
202       LFBFL_j=$((LFBFL_i+1))
203       LFBFL_current_char="${1:${LFBFL_i}:1}"
204       if [[ "${LFBFL_current_char}" == "$5" ]]; then
205         LFBFL_positions["${LFBFL_j}"]="-1"
206       fi
207     done
208   fi
209
210   if [[ "$4" == "7" ]]; then
211     for ((LFBFL_j=LFBFL_i_max; LFBFL_j>0; --LFBFL_j)) do
212       LFBFL_i=$((LFBFL_j-1))
213       LFBFL_current_char="${1:${LFBFL_i}:1}"
214       eval "$3 '${LFBFL_current_char}' 0"
215       # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
216       if [[ split_score_result -ge 1 ]]; then
217         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
218         if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
219           LFBFL_positions["${LFBFL_i}"]=$(
220             max "${LFBFL_sort_command}"\
221               "${LFBFL_positions["${LFBFL_i}"]}"\
222               "${split_score_result}"
223           )
224         fi
225       fi
226       eval "$3 '${LFBFL_current_char}' 1"
227       # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
228       if [[ split_score_result -ge 1 ]]; then
229         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
230         if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
231           LFBFL_positions["${LFBFL_j}"]=$(
232             max "${LFBFL_sort_command}"\
233               "${LFBFL_positions["${LFBFL_j}"]}"\
234               "${split_score_result}"
235           )
```

```
236           break
237         fi
238       fi
239     done
240   else
241     for ((LFBFL_i=0; LFBFL_i<LFBFL_i_max; ++LFBFL_i)) do
242       LFBFL_j=$((LFBFL_i+1))
243       LFBFL_current_char="${1:${LFBFL_i}:1}"
244       eval "$3 '${LFBFL_current_char}' ${LFBFL_i} 0"
245       # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
246       if [[ split_score_result -ge 1 ]]; then
247         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_i}"
248         if [[ ${LFBFL_positions["${LFBFL_i}"]} != "-1" ]]; then
249           LFBFL_positions["${LFBFL_i}"]=$(
250             max "${LFBFL_sort_command}"\
251               "${LFBFL_positions["${LFBFL_i}"]}"\
252               "${split_score_result}"
253           )
254         fi
255       fi
256       eval "$3 '${LFBFL_current_char}' ${LFBFL_j} 1"
257       # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
258       if [[ split_score_result -ge 1 ]]; then
259         # printf "%s|%s\n" "${split_score_result}" "${LFBFL_j}"
260         if [[ ${LFBFL_positions["${LFBFL_j}"]} != "-1" ]]; then
261           LFBFL_positions["${LFBFL_j}"]=$(
262             max "${LFBFL_sort_command}"\
263               "${LFBFL_positions["${LFBFL_j}"]}"\
264               "${split_score_result}"
265           )
266         fi
267       fi
268     done
269   fi
270   declare -ir LFBFL_max_score=$(
271     max 'sort --numeric-sort' "${LFBFL_positions[@]}"
272   )
273   local LFBFL_value
274   declare -i LFBFL_best_position
275   for ((LFBFL_i=LFBFL_i_max; LFBFL_i>=0; --LFBFL_i)) do
276     if [[ LFBFL_i -eq 0 ]]; then
277       printf $'/!\\split_line_at_most bug: no split position found./!\\\n'
278     fi
279     local LFBFL_value=${LFBFL_positions[${LFBFL_i}]}
280     # printf "%s %s\n" "${LFBFL_i} ${LFBFL_value}"
281     if [[ "${LFBFL_value}" == "${LFBFL_max_score}" ]]; then
```

```
      LFBFL_best_position=$((LFBFL_i))
      break
    fi
  done
  # shellcheck disable=SC2248
  split_line_at "$1" ${LFBFL_best_position}
  split_line_at_most_result_start="${split_line_at_result_beginning}"
  split_line_at_most_result_end="${split_line_at_result_end}"
  # printf "%s\n" "$split_line_at_most_result_start"
  # printf "%s\n" "$split_line_at_most_result_end"
}

split_last_line(){
  # $1=$new_lines
  # $2=$prefix
  # $3=$max_length
  # $4=$suffix
  # $5=$split_score_command
  # $6=$split_score_command_properties
  # $7=$forbidden_previous_character ('\' usually)
  # In general, a prefix/suffix can be a required line prefix/suffix
  # for all final lines like a comment prefix,
  # or it can be a continuation line prefix/suffix applied only
  # when splitting a line in 2 lines,
  # or it can be a prefix/suffix for the string made of all the lines.
  # Clearly, there is no gain to couple handling a prefix/suffix for
  # the string made of all the lines in this function.
  # Since we add a prefix/suffix when splitting,
  # it is easy to see that we do not need to distinguish between
  # required line prefix/suffix and continuation line prefix/suffix.
  # The $2 and $4 arguments are given as concatenation of the
  # required and continuation line prefixes/suffixes as needed by the
  # user.
  # Thus lines arguments should be given already prefixed
  # and suffixed.
  # This is logic, since it would be cumbersome to handle prefixing
  # or not already splitted lines in this function
  # (see repeated_split_last_line).
  # The given max_length is the goal.
  # This function adapts the effective max position for the split by
  # taking into account the length of the suffix.
  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -g split_last_line_result="$1"
  declare -ir LFBFL_max_length_plus=$(($3 + 1))
```

```
  declare -ir LFBFL_length2=$(($3 - ${#4}))
  declare -r LFBFL_regexp='.\{'"${LFBFL_max_length_plus}"'\}$'

  # Testing if some line is too long.
  printf "%s" "$1"\
    | grep -q "${LFBFL_regexp}";
  if [[ ${PIPESTATUS[1]} -eq 1 ]]; then
    return
  fi

  declare -r LFBFL_start=$(
    printf "%s" "$1"\
    | head --lines=-1
  )
  # printf "start: %s\n" "${LFBFL_start}"
  declare -r LFBFL_last_line=$(
    printf "%s" "$1"\
    | tail --lines=1
  )
  # printf "last_line: %s\n" "${LFBFL_last_line}"
  split_last_line_result=""
  if [[ -n "${LFBFL_start}" ]]; then
    split_last_line_result="${LFBFL_start}"
    # Last line-return of LFBFL_start seems dropped by $() and not ="${}".
    # printf "%s %s\n" "${#split_last_line_result}"
    #   "${split_last_line_result}"
    split_last_line_result+=$'\n'
    # something="${split_last_line_result}"
    # printf "%s %s\n" "${#something}" "${something}"
  fi
  if [[ -n "$5" ]]; then
    split_line_at_most "${LFBFL_last_line}" "${LFBFL_length2}"\
      "$5" "$6" "$7"
    split_last_line_result+="${split_line_at_most_result_start}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${split_line_at_most_result_end}"
  else
    split_last_line_result+="${LFBFL_last_line:0:${LFBFL_length2}}"
    split_last_line_result+="$4"
    split_last_line_result+=$'\n'
    split_last_line_result+="$2${LFBFL_last_line:${LFBFL_length2}}"
  fi
  # printf "result: %s\n" "${split_last_line_result}"
}
```

```
repeated_split_last_line(){
  # See split_last_line for the first arguments.
  # $8=final_suffix to handle when the end of the original string
  #    is the suffix.
  declare -g repeated_split_last_line_result="$1"
  local LFBFL_result="$1x"
  until
    [[ "${LFBFL_result}" == "${repeated_split_last_line_result}" ]];
  do
    split_last_line "${repeated_split_last_line_result}" "$2" "$3"\
      "$4" "$5" "$6" "$7"
    LFBFL_result="${repeated_split_last_line_result}"
    repeated_split_last_line_result="${split_last_line_result}"
    # printf "%s\n" "${repeated_split_last_line_result}"
  done
  if [[ -n "$4" && -n "$8" ]]; then
    declare -ir LFBFL_offset=$((-${#4}))
    local LFBFL_lines_end
    LFBFL_lines_end="${repeated_split_last_line_result: ${LFBFL_offset}}"
    if [[ "${LFBFL_lines_end}" == "$4" ]]; then
      repeated_split_last_line_result+="$8"
    fi
  fi
}
```

2.29 build_and_checks_dependencies/too_long_code_lines.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

```
20 # along with DevOrSysAdminScripts.
21 # If not, see <https://www.gnu.org/licenses/>.
22 #
23 # ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
24 # This file was renamed from "too_long_code_lines.sh"
25 # to "too_long_code_lines.libr.sh".
26
27 LFBFL_subdir="build_and_checks_dependencies"
28 # shellcheck source=common_options.libr.sh
29 source "./${LFBFL_subdir}/common_options.libr.sh"
30 # shellcheck source=get_common_text_glob_patterns.libr.sh
31 source "./${LFBFL_subdir}/get_common_text_glob_patterns.libr.sh"
32 # shellcheck source=lines_filters.libr.sh
33 source "./${LFBFL_subdir}/lines_filters.libr.sh"
34
35 too_long_code_lines(){
36   # Options:
37   #   --verbose
38   #   --work-directory=""
39   #   --max-line-length
40   declare -i LFBFL_i_verbose=0
41   get_verbose_option "$@"
42   local LFBFL_work_directory=""
43   get_work_directory_option "$@"
44   pushd_to_work_directory\
45     && trap 'popd_from_work_directory' RETURN
46   can_continue_after_enhanced_pushd || return 1
47
48   local LFBFL_arg
49   declare -i LFBFL_max_line_length=70
50   for LFBFL_arg in "$@"; do
51     if [[ "${LFBFL_arg}" == --max-line-length=* ]]; then
52       LFBFL_arg=${LFBFL_arg#--max-line-length=}
53       LFBFL_max_line_length=$((LFBFL_arg))
54       break
55     fi
56   done
57
58   enhanced_set_shell_option pipefail\
59     && trap 'enhanced_unset_shell_option pipefail' RETURN
60
61   get_COMMON_TEXT_FILES_GLOB_PATTERNS
62   local LFBFL_pattern
63   local LFBFL_long_line
64   local LFBFL_file_name
65   local LFBFL_line
```

```
  local LFBFL_extension
  local LFBFL_base_name
  declare -ir LFBFL_overlength=$((LFBFL_max_line_length+1))
  local LFBFL_s_long_lines
  declare -a LFBFL_arr_long_lines
  for LFBFL_pattern in "${COMMON_TEXT_FILES_GLOB_PATTERNS[@]}"; do
    [[ LFBFL_i_verbose -eq 1 ]]\
      && printf "Iterating on pattern: %s.\n" "${LFBFL_pattern}"
    # shellcheck disable=SC2248
    LFBFL_s_long_lines=$(
      find . -type f -name "${LFBFL_pattern}" -printf '%P\n'\
      | relevant_find\
      | not_license_find\
      | xargs grep -H '.\{'${LFBFL_overlength}'\}'
    )
    if [[ -z "${LFBFL_s_long_lines}" ]]; then
      [[ LFBFL_i_verbose -eq 1 ]]\
        && printf "No file found with long lines.\n"
      continue
    fi
    mapfile -t LFBFL_arr_long_lines <<< "${LFBFL_s_long_lines}"
    for LFBFL_long_line in "${LFBFL_arr_long_lines[@]}"; do
      LFBFL_file_name=${LFBFL_long_line%%:*} # Drop long ':*' suffix
      LFBFL_line=${LFBFL_long_line#*:} # Drop short '*:' prefix
      # LFBFL_line="${LFBFL_line/%\\/}" # Drop/Replace '\' suffix by ''
      LFBFL_extension=${LFBFL_file_name##*.} # Drop long '*.' prefix
      LFBFL_base_name=${LFBFL_file_name%.*} # Drop short '.*' suffix
      if [[ "${LFBFL_extension}" == "html" ]]; then
        if [[ -f "${LFBFL_base_name}.md" ]]; then
          continue
        fi
        if grep --quiet --fixed-strings -- "${LFBFL_line}"\
          "./build_and_checks_variables/temp/files_listing.html.tpl"
        then
          continue
        fi
        printf "%s\n" "${LFBFL_long_line}"
      elif [[ "${LFBFL_extension}" == "md" ]]; then
        if ! [[ -f "${LFBFL_base_name}.md.tpl" ]]; then
          printf "%s\n" "${LFBFL_long_line}"
        fi
      else
        printf "%s\n" "${LFBFL_long_line}"
      fi
    done
  done
```

```
}
```

2.30 build_and_checks_dependencies/update_common_build_and_checks.exec.sh

```
#!/usr/bin/env bash
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# This file was renamed from "update_common_build_and_checks.sh"
# to "update_common_build_and_checks.exec.sh".

source "overwrite_if_not_equal.libr.sh"

update_common_build_and_checks(){
  declare -i LFBFL_i_verbose=0
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_i_verbose=1
  fi
  readonly LFBFL_i_verbose

  if [[ ! -o pipefail ]]; then
    [[ LFBFL_i_verbose -eq 1 ]] && printf "pipefail option activated\n"
    set -o pipefail
    trap 'set +o pipefail' RETURN
  fi

```

```
43    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
44    cp "./${LFBFL_common_file_name}.tpl"\
45      "./${LFBFL_common_file_name}.temp"
46
47    declare -ar LFBFL_file_names=(
48      "build_md_from_printable_md.exec.sh"
49      "check_shell_scripts_beginnings.libr.sh"
50      "check_shell_scripts_indentation.libr.sh"
51      "check_URLs.libr.sh"
52      "common_options.libr.sh"
53      "comparisons.libr.sh"
54      "create_PDF.exec.sh"
55      "generate_from_template.libr.sh"
56      "get_common_text_glob_patterns.libr.sh"
57      "grammar_and_spell_check.libr.sh"
58      "licenses_templates/build_licenses_templates.exec.sh"
59      "licenses_templates/license_file_header_AGPLv3+.tpl"
60      "licenses_templates/license_file_header_GPLv3+.tpl"
61      "licenses_templates/license_file_header_LGPLv3+.tpl"
62      "lines_counts.libr.sh"
63      "lines_filters.libr.sh"
64      "lines_maps.libr.sh"
65      "update_or_check_files_names_listing.exec.sh"
66      "overwrite_if_not_equal.libr.sh"
67      "python_black_complement.libr.sh"
68      "python_isort_complement.libr.sh"
69      "repository_data.libr.sh"
70      "shell_checks_complement.libr.sh"
71      "split_score.exec.php"
72      "split_score.libr.php"
73      "strings_functions.libr.sh"
74      "too_long_code_lines.libr.sh"
75    )
76
77    local LFBFL_file_name
78    local LFBFL_file_sha512
79    local LFBFL_file_sha512_all
80    local LFBFL_base_file_name
81    for LFBFL_file_name in "${LFBFL_file_names[@]}"; do
82      LFBFL_file_sha512=$(
83        sha512sum "./${LFBFL_file_name}"\
84        | cut -f1 -d' '
85      )
86      LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
87      LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
88      LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
```

```
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
    LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
    LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
    LFBFL_base_file_name=$(basename "${LFBFL_file_name}")
    sed -i\
      "s|@sha512_${LFBFL_base_file_name}@|${LFBFL_file_sha512_all}|g"\
      "./${LFBFL_common_file_name}.temp"
  done

  overwrite_if_not_equal "./${LFBFL_common_file_name}"\
    "./${LFBFL_common_file_name}.temp"

  declare -r LFBFL_main_file_name="build_and_checks.exec.sh"
  cp "./${LFBFL_main_file_name}.tpl" "../${LFBFL_main_file_name}.temp"

  LFBFL_file_sha512=$(
    sha512sum "./${LFBFL_common_file_name}"\
    | cut -f1 -d' '
  )
  LFBFL_file_sha512_all="local LFBFL_correct_sha512\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:0:45}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:45:44}'\n"
  LFBFL_file_sha512_all+="  LFBFL_correct_sha512\+="
  LFBFL_file_sha512_all+="'${LFBFL_file_sha512:89}'"
  sed -i\
    "s|@sha512_${LFBFL_common_file_name}@|${LFBFL_file_sha512_all}|g"\
    "../${LFBFL_main_file_name}.temp"

  overwrite_if_not_equal "../${LFBFL_main_file_name}"\
    "../${LFBFL_main_file_name}.temp"
}

update_common_build_and_checks "$@"
```

2.31 build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh

```
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

# 2025-12-30T21:33+01:00: This file was moved from
# "./build_and_checks_dependencies/listings"
# to
# "./build_and_checks_dependencies".
# This file was renamed from "update_or_check_files_names_listing.sh"
# to "update_or_check_files_names_listing.exec.sh".
# This file was renamed from "update_or_check_file_names_listing.sh"
# to "update_or_check_files_names_listing.sh"?

LFBFL_subdir="build_and_checks_dependencies"
# shellcheck source=common_options.libr.sh
source "./${LFBFL_subdir}/common_options.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "./${LFBFL_subdir}/lines_filters.libr.sh"
# shellcheck source=strings_functions.libr.sh
source "./${LFBFL_subdir}/strings_functions.libr.sh"

update_or_check_files_names_listing(){
  # Options:
  #   --verbose
  #   --work-directory=""
  declare -i LFBFL_i_verbose=0
  get_verbose_option "$@"
  local LFBFL_work_directory=""
  get_work_directory_option "$@"
  pushd_to_work_directory\
    && trap 'popd_from_work_directory' RETURN
  can_continue_after_enhanced_pushd || return 1

```

```
54    declare -i LFBFL_write=0
55    if [[ "$*" == *--write* ]]; then
56      LFBFL_write=1
57    fi
58    readonly LFBFL_write
59
60    declare -i LFBFL_append=0
61    if [[ "$*" == *--append* ]]; then
62      LFBFL_append=1
63    fi
64    readonly LFBFL_append
65
66    enhanced_set_shell_option pipefail\
67      && trap 'enhanced_unset_shell_option pipefail' RETURN
68
69    LFBFL_subdir2="build_and_checks_variables"
70    local LFBFL_listing="./${LFBFL_subdir2}/files_names_listing.txt"
71    readonly LFBFL_listing
72    local LFBFL_data_file_name="./${LFBFL_subdir2}/repository_data.txt"
73    readonly LFBFL_data_file_name
74
75    declare -i LFBFL_max_line_length
76    grep_variable "${LFBFL_data_file_name}" max_line_length\
77      --result-variable-prefix="LFBFL_"
78
79    if [[ LFBFL_write -eq 1 ]]; then
80      : > "${LFBFL_listing}"
81    fi
82    # Remove line returns that are here to keep lines short.
83    sed -Ez 's/(.)\\\n/\1/Mg' "${LFBFL_listing}"\
84      > "${LFBFL_listing}.temp"
85    # shopt -s dotglob was needed at some point but I don't see why now.
86    # shellcheck disable=SC2248
87    get_split_score_simple 1 ${LFBFL_max_line_length} /
88    declare -r LFBFL_split_score_command="${get_split_score_result}"
89    local LFBFL_split_score_command_properties
90    LFBFL_split_score_command_properties="${get_split_score_result2}"
91    readonly LFBFL_split_score_command_properties
92    local LFBFL_suffix=\\ # instead of '\' to avoid shellcheck SC1003
93    readonly LFBFL_suffix
94    # There is always a $'\n' final suffix from >>.
95    # But when origin string ends with the suffix we must add something.
96
97    local LFBFL_s_file_paths
98    LFBFL_s_file_paths=$(
99      find . -type f -printf '%P\n'\
```

```
    | relevant_find\
    | sort
  )
  declare -a LFBFL_arr_file_paths
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
  fi

  declare -r LFBFL_final_suffix=$'\n\\'
  local LFBFL_file_path
  local LFBFL_base_file_name
  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    git check-ignore -q "${LFBFL_file_path}" && continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    [[ "${LFBFL_base_file_name}" != files_names_listing.txt* ]]\
      || continue
    if [[ "${LFBFL_base_file_name}" == *.md ]]; then
      if [[ -f "${LFBFL_file_path}.tpl" ]]; then
        continue
      fi
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "Non-ignored file: %s\n" "${LFBFL_file_path}"
    fi
    # shellcheck disable=SC2248
    repeated_split_last_line "${LFBFL_file_path}"\
      ""\
      ${LFBFL_max_line_length}\
      "${LFBFL_suffix}"\
      "${LFBFL_split_score_command}"\
      "${LFBFL_split_score_command_properties}"\
      ""\
      "${LFBFL_final_suffix}"
    if [[ LFBFL_write -eq 1 ]]; then
      # shellcheck disable=SC2001
      printf "%s\n" "${repeated_split_last_line_result}"\
        >> "${LFBFL_listing}"
      continue
    fi
    if grep_fixed_string_with_anchor "${LFBFL_listing}.temp"\
        "${LFBFL_file_path}" --quiet --enforce-line-ends-with-fixed-string;
    then
      :
    else
      printf "The file %s is not listed in %s.\n"\
        "${LFBFL_file_path}"\
```

```
        "${LFBFL_listing}"
      if [[ LFBFL_append -eq 1 ]]; then
        # shellcheck disable=SC2001
        printf "%s\n" "${repeated_split_last_line_result}"\
          >> "${LFBFL_listing}"
      fi
    fi
  done

  LFBFL_s_file_paths=$(cat "${LFBFL_listing}.temp")
  if [[ -n "${LFBFL_s_files_paths}" ]]; then
    mapfile -t LFBFL_arr_file_paths <<< "${LFBFL_s_file_paths}"
  fi

  for LFBFL_file_path in "${LFBFL_arr_file_paths[@]}"; do
    [[ "${LFBFL_file_path}" != '//'* ]] || continue
    LFBFL_base_file_name=$(basename "${LFBFL_file_path}")
    if ! [[ -f "${LFBFL_file_path}" ]]; then
      printf "The non-file %s is listed in %s.\n"\
        "${LFBFL_file_path}"\
        "${LFBFL_listing}"
    fi
  done

  rm "${LFBFL_listing}.temp"

  if [[ LFBFL_i_verbose -eq 1 ]]; then
    cat "${LFBFL_listing}"
  fi
}

update_or_check_files_names_listing "$@"
```

2.32 build_and_checks_variables/.gitignore

```
temp/
upgrade_venvs_ts

```

2.33 build_and_checks_variables/bandit.ini

```
# FILE: .bandit
[bandit]
```

2.34 build_and_checks_variables/bandit_baseline.json

```
{"results":[]}
```

2.35 build_and_checks_variables/black.toml

```
[tool.black]
line-length = 75
```

2.36 build_and_checks_variables/DevOrSysAdminScripts.html.tpl

```
<!DOCTYPE html>
<html lang="en">
<!--
This file is part of DevOrSysAdminScripts library.

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you can redistribute it and/or modify it under the terms
of the GNU Lesser General Public License
as published by the Free Software Foundation,
either version 3 of the License,
or (at your option) any later version.

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that it will be useful,
but WITHOUT ANY WARRANTY;
without even the implied warranty of
MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
See the GNU Lesser General Public License for more details.

You should have received a copy of
the GNU Lesser General Public License
along with DevOrSysAdminScripts.
If not, see <https://www.gnu.org/licenses/>.

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-->
<head>
<meta http-equiv="Content-Type" content="text/html; charset=UTF-8">
<meta name="generator" content="DevOrSysAdminScripts">
<style>
.title{
  margin: 2ex auto;
  text-align: center;
}
.titlemain{
```

```
  margin: 1ex 2ex 2ex 1ex;
}
.titlerest{
  font-size: 1.17em;
  margin: 0ex 2ex;
}
table{
  border-collapse: collapse;
}
td{
  padding: 0;
}
pre{
  text-align: left;
  margin-left: 0ex;
  margin-right: auto;
}
pre.numbered_lines:before{
  background: snow;
  border-right: 1px black solid;
  color: #010101;
  content: attr(line-numbers);
  float: left;
  padding: 0 1ch;
  text-align: right;
}
blockquote{
  margin-left: 4ex;
  margin-right: 4ex;
  text-align: left;
}
ol.level1, ol.level2{
  list-style-type: none;
  /*
  list-style-type: decimal;
  Previous rule is shorter and more efficient if you need only level1.
  But you don't have accessible counter if you use decimal.
  */
}
ol.level1{
  counter-reset: level1;
}
ol.level2{
  counter-reset: level2;
}
ol.level1 li:before{
```

```
  content: counter(level1) ". ";
  counter-increment: level1;
}
ol.level2 li:before{
  content: counter(level1) "." counter(level2) " ";
  counter-increment: level2;
}
</style>
<title>@repository_name@</title>
</head>
<body >
<table class="title">
  <tr>
    <td style="padding:1ex">
      <h1 class="titlemain">@repository_name@</h1>
      <h2 class="titlerest">
        @author_full_name@
        <br>
        <a href="@author_website@">
          <span style="font-family:monospace">
          @author_website@
          </span>
        </a>
        <br>
        <a href="mailto:@author_email@">
          <span style="font-family:monospace">
          @author_email@
          </span>
        </a>
      </h2>
    </td>
  </tr>
</table>

<blockquote class="abstract">
  <span style="font-weight:bold">Abstract:</span>
  @abstract@
</blockquote>
<p>Current version: @current_date@</p>
<p>Current number of commits: @number_of_commits@</p>
<p>Current git SHA1: @current_git_SHA1@</p>
<p>
Code lines: @number_of_lines@
</p>


```

```
<h2>Contents</h2>

<ol class="level1">
  <li><a href="#section1">Files Tree</a></li>
  <li><a href="#section2">Listing of files</a><br>
    <ol class="level2">
      @files_lis@
    </ol>
  </li>
</ol>


<h2 id="section1">1 Files tree</h2>
<pre class="verbatim">
@current_tree_light@
</pre>
<pre class="verbatim">
@current_tree@
</pre>


<h2 id="section2">2 Listing of files</h2>

<p>
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.
</p>

<p>
The text of the licenses is available at:
<a href="https://www.gnu.org/licenses/"><span
  style="font-family:monospace"
>https://www.gnu.org/licenses/</span></a>.
The git repository of this source code is also available at:
<a href="@repository_web_url@"><span
 style="font-family:monospace"
>@repository_web_url@</span></a>.
Clone it with this URL:
<span style="font-family:monospace"
>@repository_git_url@</span>.
</p>

<div>
<p>
```

```
Here is a listing of the dependencies of the code in this repository:
</p>
<ul>
<li> The following dependencies are GPLv3+ licensed:<br>
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
<li> The following dependencies are GPLv2+ licensed:<br>
more, pandoc, pdftex, pdflatex, tree;
<li> The following dependencies are GPLv2 licensed:<br>
git, pylint;
<li> The following dependencies are LGPLv2.1+ licensed:<br>
notify-send;
<li> The following dependencies are MIT licensed:<br>
black, isort, mypy, pip, ruff;
<li> The following dependencies are Apache-2.0 licensed:<br>
bandit;
<li> The following dependencies are BSD-3-Clause licensed:<br>
phpmd;
<li> php is licensed under PHP License v3.01;
<li> tex is licensed under Knuth license;
<li> pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception.
<li> python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
</ul>
</div>

<p>
Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.
</p>

<p>
Bash can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/bash/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/bash/</span></a>,<br>
<a href="https://tiswww.case.edu/php/chet/bash/bashtop.html"><span
  style="font-family:monospace"
>https://tiswww.case.edu/php/chet/bash/bashtop.html</span></a>,<br>
<a href="https://savannah.gnu.org/projects/bash/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/bash/</span></a>.
```

```
</p>

<p>
GNU coreutils can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/coreutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/coreutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/coreutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/coreutils/</span></a>.
</p>

<p>
uutils coreutils can be found at the following URLs:<br>
<a href="https://uutils.github.io/coreutils/"><span
  style="font-family:monospace"
>https://uutils.github.io/coreutils/</span></a>,<br>
<a href="https://github.com/uutils/coreutils"><span
  style="font-family:monospace"
>https://github.com/uutils/coreutils</span></a>.
</p>

<p>
diff (GNU diffutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/diffutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/diffutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/diffutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/diffutils/</span></a>.
</p>

<p>
find, xargs (GNU findutils) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/findutils/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/findutils/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/findutils/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/findutils/</span></a>.
</p>

<p>
grep (GNU grep) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/grep/"><span
  style="font-family:monospace"
```

```
>https://www.gnu.org/software/grep/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/grep/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/grep/</span></a>.
</p>

<p>
sed (GNU sed) can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/sed/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/sed/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/sed/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/sed/</span></a>.
</p>

<p>
ShellCheck can be found at the following URLs:<br>
<a href="https://www.shellcheck.net/"><span
  style="font-family:monospace"
>https://www.shellcheck.net/</span></a>,<br>
<a href="https://github.com/koalaman/shellcheck"><span
  style="font-family:monospace"
>https://github.com/koalaman/shellcheck</span></a>.
</p>

<p>
GNU Wget can be found at the following URLs:<br>
<a href="https://www.gnu.org/software/wget/"><span
  style="font-family:monospace"
>https://www.gnu.org/software/wget/</span></a>,<br>
<a href="https://savannah.gnu.org/projects/wget/"><span
  style="font-family:monospace"
>https://savannah.gnu.org/projects/wget/</span></a>.
</p>

<p>
more of util-linux can be found at the following URLs:<br>
<a href="https://www.kernel.org/pub/linux/utils/util-linux/"><span
  style="font-family:monospace"
>https://www.kernel.org/pub/linux/utils/util-linux/</span></a>,<br>
<a
href="https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/"
><span
  style="font-family:monospace"
>https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/</span
```

```
></a>,<br>
<a href="https://github.com/util-linux/util-linux"><span
  style="font-family:monospace"
>https://github.com/util-linux/util-linux</span></a> official?
</p>

<p>
pandoc can be found at the following URLs:<br>
<a href="https://pandoc.org/"><span
  style="font-family:monospace"
>https://pandoc.org/</span></a>,<br>
<a href="https://github.com/jgm/pandoc"><span
  style="font-family:monospace"
>https://github.com/jgm/pandoc</span></a>.
</p>

<p>
pdftex/pdflatex can be found at the following URLs:<br>
<a href="https://www.tug.org/applications/pdftex/"><span
  style="font-family:monospace"
>https://www.tug.org/applications/pdftex/</span></a>,<br>
<a href="https://tug.org/texlive/svn/"><span
  style="font-family:monospace"
>https://tug.org/texlive/svn/</span></a>.
</p>

<p>
tree can be found at the following URLs:<br>
<a href="https://oldmanprogrammer.net/source.php?dir=projects/tree"
><span style="font-family:monospace"
>https://oldmanprogrammer.net/source.php?dir=projects/tree</span></a>,<br>
<a href="https://gitlab.com/OldManProgrammer/unix-tree"><span
  style="font-family:monospace"
>https://gitlab.com/OldManProgrammer/unix-tree</span></a>,<br>
<a href="https://github.com/Old-Man-Programmer/tree"><span
  style="font-family:monospace"
>https://github.com/Old-Man-Programmer/tree</span></a>.
</p>

<p>
git can be found at the following URLs:<br>
<a href="https://git-scm.com/about"><span
  style="font-family:monospace"
>https://git-scm.com/about</span></a>,<br>
<a href="https://github.com/git/git"><span
  style="font-family:monospace"
```

```
>https://github.com/git/git</span></a>.
</p>

<p>
pylint can be found at the following URLs:<br>
<a href="https://www.pylint.org/"><span
  style="font-family:monospace"
>https://www.pylint.org/</span></a>,<br>
<a href="https://pylint.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://pylint.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/pylint-dev/pylint"><span
  style="font-family:monospace"
>https://github.com/pylint-dev/pylint</span></a>,<br>
<a href="https://pypi.org/project/pylint/"><span
  style="font-family:monospace"
>https://pypi.org/project/pylint/</span></a>.
</p>

<p>
notify-send can be found at the following URLs:<br>
<a href="https://gitlab.gnome.org/GNOME/libnotify/"><span
  style="font-family:monospace"
>https://gitlab.gnome.org/GNOME/libnotify/</span></a>,<br>
<a href="https://gnome.pages.gitlab.gnome.org/libnotify/"><span
  style="font-family:monospace"
>https://gnome.pages.gitlab.gnome.org/libnotify/</span></a>.
</p>

<p>
black can be found at the following URLs:<br>
<a href="https://github.com/psf/black"><span
  style="font-family:monospace"
>https://github.com/psf/black</span></a>,<br>
<a href="https://black.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://black.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://pypi.org/project/black/"><span
  style="font-family:monospace"
>https://pypi.org/project/black/</span></a>.
</p>

<p>
isort can be found at the following URLs:<br>
<a href="https://pycqa.github.io/isort/"><span
  style="font-family:monospace"
```

```
>https://pycqa.github.io/isort/</span></a>,<br>
<a href="https://github.com/pycqa/isort/"><span
  style="font-family:monospace"
>https://github.com/pycqa/isort/</span></a>,<br>
<a href="https://pypi.org/project/isort/"><span
  style="font-family:monospace"
>https://pypi.org/project/isort/</span></a>.
</p>

<p>
mypy can be found at the following URLs:<br>
<a href="https://mypy-lang.org/"><span
  style="font-family:monospace"
>https://mypy-lang.org/</span></a>,<br>
<a href="https://mypy.readthedocs.io/en/stable/"><span
  style="font-family:monospace"
>https://mypy.readthedocs.io/en/stable/</span></a>,<br>
<a href="https://github.com/python/mypy"><span
  style="font-family:monospace"
>https://github.com/python/mypy</span></a>,<br>
<a href="https://pypi.org/project/mypy/"><span
  style="font-family:monospace"
>https://pypi.org/project/mypy/</span></a>.
</p>

<p>
pip can be found at the following URLs:<br>
<a href="https://pip.pypa.io/en/stable/"><span
  style="font-family:monospace"
>https://pip.pypa.io/en/stable/</span></a>,<br>
<a href="https://github.com/pypa/pip"><span
  style="font-family:monospace"
>https://github.com/pypa/pip</span></a>,<br>
<a href="https://pypi.org/project/pip/"><span
  style="font-family:monospace"
>https://pypi.org/project/pip/</span></a> (egg and chicken ;) ).
</p>

<p>
ruff can be found at the following URLs:<br>
<a href="https://docs.astral.sh/ruff/"><span
  style="font-family:monospace"
>https://docs.astral.sh/ruff/</span></a>,<br>
<a href="https://github.com/astral-sh/ruff"><span
  style="font-family:monospace"
>https://github.com/astral-sh/ruff</span></a>,<br>
```

```
<a href="https://pypi.org/project/ruff/"><span
  style="font-family:monospace"
>https://pypi.org/project/ruff/</span></a>.
</p>

<p>
bandit can be found at the following URLs:<br>
<a href="https://bandit.readthedocs.io/en/latest/"><span
  style="font-family:monospace"
>https://bandit.readthedocs.io/en/latest/</span></a>,<br>
<a href="https://github.com/PyCQA/bandit"><span
  style="font-family:monospace"
>https://github.com/PyCQA/bandit</span></a>,<br>
<a href="https://pypi.org/project/bandit/"><span
  style="font-family:monospace"
>https://pypi.org/project/bandit/</span></a>.
</p>

<p>
phpmd can be found at the following URLs:<br>
<a href="https://phpmd.org/"><span
  style="font-family:monospace"
>https://phpmd.org/</span></a>,<br>
<a href="https://github.com/phpmd/phpmd"><span
  style="font-family:monospace"
>https://github.com/phpmd/phpmd</span></a>,<br>
<a href="https://packagist.org/packages/phpmd/phpmd"><span
  style="font-family:monospace"
>https://packagist.org/packages/phpmd/phpmd</span></a>.
</p>

<p>
php can be found at the following URLs:<br>
<a href="https://www.php.net/"><span
  style="font-family:monospace"
>https://www.php.net/</span></a>,<br>
<a href="https://github.com/php/php-src"><span
  style="font-family:monospace"
>https://github.com/php/php-src</span></a>.
</p>

<p>
tex can be found at the following URL:<br>
<a href="https://ctan.org/tex-archive/systems/knuth/dist/tex"><span
  style="font-family:monospace"
>https://ctan.org/tex-archive/systems/knuth/dist/tex</span></a>.
```

```
</p>

<p>
pcregrep, pcre2grep can be found at the following URLs:<br>
<a href="https://www.pcre.org/"><span
  style="font-family:monospace"
>https://www.pcre.org/</span></a>,<br>
<a href="https://pcre2project.github.io/pcre2/"><span
  style="font-family:monospace"
>https://pcre2project.github.io/pcre2/</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre1"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre1</span></a>,<br>
<a href="https://github.com/PCRE2Project/pcre2"><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2</span></a>,<br>
<a
href="https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c"
><span
  style="font-family:monospace"
>https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c</span
></a>.
</p>

<p>
python3 can be found at the following URLs:<br>
<a href="https://www.python.org/"><span
  style="font-family:monospace"
>https://www.python.org/</span></a>,<br>
<a href="https://docs.python.org/3/"><span
  style="font-family:monospace"
>https://docs.python.org/3/</span></a>.
</p>

<p>
If you see something that I missed regarding my dependencies,
please tell me/email me :).
</p>

@files_listing_HTMLPreInput@

<p>
@acknowledgments@
</p>

<script>
```

```
"use strict";
const pres = Array.from(
  document.getElementsByClassName('numbered_lines')
);
pres.forEach(
  (pre) => {
    const nb = Array.from(pre.innerText.matchAll("\n")).length + 1;
    let numbers = "";
    for(let i = 1; i <= nb; ++i){
      numbers += i.toString();
      if(i < nb){ numbers += "\n "; }
    }
    pre.setAttribute('line-numbers', numbers);
  }
);
</script>
</html>
```

2.37 build_and_checks_variables/DevOrSysAdminScripts.tex.tpl

```
% This file is part of DevOrSysAdminScripts library.
%
% DevOrSysAdminScripts is free software:
% you can redistribute it and/or modify it under the terms
% of the GNU Lesser General Public License
% as published by the Free Software Foundation,
% either version 3 of the License,
% or (at your option) any later version.
%
% DevOrSysAdminScripts is distributed in the hope
% that it will be useful,
% but WITHOUT ANY WARRANTY;
% without even the implied warranty of
% MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
% See the GNU Lesser General Public License for more details.
%
% You should have received a copy of
% the GNU Lesser General Public License
% along with DevOrSysAdminScripts.
% If not, see <https://www.gnu.org/licenses/>.
%
% ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet

% 2025-12-29T17:08+01:00: This file was moved from
% "./latex"
% to
```

```
% "./build_and_checks_variables".

\documentclass{article}

\usepackage[utf8]{inputenc}
\usepackage[T1]{fontenc}
\usepackage{subfigure}
\usepackage{amsmath}
\usepackage{amssymb}
\usepackage{amsthm}
\usepackage[pdftex]{hyperref}
\usepackage{tikz}
\usepackage{caption}
\usepackage[round]{natbib}
\usepackage{fancyhdr}
\usepackage{amsfonts}
\usepackage{times}
\usepackage{ifpdf}
\usepackage{latexsym}
\usepackage{graphicx}
\usepackage{enumerate}
\usepackage{pmboxdraw}
\usepackage{fancyvrb}
\usepackage[french,english]{babel}

% *** les environnements ***
%\theoremstyle{break}
\newtheorem{definition}{Definition}[section]
\newtheorem{proposition}[definition]{Proposition}
\newtheorem{theorem}[definition]{Theorem}
\newtheorem{lemma}[definition]{Lemma}
\newtheorem{corollary}[definition]{Corollary}
\newtheorem{remark}[definition]{Remark}
\newtheorem{openproblem}[definition]{Open problem}

\begin{document}

\author{
  @author_full_name@\\
  \url{@author_website@}\\
  \href{mailto:@author_email@}{@author_email@}
}
\title{@repository_name@}

\maketitle
\begin{abstract}
```

```
@abstract@
\end{abstract}

Current version: @current_date@

Current number of commits: @number_of_commits@

Current git SHA1: @current_git_SHA1@

Code lines: @number_of_lines@

\tableofcontents
\newpage

\section{Files tree}
\label{section:tree}

\begin{verbatim}
@current_tree_light@
\end{verbatim}

\begin{verbatim}
@current_tree@
\end{verbatim}

\section{Listing of files}
\label{section:listing}

This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

The text of the licenses is available at:
\url{https://www.gnu.org/licenses/}.
The git repository of this source code is also available at:
\url{@repository_web_url@}.
Clone it with this URL:
\url{@repository_git_url@}.

Here is a listing of the dependencies of the code in this repository:
\begin{itemize}
\item The following dependencies are GPLv3+ licensed:\newline
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
```

```
\item The following dependencies are GPLv2+ licensed:\newline
more, pandoc, pdftex, pdflatex, tree;
\item The following dependencies are GPLv2 licensed:\newline
git, pylint;
\item The following dependencies are LGPLv2.1+ licensed:\newline
notify-send;
\item The following dependencies are MIT licensed:\newline
black, isort, mypy, pip, ruff;
\item The following dependencies are Apache-2.0 licensed:\newline
bandit;
\item The following dependencies are BSD-3-Clause licensed:\newline
phpmd;
\item php is licensed under PHP License v3.01;
\item tex is licensed under Knuth license;
\item pcregrep, pcre2grep has a custom license from Cambridge
University, but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
\item python3 is licensed under PSF License Version 2 and the Zero-Clause
BSD license.
\end{itemize}

Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:\newline
\url{https://www.gnu.org/software/bash/},\newline
\url{https://tiswww.case.edu/php/chet/bash/bashtop.html},\newline
\url{https://savannah.gnu.org/projects/bash/}.

GNU coreutils can be found at the following URLs:\newline
\url{https://www.gnu.org/software/coreutils/},\newline
\url{https://savannah.gnu.org/projects/coreutils/}.

uutils coreutils can be found at the following URLs:\newline
\url{https://uutils.github.io/coreutils/},\newline
\url{https://github.com/uutils/coreutils}.

diff (GNU diffutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/diffutils/},\newline
\url{https://savannah.gnu.org/projects/diffutils/}.

find, xargs (GNU findutils) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/findutils/},\newline
\url{https://savannah.gnu.org/projects/findutils/}.
```

```
grep (GNU grep) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/grep/},\newline
\url{https://savannah.gnu.org/projects/grep/}.

sed (GNU sed) can be found at the following URLs:\newline
\url{https://www.gnu.org/software/sed/},\newline
\url{https://savannah.gnu.org/projects/sed/}.

ShellCheck can be found at the following URLs:\newline
\url{https://www.shellcheck.net/},\newline
\url{https://github.com/koalaman/shellcheck}.

GNU Wget can be found at the following URLs:\newline
\url{https://www.gnu.org/software/wget/},\newline
\url{https://savannah.gnu.org/projects/wget/}.

more of util-linux can be found at the following URLs:\newline
\url{https://www.kernel.org/pub/linux/utils/util-linux/},\newline
{\small
  \url{https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/}
},
\newline
\url{https://github.com/util-linux/util-linux} official?

pandoc can be found at the following URLs:\newline
\url{https://pandoc.org/},\newline
\url{https://github.com/jgm/pandoc}.

pdftex/pdflatex can be found at the following URLs:\newline
\url{https://www.tug.org/applications/pdftex/},\newline
\url{https://tug.org/texlive/svn/}.

tree can be found at the following URLs:\newline
\url{https://oldmanprogrammer.net/source.php?dir=projects/tree},\newline
\url{https://gitlab.com/OldManProgrammer/unix-tree},\newline
\url{https://github.com/Old-Man-Programmer/tree}.

git can be found at the following URLs:\newline
\url{https://git-scm.com/about},\newline
\url{https://github.com/git/git}.

pylint can be found at the following URLs:\newline
\url{https://www.pylint.org/},\newline
\url{https://pylint.readthedocs.io/en/stable/},\newline
\url{https://github.com/pylint-dev/pylint},\newline
\url{https://pypi.org/project/pylint/}.
```

```
notify-send can be found at the following URLs:\newline
\url{https://gitlab.gnome.org/GNOME/libnotify/},\newline
\url{https://gnome.pages.gitlab.gnome.org/libnotify/}.

black can be found at the following URLs:\newline
\url{https://github.com/psf/black},\newline
\url{https://black.readthedocs.io/en/stable/},\newline
\url{https://pypi.org/project/black/}.

isort can be found at the following URLs:\newline
\url{https://pycqa.github.io/isort/},\newline
\url{https://github.com/pycqa/isort/},\newline
\url{https://pypi.org/project/isort/}.

mypy can be found at the following URLs:\newline
\url{https://mypy-lang.org/},\newline
\url{https://mypy.readthedocs.io/en/stable/},\newline
\url{https://github.com/python/mypy},\newline
\url{https://pypi.org/project/mypy/}.

pip can be found at the following URLs:\newline
\url{https://pip.pypa.io/en/stable/},\newline
\url{https://github.com/pypa/pip},\newline
\url{https://pypi.org/project/pip/} (egg and chicken ;) ).

ruff can be found at the following URLs:\newline
\url{https://docs.astral.sh/ruff/},\newline
\url{https://github.com/astral-sh/ruff},\newline
\url{https://pypi.org/project/ruff/}.

bandit can be found at the following URLs:\newline
\url{https://bandit.readthedocs.io/en/latest/},\newline
\url{https://github.com/PyCQA/bandit},\newline
\url{https://pypi.org/project/bandit/}.

phpmd can be found at the following URLs:\newline
\url{https://phpmd.org/},\newline
\url{https://github.com/phpmd/phpmd},\newline
\url{https://packagist.org/packages/phpmd/phpmd}.

php can be found at the following URLs:\newline
\url{https://www.php.net/},\newline
\url{https://github.com/php/php-src}.

tex can be found at the following URL:\newline
```

```
\url{https://ctan.org/tex-archive/systems/knuth/dist/tex}.

pcregrep, pcre2grep can be found at the following URLs:\newline
\url{https://www.pcre.org/},\newline
\url{https://pcre2project.github.io/pcre2/},\newline
\url{https://github.com/PCRE2Project/pcre1},\newline
\url{https://github.com/PCRE2Project/pcre2},\newline
{\small
  \url{https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c}
}.

python3 can be found at the following URLs:\newline
\url{https://www.python.org/},\newline
\url{https://docs.python.org/3/}.

If you see something that I missed regarding my dependencies,
please tell me/email me :).

@files_listing_VerbatimInput@

@acknowledgments@

\end{document}
```

2.38 build_and_checks_variables/files_listing.tex.tpl

```
\subsection{
  build\_and\_checks\_dependencies/build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/build\_md\_from\_printable\_md.exec.sh
}
\label{
  build_and_checks_dependencies:build_md_from_printable_mdexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
```

```
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_beginnings.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_beginningslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
check\_shell\_scripts\_indentation.libr.sh
}
\label{
  build_and_checks_dependencies:check_shell_scripts_indentationlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/check\_URLs.libr.sh
}
\label{
  build_and_checks_dependencies:check_URLslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/check_URLs.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh
}
\label{
```

```
  build_and_checks_dependencies:common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/common\_build\_and\_checks.exec.sh.tpl
}
\label{
  build_and_checks_dependencies:common_build_and_checksexecshtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
}


\subsection{
  build\_and\_checks\_dependencies/common\_options.libr.sh
}
\label{
  build_and_checks_dependencies:common_optionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/common_options.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/comparisons.libr.sh
}
\label{
  build_and_checks_dependencies:comparisonslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/comparisons.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/create\_PDF.exec.sh
```

```
}
\label{
  build_and_checks_dependencies:create_PDFexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/create_PDF.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/generate\_from\_template.libr.sh
}
\label{
  build_and_checks_dependencies:generate_from_templatelibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/generate_from_template.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
get\_common\_text\_glob\_patterns.libr.sh
}
\label{
  build_and_checks_dependencies:get_common_text_glob_patternslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/grammar\_and\_spell\_check.libr.sh
}
\label{
  build_and_checks_dependencies:grammar_and_spell_checklibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
}

```

```
\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
build\_licenses\_templates.exec.sh
}
\label{
  build_and_checks_dependencies:licenses_templates:%
build_licenses_templatesexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
build_licenses_templates.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_AGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_AGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_AGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_GPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_GPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_GPLv3+.tpl
}
```

```
\subsection{
  build\_and\_checks\_dependencies/licenses\_templates/%
license\_file\_header\_LGPLv3+.tpl
}
\label{
  build_and_checks_dependencies:licenses_templates:%
license_file_header_LGPLv3+tpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/licenses_templates/%
license_file_header_LGPLv3+.tpl
}


\subsection{
  build\_and\_checks\_dependencies/lines\_counts.libr.sh
}
\label{
  build_and_checks_dependencies:lines_countslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_counts.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_filters.libr.sh
}
\label{
  build_and_checks_dependencies:lines_filterslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_filters.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/lines\_maps.libr.sh
}
\label{
  build_and_checks_dependencies:lines_mapslibrsh
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/lines_maps.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/overwrite\_if\_not\_equal.libr.sh
}
\label{
  build_and_checks_dependencies:overwrite_if_not_equallibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_black\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_black_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_black_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/python\_isort\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:python_isort_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/python_isort_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/repository\_data.libr.sh
}
\label{
  build_and_checks_dependencies:repository_datalibrsh
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/repository_data.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/shell\_checks\_complement.libr.sh
}
\label{
  build_and_checks_dependencies:shell_checks_complementlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/shell_checks_complement.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.exec.php
}
\label{
  build_and_checks_dependencies:split_scoreexecphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.exec.php
}


\subsection{
  build\_and\_checks\_dependencies/split\_score.libr.php
}
\label{
  build_and_checks_dependencies:split_scorelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/split_score.libr.php
}


\subsection{
  build\_and\_checks\_dependencies/strings\_functions.libr.sh
}
```

```
\label{
  build_and_checks_dependencies:strings_functionslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/strings_functions.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/too\_long\_code\_lines.libr.sh
}
\label{
  build_and_checks_dependencies:too_long_code_lineslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/too_long_code_lines.libr.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_common\_build\_and\_checks.exec.sh
}
\label{
  build_and_checks_dependencies:update_common_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
}


\subsection{
  build\_and\_checks\_dependencies/%
update\_or\_check\_files\_names\_listing.exec.sh
}
\label{
  build_and_checks_dependencies:update_or_check_files_names_listingexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_dependencies/update_or_check_files_names_listing.exec.sh
}

```

```latex

\subsection{
  build\_and\_checks\_variables/.gitignore
}
\label{
  build_and_checks_variables:gitignore
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/.gitignore
}


\subsection{
  build\_and\_checks\_variables/bandit.ini
}
\label{
  build_and_checks_variables:banditini
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build_and_checks_variables/bandit.ini
}


\subsection{
  build\_and\_checks\_variables/bandit\_baseline.json
}
\label{
  build_and_checks_variables:bandit_baselinejson
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/bandit_baseline.json
}


\subsection{
  build\_and\_checks\_variables/black.toml
}
\label{
  build_and_checks_variables:blacktoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/black.toml
```

```
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.html.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptshtmltpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
}


\subsection{
  build\_and\_checks\_variables/DevOrSysAdminScripts.tex.tpl
}
\label{
  build_and_checks_variables:DevOrSysAdminScriptstextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_listing.tex.tpl
}
\label{
  build_and_checks_variables:files_listingtextpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_listing.tex.tpl
}


\subsection{
  build\_and\_checks\_variables/files\_names\_listing.txt
}
\label{
  build_and_checks_variables:files_names_listingtxt
}

```

```
\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/files_names_listing.txt
}


\subsection{
  build\_and\_checks\_variables/licenses\_notes.txt
}
\label{
  build_and_checks_variables:licenses_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/licenses_notes.txt
}


\subsection{
  build\_and\_checks\_variables/phpmd\_baseline.xml
}
\label{
  build_and_checks_variables:phpmd_baselinexml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/phpmd_baseline.xml
}


\subsection{
  build\_and\_checks\_variables/pylintrc.toml
}
\label{
  build_and_checks_variables:pylintrctoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/pylintrc.toml
}


\subsection{
  build\_and\_checks\_variables/repository\_data.txt
}
\label{
  build_and_checks_variables:repository_datatxt
```

```
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/repository_data.txt
}


\subsection{
  build\_and\_checks\_variables/ruff.toml
}
\label{
  build_and_checks_variables:rufftoml
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/ruff.toml
}


\subsection{
  build\_and\_checks\_variables/shellcheck.ini
}
\label{
  build_and_checks_variables:shellcheckini
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck.ini
}


\subsection{
  build\_and\_checks\_variables/shellcheck\_notes.txt
}
\label{
  build_and_checks_variables:shellcheck_notestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks_variables/shellcheck_notes.txt
}


\subsection{
  build\_and\_checks.exec.sh
}
```

```
\label{
  build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
build_and_checks.exec.sh
}


\subsection{
  COPYING
}
\label{
  COPYING
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING
}


\subsection{
  COPYING.LESSER
}
\label{
  COPYINGLESSER
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
COPYING.LESSER
}


\subsection{
  git\_helpers/git\_checks.libr.sh
}
\label{
  git_helpers:git_checkslibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/git_checks.libr.sh
}


\subsection{
```

```
  git\_helpers/post-commit
}
\label{
  git_helpers:post-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post-commit
}


\subsection{
  git\_helpers/post\_commit\_renaming\_comment.libr.sh
}
\label{
  git_helpers:post_commit_renaming_commentlibrsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/post_commit_renaming_comment.libr.sh
}


\subsection{
  git\_helpers/pre-commit
}
\label{
  git_helpers:pre-commit
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
git_helpers/pre-commit
}


\subsection{
  IMAP\_transfert.py
}
\label{
  IMAP_transfertpy
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
IMAP_transfert.py
}

```

```
\subsection{
  LiensUtiles.txt
}
\label{
  LiensUtilestxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
LiensUtiles.txt
}


\subsection{
  miscellaneous/funny.sh
}
\label{
  miscellaneous:funnysh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/funny.sh
}


\subsection{
  miscellaneous/spampty\_data.txt
}
\label{
  miscellaneous:spampty_datatxt
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
miscellaneous/spampty_data.txt
}


\subsection{
  README.md.tpl
}
\label{
  READMEmdtpl
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
README.md.tpl
```

```
}


\subsection{
  unicode.libr.php
}
\label{
  unicodelibrphp
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
unicode.libr.php
}


\subsection{
  upgrade\_build\_and\_checks.exec.sh
}
\label{
  upgrade_build_and_checksexecsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
upgrade_build_and_checks.exec.sh
}


\subsection{
  wget\_sha512.libr.sh
}
\label{
  wget_sha512librsh
}

\VerbatimInput[numbers=left,xleftmargin=-5mm]{
wget_sha512.libr.sh
}


```

2.39 build_and_checks_variables/files_names_listing.txt

```
// This file is part of DevOrSysAdminScripts library.
//
// DevOrSysAdminScripts is free software:
// you can redistribute it and/or modify it under the terms
```

```
// of the GNU Lesser General Public License
// as published by the Free Software Foundation,
// either version 3 of the License,
// or (at your option) any later version.
//
// DevOrSysAdminScripts is distributed in the hope
// that it will be useful,
// but WITHOUT ANY WARRANTY;
// without even the implied warranty of
// MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.
// See the GNU Lesser General Public License for more details.
//
// You should have received a copy of
// the GNU Lesser General Public License
// along with DevOrSysAdminScripts.
// If not, see <https://www.gnu.org/licenses/>.
//
// ©Copyright 2023-2026 Laurent Frédéric Bernard François Lyaudet
// 2024-10-31T21:37+01:00: This file was moved from
// "./build_and_checks_dependencies/listings"
// to
// "./build_and_checks_variables".
// Blasé20240517Résultat.txt
// Blasé20240517.sh
// Blasé20240518_2.txt
// Blasé20240518_3.txt
build_and_checks_dependencies/build_and_checks.exec.sh.tpl
build_and_checks_dependencies/build_md_from_printable_md.exec.sh
build_and_checks_dependencies/check_shell_scripts_beginnings.libr.sh
build_and_checks_dependencies/check_shell_scripts_indentation.libr.sh
build_and_checks_dependencies/check_URLs.libr.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh
build_and_checks_dependencies/common_build_and_checks.exec.sh.tpl
build_and_checks_dependencies/common_options.libr.sh
build_and_checks_dependencies/comparisons.libr.sh
build_and_checks_dependencies/create_PDF.exec.sh
build_and_checks_dependencies/generate_from_template.libr.sh
build_and_checks_dependencies/get_common_text_glob_patterns.libr.sh
build_and_checks_dependencies/grammar_and_spell_check.libr.sh
build_and_checks_dependencies/licenses_templates/\
build_licenses_templates.exec.sh
build_and_checks_dependencies/licenses_templates/\
license_file_header_AGPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
license_file_header_GPLv3+.tpl
build_and_checks_dependencies/licenses_templates/\
```

```
license_file_header_LGPLv3+.tpl
build_and_checks_dependencies/lines_counts.libr.sh
build_and_checks_dependencies/lines_filters.libr.sh
build_and_checks_dependencies/lines_maps.libr.sh
// build_and_checks_dependencies/listings/files_listing.tex.tpl
build_and_checks_dependencies/overwrite_if_not_equal.libr.sh
build_and_checks_dependencies/python_black_complement.libr.sh
build_and_checks_dependencies/python_isort_complement.libr.sh
build_and_checks_dependencies/repository_data.libr.sh
build_and_checks_dependencies/shell_checks_complement.libr.sh
build_and_checks_dependencies/split_score.exec.php
build_and_checks_dependencies/split_score.libr.php
build_and_checks_dependencies/strings_functions.libr.sh
build_and_checks_dependencies/too_long_code_lines.libr.sh
build_and_checks_dependencies/update_common_build_and_checks.exec.sh
build_and_checks_dependencies/\
update_or_check_files_names_listing.exec.sh
build_and_checks_variables/.gitignore
build_and_checks_variables/bandit.ini
build_and_checks_variables/bandit_baseline.json
build_and_checks_variables/black.toml
build_and_checks_variables/DevOrSysAdminScripts.html.tpl
// build_and_checks_variables/DevOrSysAdminScripts.tex
build_and_checks_variables/DevOrSysAdminScripts.tex.tpl
build_and_checks_variables/files_listing.tex.tpl
build_and_checks_variables/files_names_listing.txt
build_and_checks_variables/licenses_notes.txt
build_and_checks_variables/phpmd_baseline.xml
build_and_checks_variables/pylintrc.toml
build_and_checks_variables/repository_data.txt
build_and_checks_variables/ruff.toml
build_and_checks_variables/shellcheck.ini
build_and_checks_variables/shellcheck_notes.txt
// build_and_checks_variables/upgrade_venvs_ts
build_and_checks.exec.sh
COPYING
COPYING.LESSER
// DevOrSysAdminScripts.html
// DevOrSysAdminScripts.pdf
// EncoreBlasé20240517-20240518.txt
git_helpers/git_checks.libr.sh
git_helpers/post-commit
git_helpers/post_commit_renaming_comment.libr.sh
git_helpers/pre-commit
IMAP_transfert.py
LiensUtiles.txt
```

```
// Uncomment some files in miscellaneous to do some testing.
// miscellaneous/\
12345678901234567890123456789012345678901234567890123456789012345678901234 5
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
56
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
56\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
5\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123 4\
\
// miscellaneous/\
1234567890123456789012345678901234567890123456789012345678901234567890123\
\
// miscellaneous/\
a_long_long_long_long_long_long_long_long_long_filename_so_that_you_can_se\
e_that_it_works.txt
miscellaneous/funny.sh
miscellaneous/spampty_data.txt
// miscellaneous/\
yet_another_long_long_long_long_long_long_long_long_long_filename_so_that_\
you_can_see_that_it_works.txt
// README.html
// README.md
README.md.tpl
unicode.libr.php
upgrade_build_and_checks.exec.sh
wget_sha512.libr.sh
```

2.40 build_and_checks_variables/licenses_notes.txt

```
This repository is dual licensed.
All the Bash scripts source code is covered by GPLv3+.
All the PHP scripts source code is covered by LGPLv3+.
All the Python scripts source code is covered by LGPLv3+.

Here is a listing of the dependencies of the code in this repository:
- The following dependencies are GPLv3+ licensed:
basename, bash (and its builtins), cat, chmod, cp, cut, date, diff,
dirname, env, find, grep, head, mkdir, mv, realpath, rm, sed,
```

```
sha512sum, shellcheck, sort, tail, touch, wc, wget, xargs;
- The following dependencies are GPLv2+ licensed:
more, pandoc, pdftex, pdflatex, tree;
- The following dependencies are GPLv2 licensed:
git, pylint;
- The following dependencies are LGPLv2.1+ licensed:
notify-send;
- The following dependencies are MIT licensed:
black, isort, mypy, pip, ruff;
- The following dependencies are Apache-2.0 licensed:
bandit;
- The following dependencies are BSD-3-Clause licensed:
phpmd;
- php is licensed under PHP License v3.01;
- tex is licensed under Knuth license;
- pcregrep, pcre2grep has a custom license from Cambridge University,
but PCRE2 has BSD-3-Clause WITH PCRE2-exception,
- python3 is licensed under PSF License Version 2 and the Zero-Clause BSD
license.


Remark: Some commands listed under GPLv3+ are POSIX
and can have different implementations under another license;
for example, uutils coreutils is MIT licensed,
whilst GNU coreutils is GPLv3+ licensed.

Bash can be found at the following URLs:
https://www.gnu.org/software/bash/
https://tiswww.case.edu/php/chet/bash/bashtop.html
https://savannah.gnu.org/projects/bash/

GNU coreutils can be found at the following URLs:
https://www.gnu.org/software/coreutils/
https://savannah.gnu.org/projects/coreutils/

uutils coreutils can be found at the following URLs:
https://uutils.github.io/coreutils/
https://github.com/uutils/coreutils

diff (GNU diffutils) can be found at the following URLs:
https://www.gnu.org/software/diffutils/
https://savannah.gnu.org/projects/diffutils/

find, xargs (GNU findutils) can be found at the following URLs:
https://www.gnu.org/software/findutils/
https://savannah.gnu.org/projects/findutils/
```

```

grep (GNU grep) can be found at the following URLs:
https://www.gnu.org/software/grep/
https://savannah.gnu.org/projects/grep/

sed (GNU sed) can be found at the following URLs:
https://www.gnu.org/software/sed/
https://savannah.gnu.org/projects/sed/

ShellCheck can be found at the following URLs:
https://www.shellcheck.net/
https://github.com/koalaman/shellcheck

GNU Wget can be found at the following URLs:
https://www.gnu.org/software/wget/
https://savannah.gnu.org/projects/wget/

more of util-linux can be found at the following URLs:
https://www.kernel.org/pub/linux/utils/util-linux/
https://git.kernel.org/pub/scm/utils/util-linux/util-linux.git/
https://github.com/util-linux/util-linux official?

pandoc can be found at the following URLs:
https://pandoc.org/
https://github.com/jgm/pandoc

pdftex/pdflatex can be found at the following URLs:
https://www.tug.org/applications/pdftex/
https://tug.org/texlive/svn/

tree can be found at the following URLs:
https://oldmanprogrammer.net/source.php?dir=projects/tree
https://gitlab.com/OldManProgrammer/unix-tree
https://github.com/Old-Man-Programmer/tree

git can be found at the following URLs:
https://git-scm.com/about
https://github.com/git/git

pylint can be found at the following URLs:
https://www.pylint.org/
https://pylint.readthedocs.io/en/stable/
https://github.com/pylint-dev/pylint
https://pypi.org/project/pylint/

notify-send can be found at the following URLs:
```

```
https://gitlab.gnome.org/GNOME/libnotify/
https://gnome.pages.gitlab.gnome.org/libnotify/

black can be found at the following URLs:
https://github.com/psf/black
https://black.readthedocs.io/en/stable/
https://pypi.org/project/black/

isort can be found at the following URLs:
https://pycqa.github.io/isort/
https://github.com/pycqa/isort/
https://pypi.org/project/isort/

mypy can be found at the following URLs:
https://mypy-lang.org/
https://mypy.readthedocs.io/en/stable/
https://github.com/python/mypy
https://pypi.org/project/mypy/

pip can be found at the following URLs:
https://pip.pypa.io/en/stable/
https://github.com/pypa/pip
https://pypi.org/project/pip/ (egg and chicken ;) )

ruff can be found at the following URLs:
https://docs.astral.sh/ruff/
https://github.com/astral-sh/ruff
https://pypi.org/project/ruff/

bandit can be found at the following URLs:
https://bandit.readthedocs.io/en/latest/
https://github.com/PyCQA/bandit
https://pypi.org/project/bandit/

phpmd can be found at the following URLs:
https://phpmd.org/
https://github.com/phpmd/phpmd
https://packagist.org/packages/phpmd/phpmd

php can be found at the following URLs:
https://www.php.net/
https://github.com/php/php-src

tex can be found at the following URL:
https://ctan.org/tex-archive/systems/knuth/dist/tex
```

```
pcregrep, pcre2grep can be found at the following URLs:
https://www.pcre.org/
https://pcre2project.github.io/pcre2/
https://github.com/PCRE2Project/pcre1
https://github.com/PCRE2Project/pcre2
https://github.com/PCRE2Project/pcre2/blob/main/src/pcre2grep.c

python3 can be found at the following URLs:
https://www.python.org/
https://docs.python.org/3/

If you see something that I missed regarding my dependencies,
please tell me/email me :).
```

2.41 build_and_checks_variables/phpmd_baseline.xml

```
<?xml version="1.0"?>
<phpmd-baseline>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="build_and_checks_dependencies/split_score.libr.php"/>
  <violation rule="PHPMD\Rule\Design\LongMethod"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseParameterName"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Naming\LongVariable"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Controversial\CamelCaseVariableName"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CleanCode\MissingImport"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\CyclomaticComplexity"
    file="unicode.libr.php"/>
  <violation rule="PHPMD\Rule\Design\NpathComplexity"
    file="unicode.libr.php"/>
</phpmd-baseline>
```

2.42 build_and_checks_variables/pylintrc.toml

```
[tool.pylint.main]
exit-zero = true
ignore = ["node_modules"]
jobs = 0
```

```
[tool.pylint.basic]
# We use uppercase acronyms like HTML, IMAP, etc.
module-rgx = "^[_a-zA-Z0-9+]+$"
function-rgx = "^[_a-zA-Z0-9+]+$"
[tool.pylint."messages control"]
# Code like R0912 is better to order disables by category.
disable = [
  "C1804",  # use-implicit-booleaness-not-comparison-to-string
  "I0011",  # locally-disabled
  "I0020",  # suppressed-message
  "R0912",  # too-many-branches
  "R0913",  # too-many-arguments
  "R0914",  # too-many-locals
  "R0915",  # too-many-statements
  "R0917",  # too-many-positional-arguments
]
enable = "all"
[tool.pylint.reports]
output-format = "colorized"
```

2.43 build_and_checks_variables/repository_data.txt

```
Look at grep_variable to understand the structure of this file.
Basically, it has almost no structure apart from some lines
starting with a "variable_name=".
Anything else is comments.

repository_name=DevOrSysAdminScripts
repository_web_url=https://github.com/LLyaudet/DevOrSysAdminScripts/
repository_git_url=git@github.com:LLyaudet/DevOrSysAdminScripts.git

abstract=Some “small” useful scripts
abstract=for enhancing development or system-administrators tasks

license=GPLv3+
license2=LGPLv3+

author_email=Laurent.Lyaudet@gmail.com
author_full_name=Laurent Frédéric Bernard François Lyaudet
author_last_name=Lyaudet
author_name=Laurent Lyaudet
author_website=https://lyaudet.eu/laurent/

acknowledgments=Merci Dieu !
acknowledgments=Merci Père ! Merci Seigneur ! Merci Saint Esprit !

```

```
Great day! Now I allow myself lines of 75 characters instead of 70 :).
I checked how many characters fit in the PDF file generated by tex with
default paper size,
when using verbatim environment with line numbers for source code listing.
With a screen capture in GIMP,
I mesured 197px between two-digits line numbers and the left border of
the page.
(It is less than 197px for three-digits or more line numbers.)
The same distance from the right border yields 75 characters allowed.
Since when printing the HTML file, the number of allowed characters in
<pre> tags is way larger, 75 is the right limit with the tools I use.

max_line_length=75

grammar_or_spell_checker_command=more


Why one venv for each Python dependency ?
Because no language of my knowledge did it right to handle all
possible dependency conflicts.
So in case of such a conflict, one specialized venv can get you out
of any problem.
The best solution would have been to allow install of different
versions of the same package, and have a mechanism:
- first intent, use the latest available version of a package among
those that satisfy the declared dependency constraints of the package
that has this dependency;
- second intent, check a dependency overload configuration file
with tree structure: if calling python package is A version 1.1.1
with dependency B itself with dependency C, use B 2.2.2 and C 3.3.3;
you could have along that another rule saying that when called
directly B 2.2.2 should use C 4.4.4;
and even more complicated:
if calling python package is A version 1.1.1 with dependency B and D
both themselves with dependency C, use B 2.2.2, D 0.1.0,
and C 3.3.3 for B, and C 4.4.4 for D;
once you’ve seen that it can never be more complicated than that once
the same version number guarantees that the code is the same,
you create a config language "à la CSS" for cascading dependencies ;)
XD fit into JSON the dumbest possible way and you stop bothering
about dependency conflicts for the rest of history of informatics.
{
  "A":{
    "1.1.1": {
      "B": {
        "2.2.2": {
```

```
            "C": {
              // Dumb, you did not found it by yourself? Failed exam.
              "3.3.3" : {}
            }
          }
        },
        "D": {
          "0.1.0": {
            "C": {
              // Dumb, you did not found it by yourself? Failed exam.
              "4.4.4" : {}
            }
          }
        }
      }
    },
    "B": {
      "2.2.2": {
        "C": {
          // Dumb, you did not found it by yourself? Failed exam.
          "4.4.4" : {}
        }
      }
    }
  }
  Wonderful someone rediscovered nested structures XD
  Seriously so many engineers or computer scientists are dumb to cry.

  bandit_venv=../venv_bandit
  black_venv=../venv_black
  isort_venv=../venv_isort
  mypy_venv=../venv_mypy
  pylint_venv=../venv_pylint
  ruff_venv=../venv_ruff


  upgrade venvs can be auto or ask, any other value is interpreted
  as ask.

  upgrade_venvs=ask


  time interval can be 0 to always upgrade or ask to upgrade,
  or a number at most bash max int
  https://superuser.com/questions/1030122/
  what-is-the-maximum-value-of-a-numeric-bash-shell-variable
```

```
it should be enough to simulate never "upgrade or ask for upgrade"
unless you claim to live many billions years ;)
If it is wat, this is a random int XD (see RANDOM).
Note that the probability that it upgrades your dependencies is not
1/2, it increases over time, and is 1 after slightly
less than one day.
If it is watyouwant?, see SRANDOM XD.

upgrade_venvs_time_interval_in_seconds=wat


For the Nu HTML, CSS, and SVG checker.

vnu_jar_path=../jar_venv/vnu.jar

Not fully implemented yet, it will only warn that vnu jar was not
updated since some time.
"wat" works, but "watyouwant?" joke was not repeated ;).

upgrade_vnu_jar_time_interval_in_seconds=604800
```

2.44 build_and_checks_variables/ruff.toml

```
[lint]
select = [
  "ALL",
  "D211",  # blank-line-before-class no...before-class-docstring!
]
# ignore = ["D203", "D212", "D404"]
ignore = [
  "C901",  # complex-structure AKA give-me-toy-code
  "D203",  # (want-)?incorrect-blank-line-before-class
  "D205",  # missing-blank-line-after-summary So so cute|true ;) XD.
  "D212",  # multi-line-summary-first-line
  "D404",  # docstring-starts-with-this Shame! XD

  # raw-string-in-exception may be useful for a real project
  # with "dictionnaries" of errors made of constants.
  "EM101",

  # commented-out-code Of course git log is always faster...
  # Note that if you're tired or someone else in your team is,
  # it may catch some commit that lacked some review,
  # or automated tests: "Last manual test succeeded, ah but when I
  # debugged at some point I commented a part that was needed in
  # another case."...
```

```
  "ERA001",

  "N802",  # invalid-function-name IMAP, HTML, etc.
  "PLR0912",  # too-many-branches AKA ...
  "PLR0913",  # too-many-arguments AKA ...
  "PLR0915",  # too-many-statements AKA ...
  # Not enough reading when young makes brain slow to read and
  # understand lengthy functions.
  # If the project is finished, you may play with AKA rules.
]
[lint.per-file-ignores]
"build_and_checks_dependencies/licenses_templates/*.py" = [
  "INP001"  # implicit-namespace-package
]
"IMAP_transfert.py" = [
  "T201",  # print Use log instead in full applications.
  "TRY003",  # raise-vanilla-args
]
```

2.45 build_and_checks_variables/shellcheck.ini

```
enable=all
check-sourced=true
external-sources=true
source-path="."
source-path="SCRIPTDIR"
source-path="./build_and_checks_dependencies"
source-path="./git_helpers"

```

2.46 build_and_checks_variables/shellcheck_notes.txt

```
A listing of the current suppressed warnings/errors from shellcheck,
and also some of the warnings that were suppressed in some previous
state of the git repository (in that case, we use X suffix, see SC2162X):
- SC1090 : Can't follow non-constant source.
           Use a directive to specify location
           # shellcheck source=bla/bli.sh
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1091 : Not following (file not found most of the time).
           Only remaining cases are in common_build_and_checks
           when sourcing some Python venv
           source venv/bin/active
- SC1111 : Use of slanted double quotes (valid inside single quotes)
```

```
            False positive, but may catch some problem.
- SC2001 : Replace some sed with bash builtin ${a//b/c}
            False positive when c contains \n.
- SC2034 : Unused variable
            False positive for globals that are declared in functions
            that are not currently called.
            Somehow it can be used to detect dead code.
- SC2059 : Don't use variables in the printf format string.
            Use printf '..%s..' "$foo".
            False positive when using a variable to build the format string.
- SC2086 : Double quote to prevent globbing and word splitting.
            False positive when returning integer variable
            via an indirection like return ${!foo}.
            See SC2248 below and the corresponding issue on GitHub.
- SC2119 : Use foo "$@" if function's $1 should mean script's $1.
            False positive when passing arguments would yield
            a wrong result.
- SC2120 : foo references arguments, but none are ever passed.
            False positives when some arguments are meant for debug in CLI.
- SC2125 : Brace expansions and globs are literal.
            False positive when assigning a variable declared with -i.
            https://github.com/koalaman/shellcheck/issues/914
- SC2145 : Argument mixes string and array.
            False positive when using "$@" line return and pipes.
            https://github.com/koalaman/shellcheck/issues/3192
            Example:
              # shellcheck disable=SC2145
              all_self_code_lines "$@"\
                | empty_lines_after_file_name
- SC2154 : var is referenced but not assigned.
            Very useful but false positive when I play with
            variable variable names.
- SC2162X: read without -r will mangle backslashes.
            False positive when read is used without variable:
            https://github.com/koalaman/shellcheck/issues/623
- SC2248 : Prefer double quoting
            False positive on returning integer values.
            https://github.com/koalaman/shellcheck/issues/3405
- SC2309 : -eq treats this as a variable.
              Use = to compare as string (or expand explicitly with $var).
            Very useful but false positive when I play with
            variable variable names.
- SC2312 : Some command is not invoked separately and thus its return
            value is masked. Frequent with pipes.
- SCNNNNX: Use printf instead of echo.
            https://github.com/koalaman/shellcheck/issues/3419
```

```

The command to check if there is anything else:
grep -r "shellcheck disable" . | grep -v -e '.git' -e "=SC1090$"\
  -e "=SC1091$" -e "=SC1090,SC1091$" -e "=SC1111$" -e "=SC2001$"\
  -e "=SC2034$" -e "=SC2059$" -e "=SC2086$" -e "=SC2119$" -e "=SC2120$"\
  -e "=SC2125$" -e "=SC2145$" -e "=SC2154$" -e "=SC2248$" -e "=SC2309$"\
  -e "=SC2312$" -e "=SC2154,SC2309$"
```

2.47 build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# The file "build_and_checks.exec.sh.tpl" was renamed from
# "build_and_checks.sh.tpl" to "build_and_checks.exec.sh.tpl".
# The file "build_and_checks.exec.sh" generated from the file
# "build_and_checks.sh.tpl" or "build_and_checks.exec.sh.tpl"
# was renamed from
# "build_and_checks.sh" to "build_and_checks.exec.sh".

build_and_checks(){
  local LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  readonly LFBFL_work_directory
```

```
37
38    local LFBFL_verbose=""
39    if [[ "$*" == *--verbose* ]]; then
40      printf "%s %s\n" "$0" "$*"
41      LFBFL_verbose="--verbose"
42    fi
43    readonly LFBFL_verbose
44
45    source ./wget_sha512.libr.sh
46
47    LFBFL_subdir="build_and_checks_dependencies"
48    mkdir -p "${LFBFL_subdir}/licenses_templates"
49    LFBFL_variables_directory="${LFBFL_work_directory}/"
50    LFBFL_variables_directory+="build_and_checks_variables"
51    mkdir -p "${LFBFL_variables_directory}/temp"
52    if ! [[ -f "${LFBFL_variables_directory}/.gitignore" ]]; then
53      printf "temp/\n\n"\
54        > "${LFBFL_variables_directory}/.gitignore"
55    fi
56
57    # LFBFL_dependencies_raw_content_download_URL
58    local LFBFL_dependencies_URL
59    LFBFL_dependencies_URL="https://raw.githubusercontent.com/LLyaudet/"
60    LFBFL_dependencies_URL+="DevOrSysAdminScripts/main/${LFBFL_subdir}"
61    readonly LFBFL_dependencies_URL
62
63    declare -r LFBFL_common_file_name="common_build_and_checks.exec.sh"
64    # LFBFL_script_download_URL
65    declare -r\
66      LFBFL_script="${LFBFL_dependencies_URL}/${LFBFL_common_file_name}"
67    declare -r\
68      LFBFL_file_path="./${LFBFL_subdir}/${LFBFL_common_file_name}"
69    local LFBFL_correct_sha512
70    LFBFL_correct_sha512='9932636c29d6e21f609edca34de0609b8dea6e4f4ed4a'
71    LFBFL_correct_sha512+='35dd5e1d83c4f2fe9c1d47e9401f5c90c323ad699205'
72    LFBFL_correct_sha512+='02b1cedb9a3a890ecfe94500d774d08f153a835'
73    wget_sha512 "${LFBFL_file_path}" "${LFBFL_script}"\
74      "${LFBFL_correct_sha512}" "${LFBFL_verbose}"
75    chmod +x "./${LFBFL_file_path}"
76
77    "${LFBFL_file_path}" "${LFBFL_work_directory}"\
78      "${LFBFL_dependencies_URL}" "${LFBFL_verbose}"
79  }
80
81  if [[ "$*" == *--fixed-point-build* ]]; then
82    printf -- "--fixed-point-build\n"
```

```
  source "./build_and_checks_dependencies/lines_filters.libr.sh"
  LFBFL_work_directory="."
  if [[ -n "$1" ]]; then
    LFBFL_work_directory="$1"
  fi
  LFBFL_variables_directory="${LFBFL_work_directory}/"
  LFBFL_variables_directory+="build_and_checks_variables"
  LFBFL_data_file_name="${LFBFL_variables_directory}/"
  LFBFL_data_file_name+="repository_data.txt"
  LFBFL_repository_name=""
  grep_variable "${LFBFL_data_file_name}" repository_name\
    --result-variable-prefix="LFBFL_"\
    --replace-line-returns-by=""
  # Touching the 3 following files first let us iterate
  # upgrade_build_and_checks 2 times instead of 3 to reach
  # a fixed point.
  touch "${LFBFL_variables_directory}/${LFBFL_repository_name}.tex"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.pdf"
  touch "${LFBFL_work_directory}/${LFBFL_repository_name}.html"
  # The two iterations of build_and_checks needs to complete
  # during the same minute to have a fixed point.
  build_and_checks "$@"
  # build_and_checks "$@" second iteration below
  # Then calling this script without --fixed-point-build should not
  # yield any new superficial modifications.
fi

build_and_checks "$@"
```

2.48 COPYING

```
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                       Version 3, 29 June 2007

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```

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```

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```

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        Corresponding Source.

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    necessary to install and execute a modified version of the
    Combined Work produced by recombining or relinking the
```

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2.50 git_helpers/git_checks.libr.sh

```
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# 2024-05-31T01:57+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# 2024-05-31T01:49+02:00: This file was renamed from
# "git_checks.libr.sh"
# to
# "git_checks.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "git_checks.sh" to "git_checks.libr.sh".

check_files(){
  # Checks used on Django projects.
  # Use `git commit --no-verify` after check.
```

```
45    declare -r LFBFL_send_summary_1="ATTENTION : models modifiés"
46    declare -r LFBFL_send_body_1="Vérifiez si besoin de migrations"
47    local LFBFL_send_body_2="Pensez aussi aux contraintes d'unicités"
48    LFBFL_send_body_2+=" dès la création du modèle"
49    readonly LFBFL_send_body_2
50    declare -r LFBFL_send_summary_2="ATTENTION : serializers modifiés"
51    declare -r LFBFL_send_body_3=\
52  "Les nouveaux preloadings sont interdits -> #Prefetch()"
53    declare -i LFBFL_error=0
54    git diff --cached --name-only\
55      > /tmp/DOSAS_django_git_check1.temp
56    if grep models /tmp/DOSAS_django_git_check1.temp;
57    then
58      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
59      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_2}"
60      LFBFL_error=1
61    fi
62    if grep serializer /tmp/DOSAS_django_git_check1.temp;
63    then
64      notify-send "${LFBFL_send_summary_2}" "${LFBFL_send_body_3}"
65      LFBFL_error=1
66    fi
67    rm /tmp/DOSAS_django_git_check1.temp
68    # shellcheck disable=SC2248
69    return ${LFBFL_error}
70  }
71
72
73  check_no_abusive_trailing_comma(){
74    # Useful for Python code, although it regularly yields
75    # false positives in Python code
76    # (but it is easy to distinguish by hand).
77    # Use `git commit --no-verify` after check.
78    declare -r LFBFL_send_summary_1="ATTENTION"
79    local LFBFL_send_body_1="Il semblerait que vous affectiez un tuple"
80    LFBFL_send_body_1+=" au lieu d'une autre valeur dans une variable."
81    readonly LFBFL_send_body_1
82    declare -i LFBFL_error=0
83    git diff --cached -r\
84      > /tmp/DOSAS_django_git_check2.temp
85    if grep ' = .*,\s*$' /tmp/DOSAS_django_git_check2.temp;
86    then
87      notify-send "${LFBFL_send_summary_1}" "${LFBFL_send_body_1}"
88      LFBFL_error=1
89    fi
90    rm /tmp/DOSAS_django_git_check2.temp
```

```
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}


check_black_code_formatting(){
  # attention ça ne marche que sur les fichiers "stagés" avec git add
  declare -r LFBFL_files_string=$(
    git diff --cached --name-only\
    | grep '\.py'
  )
  printf "%s\n" "${LFBFL_files_string}"
  declare -a LFBFL_some_files
  mapfile -t LFBFL_some_files <<< "${LFBFL_files_string}"
  readonly LFBFL_some_files
  declare -i LFBFL_error=0
  # Not very useful since LFBFL_file should be local to the loop by
  # default.
  local LFBFL_file
  for LFBFL_file in "${LFBFL_some_files[@]}"; do
    printf "Black will check formatting on file %s\n" "${LFBFL_file}"
    if ! black --check --diff "${LFBFL_file}";
    then
      LFBFL_error=1
    fi
  done
  # shellcheck disable=SC2248
  return ${LFBFL_error}
}
```

2.51 git_helpers/post-commit

```
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post-commit"
# to
# "post-commit".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="post_commit_renaming_comment.libr.sh"
# shellcheck source=post_commit_renaming_comment.libr.sh
source "${LFBFL_library}"
LFBFL_library="../DevOrSysAdminScripts/build_and_checks_dependencies/"
LFBFL_library+="lines_filters.libr.sh"
# shellcheck source=lines_filters.libr.sh
source "${LFBFL_library}"

LFBFL_data_file_name="build_and_checks_variables/repository_data.txt"
LFBFL_max_comment_line_length=""
if [[ -f "${LFBFL_data_file_name}" ]]; then
  LFBFL_max_line_length=""
  grep_variable "${LFBFL_data_file_name}" "max_line_length"\
    --result-variable-prefix="LFBFL_"
  LFBFL_max_comment_line_length="--max-comment-line-length="
  LFBFL_max_comment_line_length+="${LFBFL_max_line_length}"
fi

commit_a_file_renamed_comment --verbose --log-directory-change\
  "${LFBFL_max_comment_line_length}"
exit 0
```

2.52 git_helpers/post_commit_renaming_comment.libr.sh

```
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# This file is part of DevOrSysAdminScripts library.
#
```

```
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# 2024-05-31T01:49+02:00: This file was renamed from
# "post_commit_renaming_comment.libr.sh"
# to
# "post_commit_renaming_comment.libr.sh".
# 2024-05-31T01:49+02:00: This file was moved from
# "."
# to
# "./git_helpers".
# This file was renamed from "post_commit_renaming_comment"
# to "post_commit_renaming_comment.libr.sh".
# (Before first commit it seems. But I had the old name in listing.)

LFBFL_this_file_path=$(realpath "${BASH_SOURCE[0]}")
LFBFL_this_file_directory=$(dirname "${LFBFL_this_file_path}")
LFBFL_some_directory="${LFBFL_this_file_directory}/../"
declare -i LFBFL_cd_result2
pushd .
cd "${LFBFL_some_directory}" || {
  LFBFL_cd_result2=$?
  printf "post_commit_renaming_comment.libr.sh no such directory"
  # shellcheck disable=SC2248
  exit ${LFBFL_cd_result2}
}
# shellcheck source=common_options.libr.sh
source "build_and_checks_dependencies/common_options.libr.sh"
# shellcheck source=strings_functions.libr.sh
```

```
50 source "build_and_checks_dependencies/strings_functions.libr.sh"
51 declare -i LFBFL_popd_result2
52 popd || {
53   LFBFL_popd_result2=$?
54   printf "post_commit_renaming_comment.libr.sh no popd"
55   # shellcheck disable=SC2248
56   exit ${LFBFL_popd_result2}
57 }
58
59 # Never forget that the worse security is the illusion of security
60 # (Kevin Mitnick). So, do not expect to have an accurate trackability
61 # of code history just because you use this post-commit hook or
62 # something else more evolved. This is useful but not a guarantee.
63 # For this reason, this function will only handle when all files
64 # impacted in the preceding commit are only renamed.
65 # It assumes that `diff` is the current default difference utility
66 # parametered with `git`. This point could be enhanced.
67 commit_a_file_renamed_comment(){
68   # Options:
69   #   --verbose
70   #   --log-directory-change
71   #   --max-comment-line-length
72   declare -i LFBFL_i_verbose=0
73   get_verbose_option "$@"
74   local LFBFL_work_directory=""
75   get_work_directory_option "$@"
76   local LFBFL_arg
77   local LFBFL_s_format
78
79   declare -i LFBFL_log_directory_change=0
80   if [[ "$*" == *--log-directory-change* ]]; then
81     LFBFL_log_directory_change=1
82   fi
83   readonly LFBFL_log_directory_change
84
85   declare -i LFBFL_max_comment_line_length=70
86   for LFBFL_arg in "$@"; do
87     if [[ "${LFBFL_arg}" == --max-comment-line-length=* ]]; then
88       LFBFL_arg=${LFBFL_arg#--max-comment-line-length=}
89       LFBFL_max_comment_line_length=$((LFBFL_arg))
90       break
91     fi
92   done
93   readonly LFBFL_max_comment_line_length
94
95   if [[ LFBFL_i_verbose -eq 1 ]]; then
```

```
    LFBFL_s_format="LFBFL_log_directory_change: %s\n"
    LFBFL_s_format+="LFBFL_max_comment_line_length: %s\n"
    # shellcheck disable=SC2059
    printf "${LFBFL_s_format}"\
      "${LFBFL_log_directory_change}"\
      "${LFBFL_max_comment_line_length}"
  fi
  #-------------------------------------------------------------------

  enhanced_set_shell_option pipefail\
    && trap 'enhanced_unset_shell_option pipefail' RETURN

  declare -r LFBFL_renaming_lines=$(
    git log -p -1\
    | grep '^diff --git' -A 1
  )
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_renaming_lines: %s\n" "${LFBFL_renaming_lines}"
  fi

  declare -a LFBFL_renaming_lines_array
  mapfile -t LFBFL_renaming_lines_array <<< "${LFBFL_renaming_lines}"
  readonly LFBFL_renaming_lines_array

  declare -r LFBFL_timestamp=$(date --iso-8601="minute")
  if [[ LFBFL_i_verbose -eq 1 ]]; then
    printf "LFBFL_timestamp: %s\n" "${LFBFL_timestamp}"
  fi
  get_split_score_simple 1 "${LFBFL_max_comment_line_length}" _
  LFBFL_split_score_command1="${get_split_score_result}"
  LFBFL_split_score_command_properties1="${get_split_score_result2}"
  get_split_score_simple 1 "${LFBFL_max_comment_line_length}" /
  LFBFL_split_score_command2="${get_split_score_result}"
  LFBFL_split_score_command_properties2="${get_split_score_result2}"
  declare -i LFBFL_renaming_happened=0
  declare -i LFBFL_i
  declare -i LFBFL_j
  # declare -r LFBFL_special_file_name="files_names_listing.txt"
  declare -r LFBFL_sfn="files_names_listing.txt"
  local LFBFL_diff_line
  local LFBFL_similarity_line
  local LFBFL_old_file_path
  local LFBFL_old_file_name
  local LFBFL_old_file_directory
  local LFBFL_new_file_path
  local LFBFL_new_file_name
```

```
  local LFBFL_new_file_directory
  local LFBFL_useful_file_name
  local LFBFL_extension
  local LFBFL_comment_prefix
  local LFBFL_comment_prefix2
  local LFBFL_old_file_name2
  local LFBFL_old_file_directory2
  local LFBFL_new_file_name2
  local LFBFL_new_file_directory2
  local LFBFL_new_comment
  declare -i LFBFL_copyright_line_number
  declare -r LFBFL_send_summary_2="ATTENTION: No line with copyright"
  declare -i LFBFL_line_count
  declare -i LFBFL_lines_after
  local LFBFL_temp_file_path
  for ((LFBFL_i=0; LFBFL_i<${#LFBFL_renaming_lines_array[@]};)); do
    LFBFL_diff_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    LFBFL_similarity_line="${LFBFL_renaming_lines_array[${LFBFL_i}]}"
    LFBFL_i=$((LFBFL_i + 1))
    # We need to increment 3 times since grep adds "--" lines between.
    LFBFL_i=$((LFBFL_i + 1))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_diff_line: %s\nLFBFL_similarity_line: %s\n"\
        "${LFBFL_diff_line}"\
        "${LFBFL_similarity_line}"
    fi
    if [[ "${LFBFL_similarity_line:17:4}" != "100%" ]]; then
      continue
    fi

    # Extract old file name from diff line. --------------------------
    LFBFL_old_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 3
    )
    # It starts by "a/".
    LFBFL_old_file_path=".${LFBFL_old_file_path:1}"
    LFBFL_old_file_name=$(basename "${LFBFL_old_file_path}")
    LFBFL_old_file_directory=$(dirname "${LFBFL_old_file_path}")
    # Extract new file name from diff line. --------------------------
    LFBFL_new_file_path=$(
      printf "%s" "${LFBFL_diff_line}"\
      | cut -d ' ' -f 4
    )
    # It starts by "b/".
```

```
    LFBFL_new_file_path=".${LFBFL_new_file_path:1}"
    LFBFL_new_file_name=$(basename "${LFBFL_new_file_path}")
    LFBFL_new_file_directory=$(dirname "${LFBFL_new_file_path}")
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      LFBFL_s_format="LFBFL_old_file_path: %s\n"
      LFBFL_s_format+="LFBFL_old_file_name: %s\n"
      LFBFL_s_format+="LFBFL_old_file_directory: %s\n"
      LFBFL_s_format+="LFBFL_new_file_path: %s\n"
      LFBFL_s_format+="LFBFL_new_file_name: %s\n"
      LFBFL_s_format+="LFBFL_new_file_directory: %s\n"
      # shellcheck disable=SC2059
      printf "${LFBFL_s_format}"\
        "${LFBFL_old_file_path}"\
        "${LFBFL_old_file_name}"\
        "${LFBFL_old_file_directory}"\
        "${LFBFL_new_file_path}"\
        "${LFBFL_new_file_name}"\
        "${LFBFL_new_file_directory}"
    fi
    # Extract "LFBFL_extension" with smart guessing for .tpl ---------
    # and set prefix for line with ©Copyright. ------------------------
    LFBFL_useful_file_name="${LFBFL_new_file_name}"
    LFBFL_extension="${LFBFL_useful_file_name##*.}"
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
        "${LFBFL_useful_file_name}"\
        "${LFBFL_extension}"
    fi
    for ((LFBFL_j=0; LFBFL_j<3; ++LFBFL_j)); do
      if [[ "${LFBFL_extension}" != "tpl" ]]; then
        break
      fi
      LFBFL_useful_file_name="${LFBFL_useful_file_name:0:-4}"
      LFBFL_extension="${LFBFL_useful_file_name##*.}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "LFBFL_useful_file_name: %s\nLFBFL_extension: %s\n"\
          "${LFBFL_useful_file_name}"\
          "${LFBFL_extension}"
      fi
    done
    LFBFL_comment_prefix=""
    if [[ "${LFBFL_extension}" == "sh" ]]; then
      LFBFL_comment_prefix="# "
    fi
    if [[ "${LFBFL_extension}" == "sql" ]]; then
      LFBFL_comment_prefix="-- "
```

```
    fi
    if [[ "${LFBFL_extension}" == "tex" ]]; then
      LFBFL_comment_prefix="% "
    fi
    if [[ "${LFBFL_useful_file_name}" == "${LFBFL_sfn}" ]]; then
      LFBFL_comment_prefix="// "
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_comment_prefix: %s\n" "${LFBFL_comment_prefix}"
    fi
    #-----------------------------------------------------------------
    # Split if necessary old_file_name, new_file_name, etc. to have
    # comment lines below max_comment_line_length.
    LFBFL_comment_prefix2="${LFBFL_comment_prefix}"'"'
    # -----
    LFBFL_old_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_name2+="${LFBFL_old_file_name}"
    repeated_split_last_line "${LFBFL_old_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_old_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_old_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_old_file_directory2+="${LFBFL_old_file_directory}"
    repeated_split_last_line "${LFBFL_old_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_old_file_directory2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_name2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_name2+="${LFBFL_new_file_name}"
    repeated_split_last_line "${LFBFL_new_file_name2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command1}"\
      "${LFBFL_split_score_command_properties1}"
    LFBFL_new_file_name2="${repeated_split_last_line_result}"
    # -----
    LFBFL_new_file_directory2="${LFBFL_comment_prefix2}"
    LFBFL_new_file_directory2+="${LFBFL_new_file_directory}"
    repeated_split_last_line "${LFBFL_new_file_directory2}"\
      "${LFBFL_comment_prefix2}" "${LFBFL_max_comment_line_length}"\
      '"' "${LFBFL_split_score_command2}"\
      "${LFBFL_split_score_command_properties2}"
    LFBFL_new_file_directory2="${repeated_split_last_line_result}"
```

```
    # Create new comments strings. -----------------------------------
    LFBFL_new_comment=""
    if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
    then
      LFBFL_new_comment="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was renamed from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_name2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_name2}"'".'
    fi
    if [[ LFBFL_log_directory_change -eq 1
      && "${LFBFL_new_file_directory}" != "${LFBFL_old_file_directory}"
      ]];
    then
      if [[ "${LFBFL_new_file_name2}" != "${LFBFL_old_file_name2}" ]];
      then
        LFBFL_new_comment+=$'\n'
      fi
      LFBFL_new_comment+="${LFBFL_comment_prefix}"
      LFBFL_new_comment+="${LFBFL_timestamp}"
      LFBFL_new_comment+=": This file was moved from"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_old_file_directory2}"'"'
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_comment_prefix}to"
      LFBFL_new_comment+=$'\n'
      LFBFL_new_comment+="${LFBFL_new_file_directory2}"'".'
    fi
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_new_comment: %s\n" "${LFBFL_new_comment}"
    fi
    # Find line with ©Copyright. -------------------------------------
    LFBFL_copyright_line_number=$(
      grep -n '©Copyright' "${LFBFL_new_file_path}"\
      | head --lines=1\
      | cut -d ':' -f 1
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_copyright_line_number: %s\n"\
        "${LFBFL_copyright_line_number}"
    fi
    if [[ LFBFL_copyright_line_number -eq 0 ]]; then
```

```
      notify-send "${LFBFL_send_summary_2}"\
        "${LFBFL_send_summary_2} for: ${LFBFL_new_comment}"
      if [[ LFBFL_i_verbose -eq 1 ]]; then
        printf "No copyright line. Skipping file.\n"
      fi
      continue
    fi
    # Get total number of lines. -----------------------------------
    LFBFL_line_count=$(
      wc -l\
      < "${LFBFL_new_file_path}"
    )
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_line_count: %s\n" "${LFBFL_line_count}"
    fi
    # Compute number of lines after. --------------------------------
    LFBFL_lines_after=$((
      LFBFL_line_count - LFBFL_copyright_line_number
    ))
    if [[ LFBFL_i_verbose -eq 1 ]]; then
      printf "LFBFL_lines_after: %s\n" "${LFBFL_lines_after}"
    fi
    # Update file. --------------------------------------------------
    LFBFL_temp_file_path="${LFBFL_new_file_path}.LFBFL.temp"
    {
      head --lines="${LFBFL_copyright_line_number}"\
        "${LFBFL_new_file_path}"
      printf "%s\n" "${LFBFL_new_comment}"
      tail --lines="${LFBFL_lines_after}" "${LFBFL_new_file_path}"
    } >> "${LFBFL_temp_file_path}"
    mv "${LFBFL_temp_file_path}" "${LFBFL_new_file_path}"
    # Add file to stage. --------------------------------------------
    git add "${LFBFL_new_file_path}"
    LFBFL_renaming_happened=1
  done
  readonly LFBFL_renaming_happened
  if [[ LFBFL_renaming_happened -eq 1 ]]; then
    # Commit --------------------------------------------------------
    local LFBFL_message="commit_a_file_renamed_comment()"
    LFBFL_message+=" post-commit hook, ${LFBFL_timestamp}."
    readonly LFBFL_message
    git commit --message="${LFBFL_message}"
  fi
}
```

2.53 git_helpers/pre-commit

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# or (at your option) any later version.
#
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# the GNU General Public License
# along with DevOrSysAdminScripts.
# If not, see <https://www.gnu.org/licenses/>.
#
# ©Copyright 2023-2024 Laurent Frédéric Bernard François Lyaudet
# 2024-05-31T01:57+02:00: This file was renamed from
# "pre-commit"
# to
# "pre-commit".
# 2024-05-31T01:57+02:00: This file was moved from
# "."
# to
# "./git_helpers".

LFBFL_library="../DevOrSysAdminScripts/git_helpers/"
LFBFL_library+="git_checks.libr.sh"
# shellcheck source=git_checks.libr.sh
source "${LFBFL_library}"

declare -ig LFBFL_git_hook_error=0
if ! check_files; then
  LFBFL_git_hook_error=1
fi
if ! check_no_abusive_trailing_comma; then
  LFBFL_git_hook_error=1
fi
```

```
if ! check_black_code_formatting; then
  LFBFL_git_hook_error=1
fi
readonly LFBFL_git_hook_error
if [[ LFBFL_git_hook_error -eq 1 ]]; then
  exit 1
fi
exit 0
```

2.54 IMAP_transfert.py

```
"""
This file is part of DevOrSysAdminScripts library.

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either version 3 of the License,
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Pour transférer d'un compte IMAP à un autre
Penser à autoriser les apps moins sûres sur Gmail si nécessaire
"""

import argparse
import email
import imaplib


class IMAPTransfertBadArgumentError(Exception):
    """
```

```
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don’t see its intent, try badminton.
    """


class SurpriseError(Exception):
    """
    A class to avoid a linter warning,
    that needs an Error suffix to avoid another linter warning,
    that needs a docstring to avoid another linter warning.
    If you don’t see its intent after seeing its use, try volleyball.
    But "Cheer up!", now you don’t need "pass" XD.
    """


def IMAP_transfert(
    from_login: str,
    from_password: str,
    from_host: str,
    from_port: int,
    to_login: str,
    to_password: str,
    to_host: str,
    to_port: int,
    *,
    to_new_folder: str = "",
    verbose: bool = False,
) -> None:
    """
    Transfert all emails from some IMAP account
    to another IMAP account.
    """
    empty = ""
    if from_login == "":
        raise IMAPTransfertBadArgumentError("From login is empty.")
    # if from_password == "":
    if from_password == empty:
        # https://github.com/PyCQA/bandit/issues/714
        # Dumb
        raise IMAPTransfertBadArgumentError("From password is empty.")
    if from_host == "":
        raise IMAPTransfertBadArgumentError("From host is empty.")
    if from_port <= 0:
        raise IMAPTransfertBadArgumentError(
```

```
            "From port is not positive.",
        )
    if to_login == "":
        raise IMAPTransfertBadArgumentError("To login is empty.")
    if to_password == empty:
        # And Dumber The infinite collection XD
        raise IMAPTransfertBadArgumentError("To password is empty.")
    if to_host == "":
        raise IMAPTransfertBadArgumentError("To host is empty.")
    if to_port <= 0:
        raise IMAPTransfertBadArgumentError(
            "To port is not positive.",
        )
    # pylint: disable=pointless-string-statement
    """
    Maybe one day Python will ship something like that in its standard
    library:
    https://www.php.net/manual/fr/function.filter-var.php
    But now urlparse doesn't do any validation.
    You may want to check that the hosts or other fields are not
    "anything".
    """

    # pylint: disable=unreachable
    raise SurpriseError(
        "I haven't tested this version of my script... Oooooh!\n"
        "But I linted it without this exception ;). Aaaaaaaah!\n"
        "XD XD XD\n"
        "Untested Scream Horror Show (Soon on BrainfuckedFlix)",
    )
    # """
    # I haven't tested any version with an empty to_new_folder...

    if verbose:
        print("Connecting to from account.")
    from_account = imaplib.IMAP4_SSL(host=from_host, port=from_port)
    from_account.login(from_login, from_password)

    if verbose:
        print('Connecting to "to account".')
    to_account = imaplib.IMAP4_SSL(host=to_host, port=to_port)
    to_account.login(to_login, to_password)

    if to_new_folder != "":
        if verbose:
            print("Creating new folder in to account.")
```

```
        to_account.create(to_new_folder)

    if verbose:
        print("Fetching all emails ids in from account.")
    from_account.select()
    result, ids = from_account.uid("search", "all")
    if verbose:
        print(f'from_account.uid("search") result:{result} ids:{ids}')
    ids = ids[0].decode().split()
    if verbose:
        print("Fetching and transfering each email in from account.")
    for my_id in ids:
        result, raw_message = from_account.uid(
            "fetch",
            my_id,
            "(RFC822)",
        )
        if verbose:
            print(f'from_account.uid("fetch", {my_id}) result:{result}')
        message = raw_message[0][1]
        my_email = email.message_from_string(message.decode("utf-8"))
        flags = imaplib.ParseFlags(raw_message[0][0])
        # flags = " ".join(flags) mypy is unhappy but IMAP flags are
        # ASCII strings, we give mypy what it wants.
        flags_as_strs = [flag.decode("ascii") for flag in flags]
        flags_str = " ".join(flags_as_strs)
        # date = imaplib.Internaldate2tuple(raw_message[0][0])
        date = '"' + my_email["date"] + '"'
        if verbose:
            print(
                f'from_account.uid("fetch", {my_id})'
                f" flags:{flags} date:{date}",
            )
        to_account.append(to_new_folder, flags_str, date, message)


def main() -> None:
    """
    Parse command-line arguments and
    transfert all emails from some IMAP account
    to another IMAP account.
    """
    parser = argparse.ArgumentParser(
        description=(
            "Transfert all emails from some IMAP account"
            " to another IMAP account"
```

```
174         ),
175     )
176     parser.add_argument(
177         "from_login",
178         type=str,
179         help="Email account login of user one.",
180     )
181     parser.add_argument(
182         "from_password",
183         type=str,
184         help="Email account password of user one.",
185     )
186     parser.add_argument(
187         "to_login",
188         type=str,
189         help="Email account login of user two.",
190     )
191     parser.add_argument(
192         "to_password",
193         type=str,
194         help="Email account password of user two.",
195     )
196     parser.add_argument(
197         "--to_new_folder",
198         type=str,
199         default="",
200         help="Transfert emails into a new folder of this name.",
201     )
202     parser.add_argument(
203         "--from_host",
204         type=str,
205         default="imap.gmail.com",
206         help="Host of first email account",
207     )
208     parser.add_argument(
209         "--from_port",
210         type=int,
211         default=993,
212         help="Port on host of first email account",
213     )
214     parser.add_argument(
215         "--to_host",
216         type=str,
217         default="imap.gmail.com",
218         help="Host of second email account",
219     )
```

```
    parser.add_argument(
        "--to_port",
        type=int,
        default=993,
        help="Port on host of second email account",
    )
    parser.add_argument(
        "-v",
        "--verbose",
        action="store_true",
        help="Print details of operations",
    )

    args = parser.parse_args()

    if args.verbose:
        print(
            "-----------------------------------------------------\n"
            "IMAP transfert:\n"
            "-----------------------------------------------------\n"
            "From account:\n"
            f" - Login: {args.from_login}\n"
            f" - Password: {args.from_password}\n"
            f" - Host: {args.from_host}\n"
            f" - Port: {args.from_port}\n"
            "To account:\n"
            f" - Login: {args.to_login}\n"
            f" - Password: {args.to_password}\n"
            f" - Host: {args.to_host}\n"
            f" - Port: {args.to_port}\n"
            "-----------------------------------------------------\n",
        )
        if args.to_new_folder is not None:
            print(
                f"Emails will go in folder {args.to_new_folder}"
                " after it is created.",
            )
    try:
        IMAP_transfert(
            args.from_login,
            args.from_password,
            args.from_host,
            args.from_port,
            args.to_login,
            args.to_password,
            args.to_host,
```

```
            args.to_port,
            to_new_folder=args.to_new_folder,
            verbose=args.verbose,
        )
    except (IMAPTransfertBadArgumentError, SurpriseError) as e:
        print(e)


if __name__ == "__main__":
    main()
```

2.55 LiensUtiles.txt

```
https://support.google.com/a/answer/9476255?hl=fr#zippy=%2C%C3%A9tape\
-configurez-le-service-de-migration-de-donn%C3%A9es
----------------------------------------------------------------------
https://stackoverflow.com/a/78413334/5796086
https://stackoverflow.com/questions/21612980/\
why-is-usr-bin-env-bash-superior-to-bin-bash/78413334
https://web.archive.org/web/20240501105801/https://stackoverflow.com\
/questions/21612980/why-is-usr-bin-env-bash-superior-to-bin-bash/\
78413334#78413334
----------------------------------------------------------------------
https://news.ycombinator.com/item?id=22214690
https://github.com/koalaman/shellcheck/issues/2005
https://opensource.stackexchange.com/questions/6271/\
calling-gpld-programs-from-a-shell-script
----------------------------------------------------------------------
```

2.56 miscellaneous/funny.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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```

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unprotected_funny(){
  # Works in a script but not in shell if job control is active.
  shopt -s lastpipe
  declare -i LFBFL_i_result=0
  local LFBFL_line
  printf "hahaha\n" | while read -r LFBFL_line; do
    if [[ "${LFBFL_line}" == "hahaha" ]]; then
      LFBFL_i_result=1
    fi
  done
  [[ LFBFL_i_result -eq 1 ]] && printf "I had a good laugh.\n"
}

protected_funny(){
  # Always works regarding the echoes.
  printf "hahaha\n" | while read -r LFBFL_line; do
    unprotected_funny
  done
}
```

2.57 miscellaneous/spampty_data.txt

```



a
 a
empty.txt:
one_space.txt:
all_space.txt:
normal.txt:a
normal_again.txt: a
```

2.58 README.md.tpl

```
# DevOrSysAdminScripts

```

```markdown
[![CodeFactor-badge]][CodeFactor-package-page]
[![Codacy-badge]][Codacy-package-page]
![GitHub-top-language-badge]
![GitHub-code-size-in-bytes-badge]
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|                **Some "small" useful scripts**               |
|:-------------------------------------------------------------:|
| **for enhancing development or system-administrators tasks** |

Currently, the biggest part of this repository is in
a suite of shell scripts about build and checks that
I use in some of my other repositories.

I tried to follow some conventions for my bash code:

- I tried to encapsulate most of the code in functions.
- Unless used as return values, variables in functions are defined
  as local variables (`local` keyword or `declare`)
  with prefix "LFBFL_".
- When used as return values, variables in functions are defined
  as global variables (`declare -g`) with prefix
  "@function_name@_result_".
- Whenever a variable is no more modified after some point,
  add the keyword `readonly` or use `declare -r`.
- Whenever a variable will only contain integer values (or boolean
  values as 0 or 1), use `declare -i`.
- I named bash files with suffixes ".exec.sh" when the script can
  be runned
  (executable, abbreviation stops before a vowel).
- I named bash files with suffixes ".libr.sh" when the script contains
  only functions definitions
  (library, abbreviation stops before a vowel).
- When some ".exec.sh" code is encapsulated into functions,
  either these functions can be reused and go in some ".libr.sh" file,
  either they are truly specific to this script and they are kept
  in ".exec.sh" file.
- When using `source` builtin, although any path/filename with a slash
  will only be searched for from the current directory (tested),
  we add the prefix "./", to make it explicit.
- We use mapfile instead of read -r most of the time, but with the
```

```
  following exceptions: real use of read -r to read the command line of the
  shell, comments (of course), and a file funny.sh where it is kind of
  required to use read -r.
- We replaced all echo calls by printf calls to follow
  "The Open Group Base Specifications" since at least 2 decades:
  "New applications are encouraged to use printf instead of echo.",
  <https://pubs.opengroup.org/onlinepubs/000095399/>.

But the files "pre-commit" and "post-commit" were not renamed,
since it is not possible to give them other names to use them as
pre-commit/post-commit hooks in `git`.

[CodeFactor-badge]: https://www.codefactor.io/repository/github/\
llyaudet/DevOrSysAdminScripts/badge

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3dd6ef85e8db4460991d1ebc404b68dd

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&utm_content=&utm_campaign=Badge_grade

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security-bandit-yellow.svg

[Bandit-GitHub.com]: https://github.com/PyCQA/bandit

[Code-style:black:badge]: https://img.shields.io/badge/\
code%20style-black-000000.svg

[Black-GitHub.com]: https://github.com/psf/black

[Imports:isort:badge]: https://img.shields.io/badge/\
%20imports-isort-%231674b1?style=flat&labelColor=ef8336

[Isort-GitHub.io]: https://pycqa.github.io/isort/
```

```
[Typecheck:mypy:badge]: https://www.mypy-lang.org/static/\
mypy_badge.svg

[Typecheck-mypy-lang.org]: https://mypy-lang.org/

[Linting:pylint:badge]: https://img.shields.io/badge/\
linting-pylint-yellowgreen

[Pylint-GitHub.com]: https://github.com/pylint-dev/pylint

[Linting:ruff:badge]: https://img.shields.io/endpoint?url=\
https://raw.githubusercontent.com/astral-sh/ruff/main/assets/\
badge/v2.json

[Ruff-GitHub.com]: https://github.com/astral-sh/ruff
```

2.59 unicode.libr.php

```
<?php
/*
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This file was renamed from "unicode.php" to "unicode.libr.php".
*/

function decimal_code_point_to_UTF8(
```

```
29    $i_code_point_in_decimal_notation
30  ){
31    // var_dump($i_code_point_in_decimal_notation);
32    // 0xxxxxxx ASCII
33    if($i_code_point_in_decimal_notation < 128){
34      return chr($i_code_point_in_decimal_notation);
35    }
36    $i_continuation_base_value = 128;
37    // 110xxxxx 10xxxxxx
38    if($i_code_point_in_decimal_notation < 256 * (2**3)){
39      $i_first_byte_base_value = 192;
40      $i_last_byte_significant_bits = (
41        $i_code_point_in_decimal_notation % (2**6)
42      );
43      $i_first_byte_significant_bits = intdiv(
44        (
45          $i_code_point_in_decimal_notation
46          - $i_last_byte_significant_bits
47        ),
48        2**6,
49      );
50      $s_result = chr(
51        $i_first_byte_base_value
52        + $i_first_byte_significant_bits
53      );
54      $s_result .= chr(
55        $i_continuation_base_value
56        + $i_last_byte_significant_bits
57      );
58      return $s_result;
59    }
60    // 1110xxxx 10xxxxxx 10xxxxxx
61    if($i_code_point_in_decimal_notation < 256 * 256){
62      $i_first_byte_base_value = 224;
63      $arr_arr_data_per_byte_reverse = [
64        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
65        ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
66        ["i_bits" => 4, "i_base_value" => $i_first_byte_base_value],
67      ];
68      $i_rest = $i_code_point_in_decimal_notation;
69      $s_result = "";
70      foreach($arr_arr_data_per_byte_reverse as $arr_data){
71        // _Pragma unroll
72        // J'aurais pu "roller" les 4 cas des ifs.
73        // Mais pour les perfs, il faudrait tout unroller vu le nombre
74        // d'itérations. Et j'aimerais bien comparer du code unrollé
```

```
      // aux 2 niveaux à la main avec celui unrollé par le compilateur
      // sur les 2 niveaux. Je préjuge/présuppose qu'il unroll
      // l'intérieur d'abord (pour ne pas refaire le même boulot)
      // puis l'extérieur et qu'il oublie de réoptimiser l'intérieur
      // en fonction de l'extérieur unrollé. Est-ce que j'ai raison ?
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
  if($i_code_point_in_decimal_notation < 256 * 256 * (2**5)){
    $i_first_byte_base_value = 240;
    $arr_arr_data_per_byte_reverse = [
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 6, "i_base_value" => $i_continuation_base_value],
      ["i_bits" => 3, "i_base_value" => $i_first_byte_base_value],
    ];
    $i_rest = $i_code_point_in_decimal_notation;
    $s_result = "";
    foreach($arr_arr_data_per_byte_reverse as $arr_data){
      // _Pragma unroll
      $i_significant_bits = $i_rest % (2**$arr_data["i_bits"]);
      $i_rest = intdiv(
        $i_rest - $i_significant_bits,
        2**$arr_data["i_bits"],
      );
      $s_result = (  // .= or =.
        chr($i_significant_bits + $arr_data["i_base_value"])
        .$s_result
      );
    }
    return $s_result;
  }
  throw new Exception(
    "UTF-8 avec jusqu'à 6 octets a été abandonné il y a longtemps."
    ." Autre anecdote \"débile\" : MySQL a de base UTF-8 avec jusqu'à"
    ." 3 octets au lieu de 4 :face-palm:."
```

```
121      ." Du coup, ça peut donner quelques \"surprises\"."
122      ." Sinon, y'a aussi ASCII dans MySQL."
123      ." Et comme ça joue sur la mémoire et les performances,"
124      ." pourquoi ils n'ont pas fait ASCII, UTF-8-2, UTF-8-3, UTF-8-4 ?"
125      ." :)"
126    );
127  }
128
129
130
131  function hexa_code_point_to_UTF8(
132    $s_code_point_in_hexadecimal_notation
133  ){
134    $s_code_point_in_hexadecimal_notation = strtoupper(
135      $s_code_point_in_hexadecimal_notation
136    );
137    $i_code_point_in_decimal_notation = 0;
138    $i_ord_0 = ord('0');
139    $i_ord_A = ord('A');
140    $i_digit_offset = -$i_ord_0;
141    $i_letter_offset = 10 - $i_ord_A;
142    for(
143      $i = 0, $i_max = strlen($s_code_point_in_hexadecimal_notation);
144      $i < $i_max;
145      ++$i
146    ){
147      $s_hexa_digit = $s_code_point_in_hexadecimal_notation[$i];
148      $i_code_point_in_decimal_notation *= 16;
149      if(ctype_digit($s_hexa_digit)){
150        $i_code_point_in_decimal_notation += (
151          ord($s_hexa_digit) + $i_digit_offset
152        );
153        continue;
154      }
155      if(ctype_xdigit($s_hexa_digit)){
156        $i_code_point_in_decimal_notation += (
157          ord($s_hexa_digit) + $i_letter_offset
158        );
159        continue;
160      }
161      throw new Exception("Not an hexadecimal digit.");
162    }
163    return decimal_code_point_to_UTF8(
164      $i_code_point_in_decimal_notation
165    );
166  }
```

```
167
168
169
170 function check_string_is_valid_ASCII($s_string){
171   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
172     if(ord($s_string[$i]) > 127){
173       throw new Exception(
174         "Non-ASCII character found at octet position "
175         .$i
176         ., value "
177         .ord($s_string[$i])
178         ."."
179       );
180     }
181   }
182   return true;
183 }
184
185
186
187 function check_file_is_valid_ASCII($s_file_name){
188   $s_string = file_get_contents($s_file_name);
189   if($s_string === false){
190     throw new Exception("File ".$s_file_name." not found.");
191   }
192   return check_string_is_valid_ASCII($s_string);
193 }
194
195
196
197 function check_string_is_valid_UTF8($s_string){
198   $i_continuation_octet_needed = 0;
199   $i_character_start_position = 0;
200   for($i = 0, $i_max = strlen($s_string); $i < $i_max; ++$i){
201     $i_current_octet = ord($s_string[$i]);
202     if($i_continuation_octet_needed > 0){
203       if($i_current_octet < 128 || $i_current_octet >= 192){
204         throw new Exception(
205           "Non-UTF8 character found at start position "
206           .$i_character_start_position
207           ., octet at position "
208           .$i
209           ." has value "
210           .$i_current_octet
211           ." which is not a continuation octet."
212         );
```

```
213         }
214         --$i_continuation_octet_needed;
215         continue;
216       }
217
218       $i_character_start_position = $i;
219       if($i_current_octet < 128){ // 0xxxxxxx ASCII
220         continue;
221       }
222       if($i_current_octet >= 128 && $i_current_octet < 192){
223         throw new Exception(
224           "Non-UTF8 character found at start position "
225           .$i_character_start_position
226           ., octet at position "
227           .$i
228           ." has value "
229           .$i_current_octet
230           ." which is a continuation octet."
231         );
232       }
233       if($i_current_octet >= 192 && $i_current_octet < 224){
234         // 110xxxxx 10xxxxxx
235         $i_continuation_octet_needed = 1;
236         continue;
237       }
238       if($i_current_octet >= 224 && $i_current_octet < 240){
239         // 1110xxxx 10xxxxxx 10xxxxxx
240         $i_continuation_octet_needed = 2;
241         continue;
242       }
243       if($i_current_octet >= 240 && $i_current_octet < 248){
244         // 11110xxx 10xxxxxx 10xxxxxx 10xxxxxx
245         $i_continuation_octet_needed = 3;
246         continue;
247       }
248       throw new Exception(
249         "Non-UTF8 character found at start position "
250         .$i_character_start_position
251         ., octet at position "
252         .$i
253         ." has value "
254         .$i_current_octet
255         ." which is invalid."
256       );
257     }
258     return true;
```

```
}



function check_file_is_valid_UTF8($s_file_name){
  $s_string = file_get_contents($s_file_name);
  if($s_string === false){
    throw new Exception("File ".$s_file_name." not found.");
  }
  return check_string_is_valid_UTF8($s_string);
}



/*
?>
<?php
require_once("./unicode.libr.php");
var_dump(hexa_code_point_to_UTF8("002B"));
var_dump(hexa_code_point_to_UTF8("00E6"));
var_dump(hexa_code_point_to_UTF8("1400"));
var_dump(hexa_code_point_to_UTF8("10111"));
// https://en.wikibooks.org/wiki/Unicode/Character_reference/2000-2FFF
// Some invisible unicode characters that are a problem:
// 2000-200F 2028-202F 205F 2060-206F
$arr = [
  '2000', '2001', '2002', '2003', '2004', '2005', '2006', '2007',
  '2008', '2009', '200A', '200B', '200C', '200D', '200E', '200F',
  '2028', '2029', '202A', '202B', '202C', '202D', '202E', '202F',
  '205F',
  '2060', '2061', '2062', '2063', '2064', '2065', '2066', '2067',
  '2068', '2069', '206A', '206B', '206C', '206D', '206E', '206F',
];
$arr_s = [];
$arr_s2 = [];
$arr_s3 = [];
foreach($arr as $s_hexa){
  $s = hexa_code_point_to_UTF8($s_hexa);
  $arr_s []= $s;
  if(strlen($s) === 2){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1]);
  }
  if(strlen($s) === 3){
    $s2 = '\x'.bin2hex($s[0]).'\x'.bin2hex($s[1])
      .'\x'.bin2hex($s[2]);
  }
```

```
  $arr_s2 []= $s2;
  $s3 = "  | sed -e 's/".$s2."/ /g'\\";
  $arr_s3 []= $s3;
  echo($s3."\n");
}
//*/

```

2.60 upgrade_build_and_checks.exec.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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# This file was renamed from "upgrade_build_and_checks.sh"
# to "upgrade_build_and_checks.exec.sh".

upgrade_build_and_checks(){
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
    printf "%s %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  local LFBFL_fixed_point_build=""
  if [[ "$*" == *--fixed-point-build* ]]; then
    LFBFL_fixed_point_build="--fixed-point-build"
  fi
```

```
  readonly LFBFL_verbose
  (\
    cd build_and_checks_dependencies/\
    && ./update_common_build_and_checks.exec.sh "${LFBFL_verbose}"\
  )
  ./build_and_checks.exec.sh "." "${LFBFL_verbose}"\
    "${LFBFL_fixed_point_build}"
}

upgrade_build_and_checks "$@"
```

2.61 wget_sha512.libr.sh

```
#!/usr/bin/env bash
# This file is part of DevOrSysAdminScripts library.
#
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#
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# This file was renamed from "wget_sha512.sh"
# to "wget_sha512.libr.sh".

wget_sha512(){
  # $1 file_path
  # $2 download_URL
  # $3 correct_sha512
  # $4 optional --verbose
  local LFBFL_verbose=""
  if [[ "$*" == *--verbose* ]]; then
```

```
    printf "%s wget_sha512 %s\n" "$0" "$*"
    LFBFL_verbose="--verbose"
  fi
  readonly LFBFL_verbose
  if [[ ! -f "$1" ]]; then
    wget "${LFBFL_verbose}" -O "$1" "$2"
  fi
  declare -r LFBFL_present_sha512=$(
    sha512sum "$1"\
    | cut -f1 -d' '
  )
  if [[ "${LFBFL_present_sha512}" != "$3" ]]; then
    printf "%s does not have correct sha512:\n - wanted %s\n - found %s\n"\
      "$1" "$3" "${LFBFL_present_sha512}"
    return 1
  fi
}
```

Merci Dieu ! Merci Père ! Merci Seigneur ! Merci Saint Esprit !